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REAL-TIME CRIME CENTERS AS FRONTIERS OF TECHNOLOGY IN POLICING: CONFIGURATIONS, FUNCTIONS, AND THEIR IMPACT ON POLICE PRACTICES AND PUBLIC PERCEPTIONS

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DEDICATION

It is with utmost gratitude that I dedicate this dissertation to my parents. As my earliest mentors, they instilled within me a profound appreciation for the transformative power of education, shaping me into the scholar I am today.

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ABSTRACT OF THE DISSERTATION

REAL-TIME CRIME CENTERS AS FRONTIERS OF TECHNOLOGY IN POLICING: CONFIGURATIONS, FUNCTIONS, AND THEIR IMPACT ON POLICE PRACTICES AND PUBLIC PERCEPTIONS

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The purpose of this dissertation was to fill a considerable void in research pertaining to the implementation and use of real-time crime centers (RTCCs) through the application of numerous analytical strategies. For one, a key contribution of this dissertation was the development of an RTCC typology that drew upon theoretical perspectives deeply rooted in organizational change literature. Specifically, the typology classified existing RTCC models into three categories: traditional, community-based, and virtual. This approach provided a systematic framework for understanding the varied structures and functions of RTCCs nationally. Further, a quasi-experimental design was employed to determine the influence of the technologies housed within RTCCs on the case processing, case clearance and time to case clearance of homicides, aggravated assaults, robberies, and property crimes. Finally, an online experimental vignette study, using a randomized controlled trial (RCT) research design, was deployed to assess the determinants that impact the public's approval of the use and deployment of technologies commonly procured by RTCC units. The overall findings revealed that the influence of RTCC technologies on the solvability of crime are highly

dependent on the inherent case characteristics or solvability factors of each crime type considered. Further, it was evidenced that the approval of police technology is highly dependent on perceptions of police legitimacy and effectiveness, as well as the neutral broadcasting of messaging related to the utility of these technologies. Policy implications are heavily geared toward law enforcement agencies.

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ACRONYMS

ALPR – Automated License Plate Reader

CAD – Computer Aided Dispatch

CCTV – Closed Circuit Television

CPD – Chicago Police Department

CSI – Crime Scene Investigation

CVPD – Chula Vista Police Department

DHS – Department of Homeland Security

DNA - Deoxyribonucleic Acid

FAA – Federal Aviation Administration

FBI – Federal Bureau of Investigation

FRT – Facial Recognition Technology

GIS – Geographic Information System

LFR – Live Facial Recognition

MDT – Mobile Data Terminal

MPS – Metropolitan Police Service's

MRTCC- Miami Real Time Crime Center

NIBIN - National Integrated Ballistic Information Network

NIBRS - National Incident-Based Reporting System

NIMS - National Incident Management System (NIMS)

NYPD – New York Police Department

POP – Problem-Oriented Policing

RCT – Randomized Controlled Trial

RTCC – Real-Time Crime Center

SDSC – Strategic District-Level Support Centers

SOP – Standard Operating Procedures

Chapter One: Introduction

As the proverb "necessity is the mother of invention" suggests, the driving force behind all technological advancement is the need to meet society's demands. From a broad perspective, the progress of technology is intimately tied to the populace's behavior and enables global progress by improving societal efficiency. At a more micro level, technology can augment the operational procedures utilized to attain specific objectives set forth by an organization. Accordingly, the success of policing hinges on the strategic adoption of technological innovations to enable the acquisition of actionable intelligence information to support short and long-term organizational goals (Kelley & Abrials, 2003). After all, as stated by Maltz, Freidman & Gordon (1991), information is the lifeblood of police work.

While it may not always align with reality, detectives and investigators are often referred to as an elite group of individuals and portrayed by the media as possessing exceptional intuition and analytical skills to solve even the most complex cases (Bayley, 1994; Cole & Porter, 2017). Thereby, the ability of law enforcement organizations to bring cases to a successful resolution is also frequently used as a metric for assessment by the populace and local leaders. This notion, combined with the true complexities of the field, becomes particularly relevant in the context of violent crime, as the rising homicide rates and the simultaneous decrease in homicide clearance rates in some municipalities have prompted growing concerns (Mancik, Parkey & Williams, 2018) Despite extensive research, however, there is a lack of consensus on the factors differentiating successful from ineffective investigative techniques. Moreover, in light of the recent advancements

in technology and the apparent fragmentation of community and police relations, there is also a pressing need to understand the interplay of these factors on the decision-making process of investigations (Colvin & Goh, 2005).

The impetus to maximize the use of emerging technologies for the safer and more efficient execution of core functions is motivated by the constantly changing landscape of challenges, harms, and threats confronted by law enforcement (Custers & Vergouw, 2015; Mastrofski & Willis, 2010; Wright & Headley, 2021). Intuitively, the exponential increase in the availability of digital evidence should expedite case resolution and augment the quality of outcomes. However, not all law enforcement agencies possess the technical expertise to handle the demands associated with these tools, and they do not always provide sufficient training to their officers to effectively utilize their potential (Strom, 2017). Nevertheless, law enforcement agencies argue that technology is necessary to combat the technological savvy of criminals, as many agencies feel they are now being *out-teched* rather than *outgunned*. As noted by a former President of the International Association of Chiefs of Police, "police departments must be up to the same technological standards as criminals" (Nunn, 2001).

Real-Time Crime Centers

This drive to enhance intelligence capabilities and expand the range of technologies available has led to the emergence of investigative units commonly referenced as real-time crime centers (RTCC) (Przeszlowski et al., 2022). RTCCs are equipped with various technologies, such as facial recognition systems and gunshot detection systems, and have diverse applications with respect to jurisdictional needs. Their primary objective, however, is to provide immediate information to officers on ongoing

or recent criminal activities, hence the “real-time” assertion. Despite relying mainly on anecdotal evidence, these units have reportedly been highly successful in helping police officers apprehend suspects within a shortened timeframe and have been able to prevent more serious offenses from taking form.

For example, a simple search engine query on their impact on policing reveals numerous news articles that underscore their benefits to society. For instance, in October of 2021, a shooting occurred in an East Birmingham, Alabama, public housing community, resulting in one death and one injury to a minor. Shortly after the call came through the computer-aided dispatch system, detectives within the RTCC unit accessed closed-circuit television (CCTV) footage of the incident and provided detailed information on the circumstances of the crime to responding officers before they arrived at the scene. This swift action led to the suspect being apprehended within minutes, leaving Birmingham Police Chief Patrick Smith to conclude, “this is how the real-world, modern-day policing should work” (Robinson, 2021).

Although RTCCs have gained significant attention in recent years, there is a dearth of empirical evidence regarding their impact and effectiveness in police operations. Currently, the Chicago Police Department's Strategic Decision Support Centers (Hollywood et al., 2019) and the Miami Police Department's centralized Real-Time Crime Center are among the few evaluated RTCCs in the country (Guerette et al., 2022). While these evaluations have demonstrated some positive impacts on crime reduction and improved clearance rates for violent and property crimes, the totality of research available on RTCCs remains in its infancy compared to the rapid diffusion of these units across the nation.

Problem Statements

Problem One: *Despite the establishment and proliferation of real-time crime centers across the United States since 2005, there has been a notable absence of scholarly research on the marked disparities in how law enforcement agencies adopt and operationalize these units.*

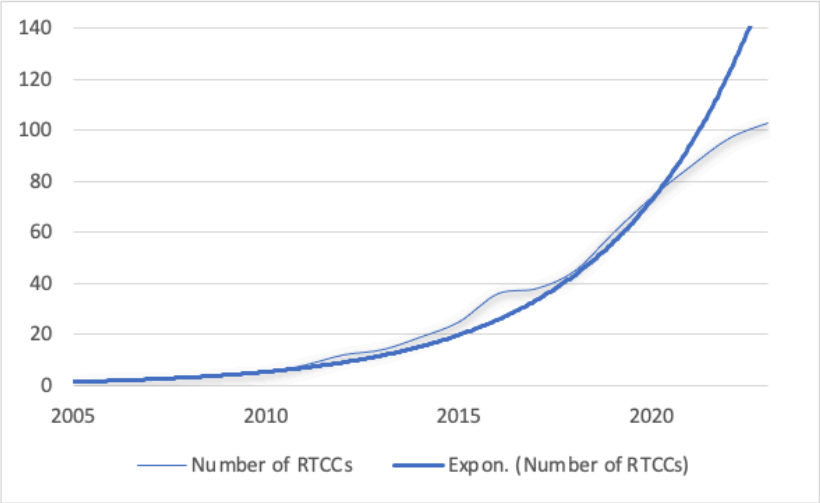
In recent decades, scholarly investigations have unveiled a consistent pattern of continuity and change in American policing. However, as noted by Mastrofski & Willis (2010), significant changes in policing may require a longer period to materialize, making it crucial to observe incremental changes over extended periods. This process of manifestation is perhaps best evidenced by real-time crime center units, which emerged nearly two decades ago but are only now receiving considerable attention.

Although grouped under an all-encompassing term, real-time crime centers are characterized by their distinct development by individual agencies in accordance with internal procedures and external stakeholder expectations. Consequently, various models have surfaced over the past two decades to accommodate this heterogeneity. Such diversity reflects the intricate nature of these centers and the diverse needs of the agencies and communities they serve. The insufficient scholarly literature dedicated to comprehensively elucidating the adoption process and decision-making involved in establishing these units nationwide is problematic for several reasons. Among them is the inability to identify universally applicable best practices, which can, in turn, lead to less than optimal outcomes and the squandering of valuable resources.

As earlier acknowledged, in 2005, the New York City Police Department (NYPD) pioneered the establishment of a real-time crime center to identify criminal patterns and

prevent emerging crimes. However, as of 2012, only ten agencies, including the Philadelphia Police Department, had access to such a comprehensive system of information and analytical tools (Gibson, 2019). Since then, the implementation of real-time crime centers has rapidly increased, with an additional 93 units established by 2013 (refer to Figure 1). This surge highlights the growing need among law enforcement agencies to enhance policing efficiency, streamline intelligence and analysis, and promptly disseminate information to officers in the field (Wuschke et al., 2018).

Figure 1. *Cumulative growth of RTCCs in the United States (2005 – 2023)*



Note. N=103[1]

RTCCs demonstrate substantial variability in their composition and operation among police agencies, lacking an established standard model (Przeszlowski et al., 2022). The successful implementation of an operational RTCC often relies on several conditions, such as the availability of resources, the nature of local crime problems, and the agency's technological complexity level. Due to the considerable differences in organizational

[1] For detailed information on data collection procedures, please refer to Przeszlowski et al. (2022), from which this figure was adapted from.

cultures between agencies, it is not recommended to merely duplicate another agency's approach or heavily rely on external evaluations without considering their similarities. For instance, a solution effective in a large city such as San Antonio, Texas, with a population of 1.5 million, may not be as effective in a smaller town like Salisbury, Maryland, with a population of 33,000, both of which have their RTCCs.

Incorporating RTCCs into police operations also necessitates a substantial financial commitment, and stakeholders such as investors, policymakers, and others interested in the RTCC's success are concerned about the return on investment. The costs of establishing a long-term infrastructure for RTCCs vary from department to department, with larger police agencies typically requiring more than a million dollars. For instance, the New York Police Department invested almost 11 million dollars in its RTCC. At the same time, the Norfolk Police Department incurred initial costs of $1.96 million and anticipated recurring expenses of $661,500 for staffing and operations. Although state and federal grants are available for funding, police agencies must carefully assess whether the technology they acquire aligns with the RTCC's mission and justifies the expenses. The experience of the Fresno Police Department's RTCC illustrates how inadequate funding and insufficient internal and external support can result in program shutdowns, as they halted operations in 2020 due to budget shortages.

Problem Two: *The existing body of literature on evaluating real-time crime centers is constrained to two assessments of RTCC units, each exhibiting notable differences in implementation and jurisdiction-specific objectives. Furthermore, the overarching research on innovative policing technologies prioritizes their capacity for crime deterrence rather than their impact on improving clearance rates.*

The policing landscape is experiencing a dynamic transformation driven by the rapid influx of technological innovations occurring at an unprecedented pace. Despite efforts to evaluate the impact of these technological advancements, the speed of progress has surpassed such endeavors, resulting in an inconclusive body of literature (Byrne & Marx, 2011; Chan, 2001). Furthermore, current attempts have mainly focused on the factors that influence homicide clearance rates, with considerably less attention paid to aggravated assaults, robberies, and property crimes, especially concerning the role played by technologies in the rapidly proliferating real-time crime centers.

While technological innovation may logically enhance organizational efficiency, the perceived benefits of speed and convenience have not led to concrete improvements in case clearance rates or enhanced community relations (Lum, Koper & Willis, 2017). Despite the growing availability of technological tools and resources for criminal investigations, clearance rates in the United States have also remained consistently low and relatively unchanged over the past four decades (Eck & Rossmo, 2019; Leipold, 2021). Homicide clearance rates have decreased by 30.56% compared to 1980, with the highest clearance rate reaching only 50%. However, clearance rates for assault and robbery cases are significantly lower, at 47% and 27%, respectively (Li & Lartey, 2022). Clearance rates for property crimes, such as burglary and larceny-theft, are even lower, with only 14% and 15%, respectively.

Despite many potential factors contributing to low clearance rates, a complete understanding of the complex interplay between technological innovation and policing outcomes remains elusive. Some argue that the investigative process has not undergone significant experimentation or innovation and that detective work remains reactive and

entrenched in the traditional professional policing model, with crime-solving objectives prioritizing justice, deterrence, and offender incapacitation (Eck & Rossmo, 2019). Thus, this area of this area of literature certainly necessitates further exploration.

Problem Three*:* *As exemplified by the 'defund the police' movement, the sentiment of the populace and their endorsement of policing methodologies carries significant implications for the future of law enforcement However, when it comes to the public's support for police technology, scant research has delved into technologies beyond the scope of facial recognition and unmanned aerial vehicles (i.e., drones).*

Technological advancements in recent decades have given police officers greater capabilities to collect information and conduct surveillance during their law enforcement duties (Heen, Liberman & Miethe, 2017). However, with these advancements, renewed concerns about privacy and ethical issues have arisen, bringing into question the appropriate use of state power and the level of transparency in police activities (Marx, 2007; Von Hirsch, Garland & Wakefield, 2000). In light of controversial police-citizen interactions recorded through "sousveillance" and in the aftermath of some of the largest protests in American history, it is apparent that police agencies now face the challenge of enhancing and repairing their relationship with members of the community (Bragias, Hine, & Fleet, 2021; Pryor, Buchanan, & Goff, 2020).

Studies have revealed varying public attitudes toward the adoption of advanced police technologies, with both negative and positive reactions reported (Mohler et al., 2021; Saulnier & Thompson, 2016). Concerns regarding excessive surveillance and violations of personal privacy are commonly cited as reasons for opposition. While individuals typically welcome technological innovations that offer convenience and efficiency to their personal

lives, such as facial recognition technology for secure access on smartphones or face authentication in banking apps, the same technology employed for policing is often met with skepticism (Bromberg et al., 2018; Hirose, 2017). Law enforcement agencies face the challenge of navigating these contrasting attitudes and striking a balance to ensure the successful adoption of advanced technologies.

Scholars suggest that the public's acceptance of police technology is dependent on the task being performed or the reason for its use (Saulnier & Thompson, 2016). When people lack information regarding the technology's specific purpose, or there is no clear victim, they may have more concerns about the motives behind surveillance (Sakiyama et al., 2020). For example, research has indicated that while people support the use of unmanned aerial drones (UAVs) in reactive policing situations such as search and rescue operations, they express caution about their deployment for proactive purposes, such as monitoring large crowds (Sakiyama, 2017). Therefore, further research is necessary to identify the factors determining public acceptance of technology deployment in communities and ensure the successful and sustainable implementation of real-time crime centers within law enforcement agencies.

Purpose and Significance

The central objective of this dissertation is to evaluate the impact of real-time crime center technologies on the handling of violent and property crime cases, with a focus on case processing, clearance rates, and time-to-case clearance, using a quasi-experimental approach. Building upon a previous evaluation of the Miami Real-Time Crime Center unit, which examined the influence of these technologies on a broader model of violent crime,

this research aims to identify specific crime features and solvability factors that may emerge when analyzing crime categories in a disaggregated manner.

In addition to the primary aim, this dissertation seeks to achieve several secondary objectives. One such objective is to develop a typology of RTCCs that accounts for the variations in their implementation across regions and jurisdictions. It is noted that the definition and implementation of RTCCs can vary considerably, with the potential for modification as technology advances and new challenges arise. To address this variation, the study will draw upon organizational perspectives previously applied to analyze the adoption of police innovations by various agencies. Through the application of these theoretical frameworks, the dissertation aims to elucidate the factors that shape the creation and operation of RTCCs and to identify the organizational features that are most closely associated with their effective implementation. Ultimately, this research will contribute to a deeper understanding of the challenges involved in establishing and managing critical law enforcement units and provide insights that can inform the development of more effective policies and strategies for crime prevention and public safety.

The final study within the dissertation will utilize a randomized controlled trial research design to assess the public's perception of police technology. Although surveillance has become a prominent institutional dimension of modern societies, the extent to which the public is informed and accepting of these systems is often unclear and not a primary concern of police organizations. Nonetheless, with the interactions and relationships between police officers and communities increasingly strained and fragmented, this has become an area of heightened interest for police organizations and policymakers. The results of this investigation are anticipated to illuminate the factors that

shape the public's acceptance of police technology and identify optimal approaches that can improve the effectiveness and legitimacy of these technologies in the eyes of the public.

Overview of Chapters

This dissertation is arranged in the following manner. Chapter one will present an overview of the existing literature on police investigations, specifically emphasizing the two main theoretical frameworks commonly employed by researchers to better comprehend the various factors that can influence investigators' ability to solve cases – namely, the devaluation thesis and the solvability perspective.[2] Additionally, chapter two will provide a detailed account of the historical progression of police practices and the cutting-edge technologies that have emerged over time, leading to the widespread integration of real-time crime centers within law enforcement agencies. The chapter will conclude by providing an overview of the concerns related to the deployment and usage of police technology, with a particular focus on the ramifications of big data policing,

Chapter three builds on the availability of anecdotal evidence documenting the adoption and implementation of RTCCs nationally to construct a typology that better denotes the distinct differences that exist. The typology consists of a traditional, community, and virtual model. Further, using organizational perspectives, the chapter posits a theory-driven framework that enhances our comprehension of the distinctions regarding the adoption, implementation, and decision-making mechanisms linked with unit structure.

[2] It is important to highlight that the dissertation will include a literature review specific to each type of crime examined within the corresponding section dedicated to analyzing that particular crime.

Following this contextual chapter, chapter four introduces the first quasi-experimental study on the influence of RTCC technologies on case processing, case clearance, and time to case clearance of homicide and aggravated assault incidents. The chapter also elaborates on the data collection process and explains the measures employed in both the treatment and control groups. It is worth noting that the data gathering methodology is consistent throughout chapters five and six, although the criminal categories being analyzed differ. Chapter five assesses incidents of robbery, while chapter six directs its focus towards property offenses.

In chapter seven, an investigation is conducted to assess the public's viewpoint on using and implementing police technology by utilizing an experimental vignette study design. The study aims to evaluate the extent of public acceptance of the technology commonly associated with the decision-making processes of law enforcement agencies that operate real-time crime center units. The research is geared towards achieving two primary goals: (1) to establish if the language and information associated with police technology use and deployment can bolster the public's support of these technologies, and (2) to determine the factors that hold the most significant sway over the public's endorsement of police technology.

The concluding chapter of this dissertation, chapter eight, will provide a comprehensive discussion of the findings obtained throughout the individual studies and derive implications firmly grounded in the criminal investigations' literature. In addition, this section will provide practical policy recommendations that can assist law enforcement in strengthening police-community relationships, as well as acknowledge any limitations of the research.

Chapter Two: Review of Foundational Literature

Introduction

This dissertation contributes to the existing body of literature on police investigations by exploring the impact of innovative police technologies on the processing of violent and property crime incidents and public perceptions regarding their deployment and use. To provide context for the ensuing studies, this chapter reviews the current state of the literature on police investigations, focusing on the two primary theoretical frameworks employed by scholars to better understand the factors that can impact the ability of investigators to resolve cases.[3]

Following this review, the chapter turns its attention to the impact of police technology on the investigation of crimes, detailing the evolution of police practices that have led to the widespread adoption of the highly tech-driven units known as real-time crime centers. Finally, the chapter concludes with an overview of the concerns associated with using and deploying police technology, with particular attention paid to the implications of big data policing.

Contextualizing Criminal Investigations

The function of criminal investigations is a critical aspect of law enforcement designed to achieve two primary objectives. The first involves identifying and apprehending criminal offenders through the information-gathering process, which establishes probable cause for their arrest (Lee, 2020). The second involves collecting and presenting evidence and testimony in court to secure convictions (Gottschalk, 2006). To

[3] Each chapter will also contain a more specific review of relevant literature on the crime type or topic under investigation.

achieve these objectives, criminal investigations are generally divided into two main domains of operation: (1) the preliminary investigation, which uniformed patrol officers typically conduct, and (2) the follow-up investigation, which is performed by detectives who possess specialized training in investigative techniques and are typically assigned to detective bureaus (Thibault, Lynch & McBride, 2004).

Criminal investigations play a crucial role in improving clearance rates. The clearance rate of a law enforcement agency is defined as the ratio between the number of cases cleared by the arrest of an offender or by an exceptional clearance within an established period (e.g., 12 months) and the number of crime incidents reported to the police (Federal Bureau of Investigation, 2017). A high clearance rate indicates that a law enforcement agency has successfully solved a significant proportion of reported cases, leading to a higher number of cleared cases. Law enforcement agencies strive to achieve good case clearance rates as it has been found to have several positive outcomes for the agencies and the communities they serve.

One favorable outcome is improving citizen perception of police efficacy and legitimacy (Scott, Wellford, Lum & Vovak, 2019). Research has shown that citizens are more likely to trust and cooperate with law enforcement agencies that have demonstrated a successful record of solving crimes, further contributing to public satisfaction and confidence in the police force (Tyler, 2004). Conversely, the inability of law enforcement agencies to clear cases can lead to unfavorable outcomes. It may tarnish the agency's reputation, leaving victims and their families without resolution or reparation (Phillips, Drake, & Altheimer, 2022; Tyler, 2004). Such ramifications can ultimately compromise public safety and hinder the agency's capacity to perform its duties efficiently. The

inability of law enforcement agencies to achieve high clearance rates may further significantly impede their effectiveness in carrying out their responsibilities and upholding community trust (Byrne & Marx, 2011; Koper, Taylor, & Kubu, 2009).

Nonetheless, police agencies face an ongoing challenge to improve the quality of investigations. Prior research suggests that potential setbacks may arise from the decentralized nature of investigative units (McCluskey et al., 2014), inadequate departmental efforts, inconsistencies in established protocols (Braga & Dusseault, 2018), the overabundance of administrative tasks and excessive workload assignment (Puckett & Lundman, 2003; Rydberg & Pizarro, 2014), and the improper utilization of innovative forensic and information technologies (Baskin & Sommers, 2012; Carter & Grommon, 2014; Schroeder & White, 2009), amongst many others.

Foundational Work on Criminal Investigations

In the wake of the President's Commission on Law Enforcement and Administration of Justice in 1965, there was a notable increase in federal funding and corresponding initiatives that led to a significant body of research dedicated to the study of law enforcement and investigative processes (Horvath, Messig & Lee, 2001). One of the most notable attempts to evaluate the effectiveness of investigative processes, the role of technological advancements in augmenting these efforts, and the overall management of investigative processes was Greenwood and colleagues (1975) seminal Rand Report. To capture the complexity of this domain, the scholars surveyed 156 police agencies and conducted observations and interviews across twenty-five different departments.

For years, this study was considered the most comprehensive conceptual framework for crime clearance and served as a model research design for investigating the

myriad factors contributing to successful case resolution. Ultimately, the report highlighted the critical role played by the victim's initial account to the first responding patrol officer in resolving a case. It also noted that detective investigative efforts were hindered by excessive administrative obligations that diverted their attention from crime-solving pursuits. Investigative "special action," or non-routine initiatives taken by investigators, only made a noticeable difference in homicide, robbery and burglary case clearances. These findings' implications for the criminology and police agencies and their resource allocation were significant, further underscoring the importance of effective and efficient investigative processes in the successful resolution of criminal cases.

Despite the significance of the Rand Report's contributions to understanding crime clearance, it has been criticized for its broad generalizations and non-representative sample selection (Gates & Knowles, 1976). To address these limitations, Horvath et al. (2001) surveyed a larger, randomly selected, and nationally representative sample of state, county, and municipal agencies (n = 3,123). In this study, Horvath and colleagues presented a list of twenty factors related to clearance rates. They asked participating agencies to indicate how implementing them would improve their clearance rates. The results indicated that the top two selected factors were increased investigator personnel and technological improvements. The third most selected option was formal training upon appointment as an investigator, followed by more time to work on unsolved cases. These findings further highlighted the importance of properly allocating resources to support personnel, technology, and training to enhance the efficacy of investigative processes and ultimately improve crime clearance rates.

Wellford and colleagues (2019) replicated the methodological approach of the RAND studies. They utilized a multiagency, multi-method, and detailed case file approach to examine the relationship between investigative effort, case characteristics, organizational factors, and homicide case clearance rates. The study also investigated the role of organizational factors in this relationship. Through this systematic examination (rather than convenience sampling as employed by the RAND study, it was discovered that the investigative effort and implementation of organizational best practices are crucial in the clearance of individual homicide cases, in addition to certain crime-specific features, such as the familiarity between victim and perpetrator, the identification of the offender, or the recovery of the weapon. Notably, the clearance probability of a case was predicated on factors such as the success of investigators in eliciting cooperation from witnesses, and the number of investigators present at the crime scene. These factors continued to exhibit statistical significance even after controlling for the level of investigative best practices implemented by the agency and, overall, contradicted the findings of the RAND report.

Further scholarship has revealed that the intricacy of detectives' roles surpasses the previously held assumptions (Prince, Lum & Koper, 2021; Worral, 2019). For instance, Wellford et al.'s (1999) research identified a total of 51 characteristics related to homicide events and investigative processes that significantly contributed to case clearance. Out of these, thirty-seven were linked to the investigative process, such as the number of detectives involved, the approach taken during interviews, and other related factors. These results suggest that the specific policies and procedures of law enforcement agencies play a critical role in solving cases, especially those related to homicides. Moreover, additional internal characteristics, such as limited capacity to carry out investigations, failure to seek

out witnesses, improper handling of evidence, and insufficient cooperation between police and forensic units, also contribute to successful case clearance.

According to Keel, Jarvis, and Muirhead's (2009) research, respondents linked higher clearance rates to investigator training and sophisticated analyses. Hsu's (2007) findings suggest that resources, particularly the number of officers, percentage of investigators, and computer use, have the potential to enhance homicide clearance rates overall. Schroeder and White's (2009) study identified that detectives generating new leads and conducting computer checks on suspects have a positive impact on clearance rates. Additionally, Fallik's (2018) study revealed that the amount of time spent interrogating suspects was the most significant factor affecting case clearance rates, with increased interview time leading to more solved cases.

Moreover, in their quest to enhance investigative and organizational practices for homicide investigations, Braga and Dusseault (2018) evaluated the problem-oriented policing approach in the Boston Police Department (BPD). The intervention, developed after a thorough examination of BPD investigative procedures, aimed to bolster the capacity and effort expended on homicide investigations, including response times, the number of detectives and officers present at the crime scene, witness involvement, forensic evidence analysis, and the application of information technology. The study's results indicated that the problem-solving intervention significantly boosted investigative activities, which, in turn, augmented the likelihood of clearing homicide cases during the intervention period compared to the pre-intervention period. Therefore, the study indirectly demonstrated that enhancing specific investigative practices and resources can shift clearance rates, highlighting the importance of organizational factors and elements

of the investigative process within a law enforcement agency, not just the elements of the crime itself.

Thus, despite the groundbreaking Rand Report being published more than four decades ago, a consensus on the defining traits that differentiate successful investigative methods from those that fail has yet to be reached. Although most research has concentrated on homicide investigations, recent discoveries indicate that organizational elements and investigative techniques may substantially impact case outcomes, extending beyond the specifics of the crime. Thus, this dissertation sheds light on an overlooked area of research, specifically, the role of centralized information technologies in investigating and resolving violent and property crime incidents and while the primary focus of this dissertation is not on theory testing, it is worthwhile to examine the theoretical frameworks that arise from subsequent case clearance literature. Among these, the devaluation thesis and the solvability perspective are particularly instructive and commonly used to scrutinize conventional accounts of case processing.

Devaluation Thesis

The devaluation thesis, also known as the discretionary framework, is a theoretical perspective in the field of criminal justice that posits that the level of investigative effort assigned to a crime is influenced by the social status and perceived respectability of the victim (e.g., such as those who have a prior arrest record) (Schroeder & White, 2009). The roots of this perspective can be traced back to Black's (1976) theory of law, which suggests that a person's social position determines the amount of "law" they receive. In the context of clearance of violent crime, this approach assumes that police discretion plays a role in investigations based on extralegal factors such as victim demographics and socioeconomic

status (Riedel, 2008), in that crimes against victims who are non-white, female, or younger, are less likely to be cleared than their counterpart. [4]

Empirical investigations into the assertion that members of racial minorities face a reduced likelihood of being cleared of criminal charges have produced varied outcomes. Some noteworthy studies have found that violent crimes perpetrated against non-white victims are less likely to be cleared than those against white victims (Addington, 2007; Roberts & Lyons, 2009). Conversely, other studies have reported no apparent racial effect (Ousey & Lee, 2010; Regoeczi & Jarvis, 2013) or suggested lower chances of clearance for violent crimes committed against white victims (Addington & Rennison, 2008; Trussler, 2010).

Despite these mixed findings, using the devaluation concept in evaluating policing practices and outcomes has illuminated areas that require improvement to ensure fair and effective policing. Racial disparities in law enforcement have been a topic of discussion for decades, with numerous studies highlighting the disproportionate impact of policing on people of color (Alpert, Dunham, & Stroshine, 2014). The influence of social status on the clearance of violent crimes emphasizes the need for a more impartial approach to law enforcement, in which the level of investigative effort assigned to a crime is primarily determined by its solvability factors rather than victim characteristics. As discussed in the subsequent section, the shift to a solvability perspective highlights the importance of

[4] Upon arriving at the scene of an incident, it is improbable for law enforcement officials to possess knowledge of the victim's socioeconomic standing. However, the socioeconomic status of the geographical area where the incident occurred is expected to be commonly known and could potentially inform the course of the investigation (Litwin, 2004).

thorough investigation and the collection of quality evidence in the criminal justice process.

Solvability Perspective

The solvability perspective, also referred to as the full investigation framework, is a theoretical construct in the field of criminal justice that advocates for comprehensive investigations of all crimes, regardless of the victim's social status or perceived reputation (Schroeder & White, 2009).[5] This viewpoint posits that the severity of a crime is directly proportional to the quantity of criminal law applied, and that non-discretionary factors do not significantly influence the clearance status of a case. In essence, the solvability perspective offers a counter-narrative to the idea that police discretion has a considerable impact on law enforcement.

Recent years have witnessed an upsurge in empirical studies examining the solvability perspective's implications for policing practices and outcomes. Research consistently shows that the quality and quantity of evidence gathered during investigations strongly correlate with crime clearance rates. Huff and Killias (2014), demonstrated the significant impact of forensic evidence on the success of investigations, while more recently Morrow, Katz & Choate (2016) established that cases with high-quality evidence were more likely to be solved. Moreover, the availability of witnesses and the cooperation of community members have also been proven crucial in solving crimes. Jackson and colleagues (2012) have emphasized the importance of community policing in fostering

[5] Olphin and Mueller-Johnson (2019) define solvability factors as information items, including leads, that are either components of the crime or available for investigators to act upon and which together determine the likelihood of solving a crime.

trust and cooperation with the public, while Lum and Nagin (2017) have highlighted the potential of social networks and community involvement in improving investigative quality.

All in all, the solvability perspective provides a critical theoretical framework for understanding the factors that contribute to the clearance of violent and property crimes. Its emphasis on thorough, systematic investigative practices, coupled with a focus on evidence-based approaches and community engagement, offers valuable insights for developing effective policing strategies and policies.

Evolution of Technology within Police Practice

The history of policing can be divided into four distinct eras: the political era, the reform or professional era, the community problem-solving era, and the homeland security era. These eras have shaped the current practices of policing and have contributed to the development of four different policing models: traditional, community, problem-oriented, and zero-tolerance policing (Oliver, 2006). The professional era is often considered the beginning of technological innovations in policing, but elements of technology can be traced back to the 1840s. To fully understand the current state of technology in policing, it's important to consider the historical journey that has influenced it.

Political Era

The advent of the political era in the mid-19th century marked the establishment of the first legitimate law enforcement agencies in the United States (Oliver, 2006). However, owing to the absence of a centralized mandate for their organization, police agencies relied on local municipal resources and authorization, resulting in corrupt relationships with politicians and a history of abusive use of power. Some scholars have even suggested that

early police agencies were essentially an extension of political machines (Fogelson, 1977; Jordan, 1972). Despite this, during this period, the primary focus of law enforcement was to offer various social services to community members and cater to the needs of local politicians. The decentralized structure of policing at the time facilitated close relationships between officers and community members, given that officers patrolled on foot or bicycle. Furthermore, police officers were equipped with two primary forms of technology: the baton and the firearm, both of which have undergone modernization but continue to be used today.

The most significant technological advancements during the political era concerning the subject matter of this investigation were the communication and criminal identification systems deployed across law enforcement agencies. The telegraph and telephone supported the patrol box system, enabling direct communication with police stations. Police officers employed call boxes to report incidents or seek assistance if required. Additionally, adopting the Bertillon system, an anthropometry method utilized to track and identify suspects, and the early classification of fingerprints significantly aided criminal investigations.

Reform Era

The emergence of the reformation era of policing in the early 1920s aimed to rectify the long-standing issue of corruption and political influence over the police (Kelling & Moore, 1998). Police professionalism was the foundation of this era, which sought to eliminate undesirable government control over law enforcement agencies. Key reformers such as August Vollmer and O.W. Wilson believed in the power of technology and the importance of operating within a legal framework to minimize discretionary excesses from

the previous era. Focusing on crime prevention and control helped police agencies become more effective, particularly as crime became a national concern. The strategic methods developed during the reformation era promoted regulation, law enforcement equality, and national policymaking (Gultekin, 2014). The use of technology had a significant impact on patrol and response strategies. The patrol car became a recognizable symbol of policing, providing officers with a professional and noticeable presence in their communities. The two-way radio also allowed for swift information exchange and coordination of field activities.

The emphasis on scientific evidence was a defining feature of this era. August Vollmer established the first crime laboratory to aid criminal investigations and introduced polygraph, fingerprint classification, and handwriting analysis technologies. This laboratory preceded the formation of the Federal Bureau of Investigations (FBI), which opened in 1932 and became one of the world's most technologically advanced forensic research and analysis facilities (Kessler, 1993). However, during the 1964 presidential campaign, crime remained a major concern, leading to the formation of the President's Crime Commission on Law Enforcement and Administration of Justice.

The Commission discovered that police agencies were not effectively utilizing technological advancements. Despite being equipped like they were 30 or 40 years ago, advancements in other technology areas had surpassed them (President's Commission, 1967). To address this issue, the Commission recommended computerizing law enforcement, including implementing the 911 and computer-aided dispatch (CAD) systems. This revived police agencies and allowed them to benefit from innovation once again.

Community Policing Era

The policing era of community problem-solving that began in the 1980s significantly differed from the reform era's original principles. The rise in crime rates and a shift in crime focus prompted police departments to revise their written policies and operational processes in various ways. Alternative approaches to traditional enforcement methods included strengthening relationships with the patrolled communities, collaborating with outside agencies to address crime and disorder, problem-solving within the organization, and resolving conflicts through mediation and arbitration (Oliver, 2006). Essentially, police organizations acknowledged that to effectively combat crime, they must do more than just arresting suspects and closing cases.

During this era, technology made incredible strides, but police agencies acknowledged the importance of working hand in hand with the community to establish a sense of law and order. From an investigative perspective, failing to gather information from the community about criminal activity in their areas severely hindered their ability to tackle crime effectively. Foot patrol seemed to succeed, with citizens opting to raise taxes to support this type of policing (Greene, 1987).

Subsequent evaluations showed that foot patrol had the potential to decrease fear of crime, enhance satisfaction with the police, and enhance police attitudes toward citizens. At the same time, Herman Goldstein's problem-oriented policing model was also being tested nationwide. The problem-oriented policing (POP) framework similarly focuses on these goals and uses various methods to tackle crime and disorder issues (Goldstein, 1990). The difference between POP and other tested strategies of this era is its limited interaction with the community.

Technology during this era was also credited with assisting in the quality of services delivered to community members and investigations (Lewis & Lewis, 2012) Dispatch centers were upgraded with advanced CAD systems that allowed for the quick population of address and telephone numbers from emergency calls. Police vehicles were equipped with mobile data terminals (MDTs), providing officers access to dispatch services, criminal databases, and tools such as mapping software. The adoption of crime mapping also increased as computer access became more affordable and geographic information system (GIS) software became easier to use and more accurate. One of the most impactful technological advancements of this time, however, was DNA technology, which revolutionized suspect identification and has remained a highly valuable tool in law enforcement.

Homeland Security Era

The emergence of the homeland security era of policing was a direct response to the escalating threats of domestic and international terrorism that followed the 9/11 attacks. Law enforcement agencies that had previously employed community-oriented problem-solving strategies to tackle conventional crime and disorder challenges were compelled to realign their strategies and undertake new roles and responsibilities to protect their communities (Jones & Supinski, 2010). This new era of policing emphasized traditional law enforcement tactics such as arrests and intelligence gathering, with greater emphasis placed on proactive interventions that rely on collaboration and information sharing between various levels of government and functional organizations.

The establishment of the Department of Homeland Security (DHS) and the introduction of the National Incident Management System (NIMS) by the Bush

Administration in the early 2000s reinforced the importance of collaboration between different levels of jurisdiction (federal, state, local, and tribal) and various functional groups (private sector and non-government organizations). These wide-ranging, nationwide approaches to managing incidents aimed to establish a foundation for efficient and effective crisis responses, regardless of the nature of the event (Birkland, 2009). This, in conjunction with the new laws addressing terrorism (such as the USA PATRIOT Act) and the emphasis on intelligence gathering to identify potential threats, led to the establishment of the National Network of Fusion Centers.

Fusion Centers

The development of fusion centers arose from concerns that law enforcement agencies were not collaborating effectively to prevent criminal or terrorist activity. These centers were established to enhance collaboration by facilitating the exchange of intelligence, expertise, and resources between agencies involved in detecting, preventing, and investigating criminal or terrorist activity (Harbisher, 2019). There are 80 fusion centers within the Department of Homeland Security (DHS) network. These centers are categorized as either primary or recognized, with primary centers providing intelligence and information sharing for entire states and recognized centers generally serving major urban areas. The key difference between the two categories lies in the level of federal funding they receive, which can amount to over $330 million per year (Sullivan & Lester, 2022).

Although "fusion center" and "RTCCs" are sometimes used interchangeably, they exhibit distinct structural and operational differences. The fusion center network is supported at the state level and plays a critical role in sharing intelligence among

government agencies and private industries. Of the 80 recognized fusion centers, 51 are at the state level, 26 are in major cities, and three are territorial (Peteritas, 2013). These centers process information horizontally across other centers and vertically to federal or local law enforcement agencies (Global Justice Information Sharing Initiative, 2008). They gather and analyze information from various databases, exchange actionable intelligence with relevant entities in accordance with departmental policies and prioritize high-level threats and hazards using a preventative approach. Although they possess strong analytical skills and access to significant intelligence data, their technological capabilities are not their primary focus.

In contrast to fusion centers, real-time crime centers are integrated into individual police agencies with the aim of real-time information sharing, surveillance, and post-incident investigative support. They represent a significant milestone in deploying actionable intelligence at a more localized level and utilize specific technologies, such as automatic license plate readers, gun detection systems, and closed-circuit surveillance camera feeds (Bureau of Justice Assistance, 2021). Unlike fusion centers, which primarily support counterterrorism investigations, RTCCs are involved in various incidents, including disorder and disturbance calls, property crimes, and violent crimes.

The overall effectiveness of fusion centers remains uncertain (Farivar, 2021). Politicians and scholars have expressed concerns for almost a decade, finding that these centers offer "little benefit" to national counterterrorism intelligence efforts (Levin & Coburn, 2012). Furthermore, there are no tracking mechanisms to quantitatively assess the impact of intelligence gathered by fusion centers on federal terrorism and criminal cases (Oliver, 2006). These findings may be influenced by various factors, including the unclear

lines of authority and broad objectives set forth by the homeland security era of policing However, as suggested by certain scholars, fusion centers will continue to operate in the background of federal and state operations for the foreseeable future, for better or worse (McQuade, 2019).

Real-Time Crime Centers

Integrating advanced technologies to enhance police services has gained significant momentum among law enforcement agencies. This is motivated by the belief that innovative technology can 1) provide officers with unprecedented access to actionable information and intelligence and 2) address the long-standing challenge of police agencies in apprehending suspects and ensuring public safety (Hummer & Byrne, 2017; Randol & Gaffney, 2014). RTCCs have emerged as a national trend, as human resources are integrated with advanced information technologies to deliver critical information to all levels of command, resulting in improved clearance rates, faster response times, and better officer safety.

Although there is no standardized definition, RTCCs are generally centralized data platforms designed to synthesize and rapidly convey operational intelligence to officers in the field (Przeszlowski et al., 202). While specific objectives may vary across jurisdictions, RTCCs facilitate collaboration between internal units and external partner agencies, produce post-incident investigative evidence, and actively monitor emerging incidents such as protests and critical infrastructure. RTCCs are seen as instrumental in crime control and investigations due to their ability to promptly process and deliver high-quality information to the field.

Diffusion

The implementation of a real-time crime center by the NYPD in 2005 marked a groundbreaking milestone in law enforcement. Despite the acknowledged advantages of the NYPD RTCC, other municipalities initially hesitated to adopt similar models within their departments. It wasn't until 2012 that the Philadelphia Police Department became the 10th agency to gain access to comparable levels of information and technology, indicating that integrating RTCCs into daily police operations is a relatively recent and intricate phenomenon.

Przeszlowski and colleagues (2022) conducted the first nationwide evaluation of RTCCs utilizing a voluntary survey disseminated via the Qualtrics software to an exhaustive sample of 61 identified police agencies currently operate RTCCs. The survey garnered a 72 percent response rate, with 44 of the 61 identified RTCCs responding. The study aimed to achieve three main objectives. Firstly, it aimed to gather a national appraisal of existing RTCCs to better inform the development and implementation of the MRTCC, which serves as a platform for enhancing responses to incidents of crime and disorder. Secondly, the survey sought to identify similarities and differences across the RTCC sites nationwide. Thirdly, this study's findings aimed to augment the limited literature concerning the structure and function of RTCCs operating throughout the country.

Several important findings emerged from the survey responses. First, from a broader perspective, when considering all police agencies nationally (N=17,985)[6], implementing RTCCs was considered a relatively new phenomenon. Despite the validity of this, the identified agencies (N=61) within their study consisted of large-sized

[6] According to the 2015 President's Task Force on 21st Century Policing final report.

organization (≥ 100 sworn and civilian personnel). This may suggest that larger agencies are the most relevant population for adopting these units. Accordingly, of the 1,468 large-sized agencies (≥ 50 sworn officers) operating within the United States, departments with an RTCC represented just four percent of those relevant organizations. Here, both interpretations indicated that despite their diffusion, real-time crime centers are in the beginning stages of development and warrant further investigation regarding the various existing modes.

Secondly, the incorporation and configuration of RTCCs displayed substantial variation across police organizations, indicating the absence of a universal model for integrating these units into regular operations. Instead, agencies customized their operations to suit their jurisdiction's distinctive demands and organizational necessities. Thirdly, RTCCs generally obtained an array of technologies and information sources, highlighting the importance of a well-defined strategic plan to delineate the purpose and function of each asset. Finally, most agencies facilitated the prompt sharing of actionable information with their local and federal partners, emphasizing their indispensable role in the police practices of efficiently collaborating with the community and governmental partners.

The recent proliferation of real-time crime centers across municipal, county, and state agencies may be driven by various factors. The anecdotal success of RTCCs serves as a de facto incentive for all agencies nationally to adopt similar strategies to keep pace with contemporary police practices. For instance, on January 20, 2023, New Haven, Connecticut's Police Chief Karl Jacobson outlined the agency's multi-pronged approach to combating violent crime by establishing an in-house RTCC (Mirmina, 2023). The Chief

cited three neighboring jurisdictions, Hartford Police Department, Bridgeport Police Department, and Waterbury Police Department, as models, despite lacking empirical evidence of success from any of these agencies.

Law enforcement agencies are also willing to adapt to the innovation process to meet the continually evolving demands for more effective and efficient policing. However, the impact of real-time crime center technologies on case processing outcomes is limited to federally funded grant initiatives in Chicago, Illinois, and Miami, Florida. Thus, further investigation is needed. Additionally, there is an indication that the law enforcement community is willing to invest in procuring various technologies and their integration into daily police operations, with many city commissions considering funding their operations.

Current Status of Literature on RTCCs

Since the initial appraisal was conducted, the Atlas of Surveillance, a collaborative effort between the Electronic Frontier Foundation and the Reynolds School of Journalism at the University of Nevada, Reno, has created the largest repository of information on the diffusion of real-time crime centers across the country. Their efforts provide a resource for academics and the public to be aware of the procurement of technologies and their distribution across other jurisdictions. However, the limitations of this repository must be addressed. The information derived from crowdsourcing and open-source intelligence is only as good as the source it is derived from, and the adoption of technologies and the implementation of RTCCs are highly dependent on external resources to sustain operations. Thus, while the information may be accurate at the point of data collection, integration efforts by police agencies may be abandoned later due to a lack of funding or other reasons. Finally, with the amount of data points collected, fact-checking the status of

each technology may be overly ambitious relative to the quantity of individuals voluntarily assigned to data collection efforts.

Empirical Assessments of RTCCs

Despite the increasing deployment of real-time crime centers, their integration within police operations remains a relatively new and understudied phenomenon. The Chicago Police Department's Strategic Decision Support Centers (SDSCs) represent the country's first evaluation of these centers in each of the city's twenty-two districts (Hollywood et al., 2019). The Bureau of Justice Assistance's Strategies for Policing Innovation program, in conjunction with the Social and Economic Well-Being Division of the RAND Corporation, utilized a mixed-methods approach to evaluate the organizational structures, processes, and technologies employed within the SDSCs, as well as their impact on crime reduction within the districts to which they are assigned.

The process evaluation involved structured interviews with SDSC personnel, site visits to each of the twenty-two district-level units, and limited ethnographic research. The evaluation revealed that integrating SDSCs into district-level operations enabled novel responses to crime incidents and problems that were not previously possible. Additionally, the SDSC supported higher levels of data-driven and structured decision-making than had been present before their implementation. However, the level of integration of SDSCs varied across districts, posing substantial risks to their long-term sustainability due to uncertain continual support from headquarters and district-level command staff, insufficient organizational learning, and knowledge materials to further evolve the centers, and a lack of budgeted processes.

The technology assessment did not focus on technical performance analysis but aimed to determine how well individual technologies employed within the SDSCs supported operations. Generally, the individual technologies were noted as enablers of real-time situational awareness and tools that supported various analyses for long-term decision-making. Cameras, camera displays, and the public safety mapping display system, Citigraf, were among the technologies that supported real-time response and received considerable attention throughout the report. However, shortfalls primarily dealt with improper integration and the procurement of predictive policing models, such as the Crime and Victimization Risk Model, which were often misunderstood and lacked operational integration, leading to unnecessary public distrust and fear.

The outcome evaluation aimed to assess the impact of SDSCs on crime reduction within select districts. The evaluation report concluded that across 40 models, the average monthly crime counts within districts declined, although only significantly in fifteen of the models. Crime reductions ranged between three to seventeen percent, indicating that the SDSCs are a promising tool for supporting crime reduction in districts with respect to crime types and time of crime occurrence. However, the changes in the outcomes of initially proposed objectives, such as improving officer safety, decreasing response times, and case clearances, were not assessed.

The Miami Real-Time Crime Center violence response initiative was launched shortly after the SDSC's evaluation, aiming to improve responses to violent crime incidents in the Miami Police Department (MPD) (Guerette, et al., 2022). The initiative included the development of a centralized real-time crime center platform and evaluating the impact of MRTCC technologies on case processing and clearance. The evaluation consisted of a

multi-method approach, including a survey of MRTCC users, a module network analysis of MRTCC-assisted case components, and a quasi-experiment with relevant controls.

The survey of MRTCC users found that the technologies were widely perceived as easily accessible, useful, and integral to their work. Most respondents believed that MRTCC technologies had improved the identification and documentation of evidence, improved the ability to clear cases, and reduced the time to clear cases. Users suggested that the MRTCC platform could be improved by expanding the technologies, particularly CCTV, to cover more operational areas 24 hours a day and allowing direct use of some technologies by detectives and commanders.

The module network analysis of MRTCC-assisted case components revealed that MRTCC technologies were being used in conjunction with other traditional varieties of evidence, such as human-based evidence, crime scene evidence, and warrant-based evidence. They were also equally related to those other forms of evidence, and they had similar and, in some cases, greater roles in processing cases. The quasi-experiment comparing MRTCC-assisted case clearances with those of a control sample found that MRTCC-assisted cases had significantly greater odds of being cleared than similar cases without MRTCC support. After controlling for the neighborhood, crime type, and case-level characteristics, the MRTCC-assisted cases had 66 percent better odds of being cleared than those not receiving MRTCC support. However, there were no significant differences in the time to-case clearance between the two samples once case-level circumstances were considered.

Overall, the evaluation of the MRTCC initiative suggests that using the technologies housed within the unit has multiple benefits in responding to violent crime

incidents, and they have become firmly integrated into case processing. MRTCC technologies improved the ability to clear cases without prolonging the time it takes to clear them. The findings also suggest that expanding the technologies, particularly CCTV, could further improve the effectiveness of the MRTCC platform.

Impact of Technology on Policing

Technological innovation in policing is a complex phenomenon, shaped by many factors that interact in intricate ways. However, three major drivers stand out: the pursuit of effectiveness and efficiency, the imperatives of police management, and the demands of external agencies and stakeholders (Ariel, Weisberg & Braga, 2019). In terms of effectiveness and efficiency, technology is viewed as a key tool to achieve desired outcomes such as reducing crime rates, improved response time, and overall crime control. The integration of information technologies can significantly enhance investigative capacities and revolutionize the delivery of police services in the field, provided that they are effectively harnessed. Indeed, many operational policing routines have been transformed to the point where officers increasingly rely on advanced systems to perform their duties (Chan, 2001; Laufs & Borrion, 2022).

From a management perspective, technological innovation is essential to maintaining the professional status and organizational authority of policing agencies, particularly in the face of ever-evolving crime and technology landscapes (Ekblom, 2005; Liou, 2018). Police agencies must continually adapt and improve their technological capabilities to remain effective and efficient in maintaining public safety and security. Finally, technology can also address external agencies' and stakeholders' needs and expectations. In particular, technology can help to enforce fair and equitable policing

practices, institutionalize accountability, and enhance the relationships between police agencies and the communities they serve (Wright & Headley, 2021). By leveraging technology to better serve their stakeholders, policing agencies can foster greater trust, collaboration, and legitimacy.

In recent years, however, diverging perspectives on the impact of technology on the policing profession have arisen. The skeptical viewpoint contends that despite technological advances, clearance rates for violent crimes, such as homicide, have steadily decreased from a peak of 86% in 1965 to approximately 50% in 2020 (Federal Bureau of Investigation, 2021). This perspective suggests that police agencies generally resist integrating innovative technologies into traditional structures and practices, resulting in less effective outcomes (Ioimo & Aronson, 2004; Manning, 2003; Manning, 2008; Gibbs Van Brunschot et al., 2022; Lum et al., 2017). On the other hand, an optimistic outlook maintains that technology has transformed traditional divisions of labor in ways that benefit the operational efficiency of agencies, officer decision-making, and investigative procedures (Chan, 2001; Chan, 2007; Cook & Berglund, 2021; Ribaux et al., 2017).

Despite these opposing views, there is a clear demand for exploiting the benefits of technological advancements in policing. While some concerns about the integration of technology are outside the scope of this paper, the available literature highlights the importance of investigating the role of technology in improving police investigative outcomes. Thus, further research is needed to examine the impact of technology housed within RTCC centers on police investigations.

Concerns Associated with Police Technology

Saulnier and Thompson's (2016) research has revealed that the public's acceptance of police technology depends on the technology's intended purpose or justification for implementation. Furthermore, when the goals of the technology are unclear, or there is no identifiable victim, concerns regarding the underlying motives behind surveillance are heightened (Sakiyama, 2016). Endorsement of law enforcement's use of technological tools is contingent upon legitimacy, accountability, and trust, with literature suggesting that perceptions of fairness and equity among law enforcement agencies foster alignment with established norms of policing (Gerber & Jackson, 2017). In light of the recent calls to defund law enforcement agencies, gaining public approval has become a top priority for police departments across the nation.

Nonetheless, contemporary society has seen a surge in surveillance activities, recognized by scholars as a significant institutional dimension (Ball & Webster, 2003). This increase in surveillance has been unevenly distributed, with certain individuals, groups, and locations subjected to greater scrutiny than others (Fiske, 1998). Unfortunately, underprivileged communities of color have been disproportionately affected by this trend (Bass, 2001). The placement and intended use of surveillance cameras can lead to discriminatory outcomes, with CCTV cameras often installed in areas with high crime rates that frequently overlap with disadvantaged and minority communities (Sampson, Wilson & Katz, 2018). This has resulted in communities of color and low-income areas having a greater prevalence of surveillance technologies than non-minority communities, which can have a chilling effect on civil liberties.

Despite the increased use of CCTV surveillance as a crime prevention tool, evaluations have shown little to no impact on reducing crime rates (Caplan, Kennedy & Petrossian, 2011; Welsh & Farrington, 2008). Additionally, intelligent surveillance systems, equipped with facial recognition technologies and license plate readers, have been increasingly requested by police departments, although the accuracy of facial recognition algorithms is often less reliable for black individuals and women (Klare et al., 2012).

A study of Amazon's Rekognition software by The American Civil Liberties Union found major discrepancies in matching individuals of color, with false matches disproportionately involving individuals of color and identifying them as other people arrested for a crime (Snow, 2018). Moreover, multiple police departments, including Oregon, Orlando, and the Washington County Sheriff's Office, have used Amazon's Rekognition software, which has come under fire for its high rates of misidentification for individuals of color (Bala & Watney, 2019).

Overall, is imperative to acknowledge the profound impact that surveillance practices can have on civil liberties, with particular attention to the disproportionate impact on disadvantaged communities of color. A comprehensive understanding of public attitudes toward these technologies and an effective educational strategy can augment trust in law enforcement and confidence in their efficacy, thereby fostering stronger community relationships and increasing public support.

Chapter Three: Contextualizing the Variation in RTCC Implementation

Introduction

The widespread adoption and implementation of real-time crime centers nationally represents the latest effort to enhance the ability of American police agencies to combat crime and ensure a degree of public safety. Often regarded as a unit equipped with advanced information technology systems and state-of-the art data processing elements, RTCC's first emerged out of the New York City Police Department in 2005. Across all subsequent models the primary objective of an RTCC is to serve as an operational center capable of delivering information to officers in rapid timeframes en route to crimes in progress or those that have recently occurred.

Secondary objectives are often contingent on the availability of resources and the extent of technological proficiency maintained by an agency but can include, (1) providing post-incident investigative support, (2) active monitoring of critical infrastructure, protests, or known hot-spot areas, and (3) acting as a command center during large-scale events or emergency situations (Bureau of Justice Assistance, 2021; Przeszlowski et al., 2022).

Despite their emergence and rapid diffusion, the current body of knowledge on the adoption and variation that exists in the implementation of these units within agencies remains absent from the literature. In an attempt to fill this void, this chapter utilized three organizational perspectives– the rational perspective, the contingency perspective, and the institutional perspective to provide a framework for understanding this process. Building on the availability of evidence documenting the establishment of these unit, a typology was likewise constructed to better denote the differences that exist. Information regarding the

evolution of these units and commonly procured information technologies will also be detailed. The main research questions that this chapter seeks to answer is the following: (1) "Are the factors associated with real-time crime center implementation and adoption more consistent with a contingency, institutional, or rational perspective?" and (2) "What are the primary or most influential real-time crime center models that exist?"

Theoretical Explanations

The literature on organizational change, commonly employed in the field of business management, encompasses three major viewpoints: rational, contingency, and institutional (Chaudhry, 2020; Chenhall & Euske, 2007; Sousa & Voss, 2008). These perspectives are grounded in the notion that organizations act in their own self-interest, striving to meet environmental demands, acquire resources, expand territorial control, and seek support that benefits them (Scott, 1987). This self-interest serves as a primary driving force behind organizational behavior, as organizations pursue strategies that enhance their competitiveness and increase their chances of survival and growth. Despite sharing this fundamental understanding, these perspectives offer distinct explanations for why organizations adopt particular structures and practices.

In the field of policing, scholars have applied these perspectives to analyze the implementation of innovative police technologies (Strom, 2016), such as COMPSTAT, a management approach used to reduce crime (Willis, Mastrofski, & Weisburd, 2007), as well as community-oriented policing, which involves building collaborative partnerships with community members to address crime and disorder issues (Ritti & Mastrofski, 2002). These studies offer valuable insights into the organizational changes that occur within police departments and the factors that influence their adoption of new technologies and

practices. Therefore, it is essential to thoroughly examine these perspectives to understand their underlying assumptions and implications as theoretical constructs for explaining the variation in the adoption and implementation of real-time crime centers.

Institutional Perspective

The institutional perspective focuses on how cultural norms, routines, and structures become established as rules for social behavior within an organization. Scott and Meyer (1994) put forth the argument that the discernible patterns and procedures comprising organizations are direct reflections of frameworks that have been ingrained within broader societal contexts. In essence, they contend that the external environment in which organizations operate shapes and influences their internal configurations and operations.

Further, the theory suggests that the institutional environment can also influence the *rate* of adoption, with greater levels of consensus and cooperation resulting in the sustained and durable diffusion of innovative structures (Peters, 2022). Institutional theorists argue that organizations face significant pressure to conform to institutional norms and practices to maintain legitimacy within their environment. Consequently, novel structures that enhance technical efficiency in early-adopting organizations gain legitimacy within the environment and eventually evolve into standard practice.

Despite its insights, the institutional perspective has some limitations (Greenwood, Hinings & Jennings, 2015). For example, it may not fully explain the factors that motivate organizations to initially adopt new structures. Additionally, the theory may overlook the importance of individual agency and power dynamics in shaping organizational behavior.

Finally, the institutional perspective may not be as useful in understanding organizations operating in non-institutional contexts or those facing rapidly changing market conditions.

Contingency Perspective

The contingency perspective posits that each organization operates within a distinct and specific context and that its decisions and actions may be contingent upon external factors and circumstances (Lawrence & Lorsch, 1967). The term "contingency" refers to the interdependence of an organizational structure with its surrounding contextual environment (Chandler, 1962). Therefore, this theory asserts that there is no single optimal structure that suits all organizations, and the most effective structure is one that aligns with particular contextual variables, such as the economy, political climate, and competitive landscape. Here, it is paramount for organizations to carefully analyze these factors and adapt their strategies accordingly. The efficacy of a particular organizational structure is, therefore, contingent upon the degree to which it aligns with the contingencies present in a given situation.

Moreover, the contingency perspective also emphasizes the importance of adaptation and flexibility within organizations. It suggests that organizations need to be responsive and adaptive to their changing contexts to remain competitive and effective (Shenkar & Ellis, 2022). Scholars who adopt the contingency perspective within the realm of policing, often view agencies as organizations that must adjust their structure and strategies to align with the changing landscape of crime, public sentiment, and the political climate (Strom, 2016). In other words, they believe that police departments must be flexible and responsive to the contingencies of their environment to improve their performance and effectiveness.

While an influential framework in organizational management, the theory also has several limitations that can restrict its applicability and effectiveness for organizations including subjectivity and limited generalizability. For example, the theory's contingency approach requires organizations to identify and analyze the contextual factors that are relevant to their particular situation, which can be subjective and difficult to interpret. Additionally, the contingency theory's principles may not be universally applicable across all organizations and industries. Its effectiveness is highly dependent on the specific contingencies present, and different organizations may face different contingencies. Thus, the theory's generalizability is limited.

Rational Perspective

The rational perspective, which was initially introduced by Cyert and March (1963), posits that organizations function in a logical and systematic manner by establishing clear objectives, formulating strategies to attain them, and utilizing technology to facilitate these strategies. This perspective emphasizes the importance of rational decision-making, where organizations carefully weigh the costs and benefits of various options before making a choice. Scholars who adopt the rational perspective often assume that organizations are rational actors that seek to maximize their efficiency and optimize their resources to achieve their goals (Cummings & Worley, 2023). They believe that organizations act in a predictable and calculative manner, based on an analysis of available information.

Furthermore, the rational perspective also focuses on the role of top management in decision-making, where executives are responsible for defining the organization's objectives and developing strategies to achieve them (Hatch, 2018). The rational

perspective has been employed in the field of policing to examine the adoption and implementation of new technologies and practices, such as predictive policing and COMPSTAT (Willis, Mastrofski, & Weisburd, 2007). Scholars who adopt this perspective often view the police department as a rational actor that seeks to maximize its resources to minimize crime and enhance public safety.

Nonetheless, this perspective is subject to constraints due to the fact that organizations frequently confront conflicting goals, possess incomplete knowledge concerning the most effective methods for achieving their objectives, and have limited resources and capabilities (Simon, 1997). Consequently, while organizations may strive for rationality, their decisions and actions may not always conform to a strictly rational approach. As stated by Daft (2013), despite the valuable insights into organizational decision-making processes that this perspective affords, no organization can solely depend on this theory for every decision.

Theoretical Application to RTCCs

Table 1 summarizes the applicability of all three theoretical perspectives on the decision to adopt/implement a real-time crime center and manage its operational structure. In applying the rational perspective, law enforcement organizations operate in a logical and structured manner by defining their objectives and crafting strategies to achieve them. In this vein, the introduction of a RTCC can be essential in advancing the agency's efforts to pursue short or long-term goals related to a specific policing model, such as community-oriented or crime-focused approaches. By utilizing an RTCC, police departments can monitor crime data in real-time, analyze patterns, and generate actionable intelligence to support their operational strategies. Additionally, the implementation of an RTCC can

foster communication and collaboration among different units within the agency, as well as enhance coordination with external stakeholders, such as other law enforcement agencies, community organizations, and private sector partners. However, it is worth noting that the rational perspective has its limitations, and police organizations may encounter challenges in implementing the RTCC in a manner that aligns with their goals and resources. Factors such as the complexity of the technology, the availability of skilled personnel, and the potential for unintended consequences must be taken into account when pursuing the adoption and implementation of an RTCC. Nonetheless, by adhering to a rational perspective, law enforcement agencies can make informed decisions that facilitate the integration of an RTCC into their overall policing strategy.

Using the contingency perspective, we can assert that law enforcement agencies operate in a manner that is contingent upon external needs and demands, and the most effective organizational structure is one that fits all relevant contingencies. A successful police agency must adapt its operations to the external environment and cater to the specific needs of its stakeholders and the community. Jurisdictions grappling with high levels of gun violence may consider implementing gunshot detection software as part of their arsenal of tools within the unit.

Meanwhile, larger jurisdictions may benefit from a more decentralized district-level RTCC model that takes into account more localized crime problems that may differ from those of a single metropolitan area. However, a significant drawback of applying the contingency theory to the structuring of an RTCC model is the limited generalizability of structural changes within the organization to other agencies.

Table 1. *Theoretical explanations for the variation in RTCC adoption and implementation*

Perspective	Idea	Characteristics	Drawbacks	Example
Institutional	Police organizations are often more dependent on social pressures for conformity and legitimacy than on technical pressures for economic performance.	Motivated by social pressures.	Failure to adopt legitimized structural changes within an agency may be seen as negligent or irrational.	An agency may implement an RTCC as a result of social pressure, conformity or in fear of being classified as a laggard (i.e., diffusion of RTCCs at a rapid scale).
Contingency	Police organizations operate in a manner that is contingent on external needs and demands; the most effective structure is the one that fits all contingencies.	Adapts to the environmental context.	Structural changes within the organization may not be generalizable to other agencies.	An agency may implement an RTCC in a way that best meets external demands of stakeholders or community members and the availability of funding.
Rational	Police organizations behave rationally by identifying goals and designing strategies to accomplish these goals.	Operates under a decision-making framework.	Goals set forward by the agency are not always straightforward; incomplete knowledge of how to manage structural change; organizations are constrained by resources.	An agency may implement an RTCC or acquire specific technologies within their RTCC to support and facilitate the achievement of a certain model of policing.

Consequently, an agency seeking to replicate the structure using the best practices of an external agency may not operate in a manner that meets its own community's needs for public safety. However, by embracing the contingency perspective, law enforcement agencies can tailor their RTCC structures to the unique challenges they face and optimize their effectiveness in maintaining public safety.

Within the institutional perspective, police agencies are found to be more reliant on social pressures for conformity and legitimacy than on technical pressures for economic performance. For example, diffusion studies often demonstrate a bell-shaped curve for the adoption of innovations over time, with innovators, mainstream adopters, laggards, and resisters contributing to the curve (Grubler, 1996). Innovators, who are considered risk-takers with ample resources to absorb setbacks, are responsible for the initial adoption period by experimenting with innovations and demonstrating their utility.

Conversely, police agencies that refuse to conform to changes in the policing landscape may be viewed as laggers or resisters. In other words, if they fail to consider implementing a real-time crime center, a rapidly diffusing police innovation across the country, they may be deemed irrational in their decision-making. The institutional perspective can be further evidenced by agencies that have embraced the RTCC and who often cite some aspect of social and institutional pressures from surrounding agencies as their motivation for adoption. For instance, the Police Chief of New Haven recently acknowledged that the agency is modeling its crime center after those used in Hartford, Bridgeport, and Waterbury, all of which have been highly successful (Mirmina, 2023).

From a rational standpoint, it is postulated that police organizations exhibit rational behavior by outlining objectives and constructing a framework for decision-making along

with strategies to achieve said objectives. Nevertheless, this perspective carries inherent limitations. The goals established by law enforcement agencies may not always be straightforward, as they might encounter obstacles due to limited knowledge on how to effectively navigate structural changes, while resource constraints may impede their endeavors. For instance, an agency might introduce an RTCC to bolster and streamline the achievement of a specific policing model (such as community-based or crime-control-oriented). However, the ever-evolving landscape may necessitate continuous adaptation and adjustment of best practices associated with operating such a unit.

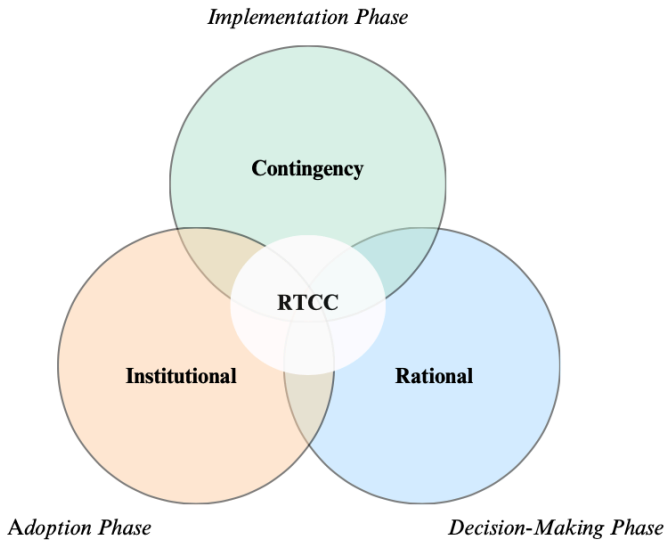
Multi-Perspective Framework

In light of the aforementioned observations, and to address the original research question, it can be argued that each distinct viewpoint presented in the previous section possesses its own inherent value throughout the various phases of adopting and implementing RTTCs.[7] This assertion is best demonstrated by referring to Figure 2, where in the initial stage of adoption, external institutional influences can act as a compelling catalyst. While not necessarily mandatory, the anecdotal success of RTCCs in neighboring jurisdictions, empirical evidence demonstrating increased crime reduction or clearance rates, and the visible display of community support all contribute to the concept's legitimacy, thus promoting acceptance among other agencies. Moreover, the possibility of collaborating with partner agencies to expand network coverage and enhance the ability to identify and apprehend transient offenders or suspects may also emerge as an institutional

[7] While beyond the confines of the research inquiry at hand, this framework can be applied to individual technology adoption, implementation, and deployment, as opposed to RTCC units as a whole.

pressure, resulting in an agency adopting additional technologies that are consistent with other systems to improve efficiency and provide officers with real-time information on demand (Avalos, 2016).

Figure 2. *Multi-perspective framework of RTCC adoption and implementation*



The second stage of the RTCC implementation process aligns with the fundamental assumptions of the contingency perspective. In this stage, the organization has already adopted an RTCC unit, but the structural and operational arrangements are contingent upon several factors. The availability of financial resources and grant funding, as well as the degree of stakeholder involvement, are among the most commonly considered factors. These elements significantly impact the development of standard operating procedures for the unit, staffing decisions, and the quality and quantity of technology acquired. Additionally, they influence the likelihood of future expansion of the RTCC unit.

The ultimate stage of the process is an ongoing one, in which the decision-making process concerning the optimal operation of the unit and the procurement of technology

may evolve to fulfill the goals established by the department and adjust to changing societal circumstances over time. Whether it be to meet heightened demands of producing post-incident investigative products or aid in other tasks that fall under the umbrella of the investigative RTCC unit, changes are likely to occur and should be anticipated. This is also true for any changes made to the budget restrictions upheld by the department.

Preliminary Typology of Existent RTCC Models

This section will expound upon the various types of RTCCs that have been deployed throughout the United States. To fully comprehend the nuances of these models, extent literature and open-source information will be utilized to develop a typology that aptly captures their salient characteristics. Essentially, a typology serves as a system of classification that arranges objects or concepts based on shared features or attributes (Nichols, 2007). In policing research, typologies have been employed to classify a diverse range of phenomena, such as cybercrime (Phillips et al., 2022), organized crime (Le, 2012), domestic violence (Johnson, 2008), and offender characteristics (Kooyman, 2012), among others. By aggregating criminal activities into categories, researchers can discern similarities and differences in offender behavior, motivations, and victim profiles.

Additionally, typologies can be instrumental in designing policing strategies and interventions. For instance, a typology of domestic violence offenders may enable law enforcement officers to detect early warning signs and devise preemptive measures to prevent further harm to victims. Similarly, in the context of RTCC implementation, a typology can provide a foundation for agencies that are considering future adoption or seeking direction in constructing an infrastructure. It can also assist in facilitating

appropriate evaluations that are consistent with the intended objectives of the RTCC of interest.

Method Used to Develop Typology

The development of this typology is an expansion of a prior inquiry conducted on the diffusion of RTCCs at a national level (Przeszlowski et al., 2022). The study referenced surveyed police agencies across the country to assess the nature of their RTCC implementation. The results indicated that 1) RTCCs are proliferating at a rapid pace but are still in their early innovation/adoption phase, 2) there is no single model of their use, 3) most employ a diverse range of technologies and information sources, and 4) most enable the sharing of information with partners in real-time.

Within the previous study, various databases were examined, encompassing Google, Google News, Google Videos, YouTube, the International Association of Crime Analysts (IACA), and LinkedIn, to accumulate an all-inclusive inventory of existing RTCCs. A forum was created on the IACA platform to facilitate the exchange of information among crime analysts regarding their real-time crime centers. Additionally, searches on LinkedIn provided insights into the deployment of these centers and individuals who work within them. Here, the results of the search identified 61 RTCCs or their equivalent.

For each of the 61 RTCCs identified, contact information was obtained by searching publicized reports and contacting the police departments directly. The survey consisted of 25 questions designed to gather information on three substantive areas, including organizational characteristics, organizational structure, and operational procedures. The survey was administered through an online platform using Qualtrics

software, which is a powerful online survey tool suitable for academic research. The survey contained both closed and open-ended questions and was distributed in May 2020. To improve response rates, two rounds of electronic reminder letters via email in three-week intervals, with a final round of targeted electronic requests sent in August 2020. At the end of the data collection period, 44 of the 61 identified RTCCs responded to the Qualtrics survey request, resulting in a response rate of 72 percent.

Here, in pursuit of developing the typology, a comparable approach was adopted. The data collection process concluded in March 2023, revealing a staggering expansion, with a total of 103 police agencies that have established operational RTCCs. Although an additional survey was not disseminated, pertinent information from available the previously mentioned sources was extracted (i.e., intended goals, structural characteristics, etc.) to create a classification system. The ensuing section outlines the findings.

Findings

As depicted in Table 2, the extant literature on the internal composition of RTCCs can be organized into three comprehensive categories, forming a preliminary typology. This typology classifies RTCC attributes according to command structure (centralized authority, regional authority, or autonomous), configurations (centralized, decentralized, or cloud-based), territory (jurisdiction, district, or digital), and premise (crime-control, crime-reduction, or incident-based), thus grouping them by type (traditional, community, or virtual).

Traditional Model

In the *traditional* model, a single command and control center assumes the responsibility of overseeing all operations. This model is distinguished by a hierarchical

structure, where power and decision-making responsibilities are centralized under a central authority. This centralized RTCC model is typically deployed in densely populated urban centers, where there is a high incidence of criminal activity and a pressing need for a coordinated response (Guerette, et al., 2022), The centralization of the RTCC model provides a superior level of control and management efficacy. This approach allows for the prompt deployment of resources, streamlined communication between relevant agencies, and a well-organized decision-making process. The success of the centralized RTCC model can be attributed to its ability to leverage cutting-edge technology and data-driven decision-making, enabling it to respond effectively to crime-related incidents.

Table 2. *Real-time crime center typology*

Type	Command	Configurations	Territory	Premise	Example
Traditional	Central Authority	Centralized; Single Control Center	Jurisdiction	Crime-Control	Miami Police Department
Community	Regional Authority	Decentralized; Multiple Control Centers	District	Crime-Reduction	Chicago Police Department
Virtual	Autonomous	Cloud-Based; Digital	Beat	Incident-Based	Chula Vista Police Department

Note. This is a preliminary typology.

To some agencies, this model may prove impractical or unfavorable for several reasons including resource constraints, dependence on technology, and limited flexibility. For example, the cost of establishing and maintaining a centralized RTCC can be prohibitive (e.g., NYPD's RTCC cost 11 million dollars to implement, recurring costs not

included), particularly for small law enforcement agencies with limited budgets (NYC Global Partners, 2010). Secondly, centralized RTCCs depend heavily on technology for the gathering, analysis, and dissemination of crime data. Any failure or malfunction of the technology infrastructure can disrupt the flow of information and adversely impact decision-making (Bureau of Justice Assistance, 2021). Thirdly, the rigid hierarchical structure can lead to delays in responding to rapidly evolving situations. This limitation can be especially problematic in situations that require quick action, such as active shooter incidents (Mastrofski & Willis, 2010).

Miami Police Department Real-Time Crime Center

Miami is a dense urban area of about 36 square miles. It has a population of roughly 450,00 individuals from various backgrounds. According to the US Census, demographically it is comprised of seventy percent (70%) Hispanic, sixteen percent (16%) African American/Black, and fourteen percent (14%) White Non-Hispanic population. The city has continuously grown as an urban area and has enjoyed a resurgence as a metropolitan hub with one of the highest tourist-to-local ratios with nearly 1,600 visitors to every 100 residents. Yet, poverty levels remain higher than the national average. Like most major urban cities, crime tends to cluster in several concentrated neighborhood areas. It is there where the RTCC technologies had been predominantly deployed. The City of Miami Police department was established in 1896 and has since grown to become one of the largest police departments in the southeastern United States, with over 1,300 sworn officers and over 400 civilian personnel.

The MPD's RTCC serves as the hub of the department's technological infrastructure, providing critical intelligence and investigative support to its officers and

detectives. Located within the central station, the MPD's RTCC is characterized by a hierarchical structure with a clear chain of command and decision-making power vested in a central authority. The center is responsible for managing all RTCC operations, including active monitoring, responding to crime-related incidents in real-time, providing, post-incident investigative support and serving as a command center (Guerette et al., 2022). To this end, the RTCC is manned by a team of nine (9) highly proficient detectives, who monitor live feeds from cameras and sensors distributed across the city, as well as social media, facial recognition, and other online platforms.

Primary Technologies: CCTV, ShotSpotter (gunshot detection system), Clarity/Vigilant (automated license plate readers), Faces/Clearview (facial recognition software).

Community-Based Model

In contrast to the centralized RTCC model, the *Community* RTCC model is highly decentralized, with multiple smaller centers distributed across a geographic area. Each center has its own command and control center and operates independently of each other. This RTCC model emphasizes engagement with the community to prevent crime and works closely with community groups and organizations to identify and address local crime issues (e.g., hosting community events). By involving the community, the RTCC can gather valuable information about local criminal activities and take preventive measures to address them. Community involvement also fosters trust and collaboration between law enforcement and citizens, creating a safer and more secure environment for all.

This model also leverages technology to enhance policing efforts, but with a different objective in mind. It employs advanced analytics and data-driven decision-making to identify crime patterns and predict potential criminal activity. By utilizing such

technologies, the RTCC can proactively target and prevent criminal activities, ultimately reducing the crime rate in the community (Hollywood et al., 2019). Moreover, having more decentralized units and less camera feeds to monitor may prove to be advantageous (Piza et al., 2019). Despite the advantages of this RTCC model, it also presents some challenges. One of the primary challenges is the need for effective communication and coordination among the different centers. Additionally, this RTCC model may require greater investment in technology and resources to achieve the same level of efficacy as a centralized model.

Chicago Police Department Strategic Decision Support Centers

Chicago, a diverse and vibrant city located in the Midwest region of the United States, boasts a population of 2.7 million people living within its 234 square miles. In accordance with the US Census, the city's demographics are diverse, with a population that is 31% White, 30% Black, 29% Hispanic or Latino, and 6% Asian, among other ethnic groups. The Chicago Police Department (CPD), with over 13,000 sworn officers, is among the largest police forces in the country and is responsible for serving 22 districts. On average, each CPD district is responsible for providing law enforcement services to about 125,000 people over an area of 11 square miles, although population and district sizes vary significantly across the city (The Civic Federation, 2022).

Since 2017, the Chicago Police Department has implemented a distinctive type of RTCC known as Strategic Decision Support Centers across 13 of its 22 districts (Hollywood, et al. 2019). The SDSCs are small conference rooms equipped with a variety of technologies, as well as three wall monitors, workstations, and a conference table. They are typically staffed with two to four individuals per shift, including both sworn officers

and civilian analysts. Analysts generally work during day shifts, while each SDSC is supervised by a sergeant or lieutenant.

The SDSCs not only provide a workspace for briefings and collaborative efforts beyond the individual work of the SDSC staff but also serve as a central hub for data analysis, surveillance, and information sharing. By leveraging cutting-edge technologies, such as license plate readers, gunshot detection systems, and predictive analytics, the SDSCs help the CPD identify crime hotspots and make data-driven decisions to reduce crime in the city. The SDSC model prioritizes not only technology but also community engagement and collaboration, with the CPD closely collaborating with community groups and organizations to identify and address local crime issues (Hollywood et al., 2019).

Primary Technologies: CCTV, Genetec Citigraf (public safety mapping display), CVRM (predictive policing software), Link Chart (social network analysis).

Virtual Model

A virtual real-time crime center is a cloud-based and fully digital model that functions as a centralized hub for gathering, analyzing, and disseminating critical data and intelligence in the law enforcement domain (Merchant, 2020). This innovative model, unlike the traditional physical facilities housing RTCCs, can operate entirely online, accessible from any location with an internet connection or through a mobile data terminal. The virtual RTCC offers an economical solution that can be accessed remotely, making it an appealing option for small to mid-sized law enforcement agencies. Furthermore, it provides an efficient and scalable solution that can improve public safety and security in communities where resources are scarce (Flock Safety, 2022).

The primary goal of the virtual RTCC is to enhance situational awareness and improve decision-making at the beat level by providing real-time information on crime trends, incidents, and threats to patrol officers in the field. This type of RTCC leverages advanced data analysis and visualization tools to discern patterns, relationships, and anomalies in crime data, thereby enabling more effective resource allocation, operational planning, and tactical response. By facilitating data-driven policing strategies and empowering officers with relevant and timely information, virtual RTCCs represent a significant step forward in the quest to improve public safety and bolster community-police relations.

Chula Vista Police Department Real-Time Crime Center

Chula Vista is a city located in the southern region of San Diego County in California, United States. The city covers an area of approximately 52 square miles and has a population of around 270,000 people, according to the latest U.S. Census Bureau (2022) estimates. The city's population is diverse, with a Hispanic or Latino population of around 59% and a non-Hispanic white population of around 27%. The city also has significant Asian and African American populations, each accounting for around 5% of the population. The Chula Vista Police Department (CVPD) is the law enforcement agency responsible for maintaining public safety and order within the city. The CVPD has approximately 260 sworn officers and 100 civilian staff, who serve the community through various divisions, including patrol, investigations, traffic, and community services.

The Chula Vista Police Department has adopted an advanced cloud-based RTCC model to boost its policing efforts. The CVPD's RTCC system leverages the CopLogic Platform, a cloud-based software that provides officers with real-time crime data and

intelligence from any device with an internet connection, such as smartphones or mobile data terminals. The CopLogic Platform includes a Live 911 software, which is a significant feature. This software allows patrol officers to monitor incoming emergency calls in real-time, providing critical information about the incident's nature and location before they reach the site (Solis, 2020). This information comprises details such as the caller's address and phone number, any relevant comments or notes from previous calls, and previous incidents at the same location. The Live 911 software also equips officers with a map of the incident location, facilitating speedy and efficient navigation to the scene.

The CVPD's cloud based RTCC model extends beyond Live 911 software, enabling officers to access a wealth of crime data and intelligence such as real-time traffic data, crime reports, and predictive analytics. This capability empowers officers to make well-informed decisions and adopt a proactive approach to policing. Moreover, since 2018, the CVPD has also implemented a “Drone as First Responder Program” started using drones to help with investigations and provide better situational awareness for officers responding to calls (Cohen, 2022). This initiative became the pioneering program in the nation to secure Federal Aviation Administration (FAA) endorsement, which permits law enforcement to observe anything beyond their visual range from aerial viewpoints. Access to drone information is further accessible to patrol.

Primary Technologies: CCTV, Drones (unmanned aerial systems), Live911 (public safety mapping display), Vigilant (automated license plate reader).

Chapter Four: The Influence of RTCC Technologies on the Case Processing and Outcomes of Homicides and Aggravated Assaults

Introduction

The primary objective of this chapter is to examine the influence of police technologies on the outcomes of homicide and aggravated assault incidents. In particular, the research assesses the extent to which RTCC technologies facilitate the investigative case processing of such violent crimes, and how they affect the clearance of cases, including whether they enhance the efficiency of investigators in resolving these cases expeditiously. The research questions guiding this study are therefore three-fold: (1) "What is the impact of RTCC technologies on the clearance of homicide and aggravated assault incidents compared to all other violent crimes?" (2) "To what degree have RTCC technologies been integrated into the case processing of this specific category of crime?" and (3) "Does RTCC support improve the time that it takes to clear these cases?"

This chapter will commence by presenting a literature review pertaining to the examination of homicides and aggravated assaults, and the utilization of technology in their investigation. A subsequent section will detail the analytical approach taken and provide a comprehensive overview of the data collection process and the measures involved within the analysis. Once all analytical approaches are exhausted, a supplemental analysis will be presented to augment the findings and provide additional insight into the impact of the technologies assessed. Limitations and policy implications for this and all subsequent studies will be located in chapter eight.

Review of Literature

To establish a foundation for this assessment, two relevant bodies of literature will be utilized. The first relates to the factors found to influence the solvability of homicides and aggravated assaults, and the second pertains to the role of technology in the investigation of these crimes.

Factors Known to Impact Case Clearance

As compared to other forms of violent crime, the study of factors influencing the case clearance of homicides and aggravated assaults have been investigated to a greater extent (Jarvis, Mancik, & Regoeczi, 2017; Scott & Wellford, 2021). This research is largely motivated by several things. Firstly, homicides and aggravated assaults are regarded as the most heinous and brutal crimes, and their resolution is a paramount objective for law enforcement agencies (Doucette, Green, Dineen, et al., 2021).[8] Secondly, these incidents frequently have a higher likelihood of being reported to law enforcement than other criminal activities, which increase the chances of an investigation taking place (Gramlich, 2021). Thirdly, the nature of these incidents, such as the existence of tangible evidence and witness accounts, can offer investigators with more leads and evidence to work with, leading to an increased likelihood of case clearance (Cook, Braga, Turchan, & Barao, 2019). Lastly, solving these types of crimes can bring a sense of closure to the victim's family and the community, which is a significant impetus for law enforcement agencies to focus on these cases (Riedel & Jarvis, 1999).

[8] The total lifetime costs of homicides are estimated to be $25.1 billion (this estimate includes medical expenses and lost economic productivity (CDC, 2020).

Homicide Case Clearance

Wellford and Cronin's (1999) scholarly inquiry unveiled a multitude of investigation-related factors that influenced the probability of arrest. Certain factors were within the purview of the police, while others were not. Within the realm of police-controlled factors, the initial actions taken by first responders, the promptness of detective arrival, the thorough documentation of the crime scene, and the level of collaboration with the crime and medical examiner's laboratory were all found to be significant. In terms of personnel, the researchers found that assigning more than one investigator to a case improved clearance rate, as did training detectives on proper investigative techniques. Factors outside of police control included the victim's race, the presence of witnesses, the type of weapon used and the whether the weapon was located at the scene of the crime.

Subsequent research has divulged that specific investigative pursuits, such as the gathering of supplementary evidence, the cultivation of cooperative informants, and the execution of search warrants, heighten the likelihood of resolving homicide cases (Alexander, 2012; Wellford et al., 2019). Furthermore, incidents involving firearms are relatively less likely to be cleared compared to those that entail the use of weapons like knives and blunt instruments, which require closer physical proximity (Addington, 2006; Litwin, 2004). Various investigations, for instance, those conducted by Lee (2000), Litwin (2004), and Regoeczi, Jarvis, and Riedel (2008), have also identified specific characteristics of individual cases, such as the demographics of the victim (age, race, and gender), which influence the probability of case resolution. In general, victim race holds less importance than the severity of the offense and the presence of evidence, and research on homicide clearance rates has produced mixed results concerning the gender of the

victim. While some scholars suggest that homicide clearance is equally probable for male and female victims (Addington, 2006; Litwin and Xu, 2007; Puckett and Lundman, 2003) others assert that homicide clearance is more likely for female victims than for males (Lee, 2005; Regoeczi, Jarvis & Mancik, 2020; Regoeczi, Jaris & Riedel., 2008).

The intricate relationship between physical evidence and the effectiveness of criminal investigations is a matter of considerable complexity, as posited by Jarvis and Regoeczi (2012). While a majority of scholars argue that forensic examination of evidence recovered from the initial crime scene carries minimal statistical significance for clearance rates (McEwen, 2010; McEwen & Regoeczi, 2015; Schroeder & White, 2009), the delayed processing of such evidence could potentially undermine its value. More recent research conducted on homicides in Boston has discovered a robust statistical association between various investigative measures, including pre- and post-arrest forensic tests, and the resolution of such cases (Braga, Turchan, & Barao, 2019).

Aggravated Assault Case Clearance

The available research on the factors that impact the successful resolution of aggravated assaults is somewhat scant. Investigations into the factors associated with the clearance of such assaults have predominantly relied on data from the National Incident-Based Reporting System (NIBRS), which fails to offer any meaningful insights into the nature of the investigation or the pertinent organizational features, implying that incident-related variables are the chief predictors (Roberts and Lyons, 2009; Taylor et al., 2009; Overstreet, McNeeley, & Lapsey, 2021). For instance, Roberts and Lyons (2009) utilized NIBRS data from 2005 to predict the time required to clear aggravated assault cases, discovering that the racial identity of both the victim and the offender influenced the case's

resolution, while age and gender were indicators of the time necessary to achieve a favorable outcome. Similarly, other studies have found that cases involving younger victims or minority groups may have lower clearance rates, and assaults committed by African American offenders are less likely to be cleared than those perpetrated by white offenders (Taylor, Holleran & Topalli, 2009). Stolzenberg and D'Alessio's (2004) research has also revealed that African American females are more prone to being arrested in connection with aggravated assault incidents than their white counterparts.

While prior research has shed some light on the clearance of aggravated assaults, its methodological limitations have hindered the incorporation of a comprehensive range of explanatory variables. To address this gap, Baskin and Sommers (2012) undertook an analysis of official record data from five U.S. jurisdictions (namely, Los Angeles County, Indianapolis, Evansville, and South Bend) to identify the factors that impact the charging and conviction of aggravated assault. Their findings indicate that extralegal factors, such as victim demographics and suspect-victim relationships, significantly impact the probability of arrest. Specifically, arrests were more likely in cases involving white victims, black or Latina female suspects, and intimate relationships. Legal variables, such as the presence of victim accounts or crime scene evidence, also increased the likelihood of arrest. However, the study discovered that physical evidence was only examined before arrest in a meager four percent of cases.

Moreover, Scott and Wellford's (2021) investigation delved into the organizational and investigative characteristics that predicted the successful resolution of aggravated assault cases, while controlling for factors at the incident level. The study's principal findings suggest that the relationship between investigative effort and case clearance is

significantly influenced by the organizational practices of an agency, including the number of best practices implemented. However, this correlation does not consistently result in favorable outcomes for case clearance. Rather, the likelihood of case clearance is more significantly influenced by case-specific factors, such as the recovery of a weapon, the presence of a cooperative witness, and incidents involving white victims. Additionally, after considering all variables in their models, the study found no significant connection between crime scene processing and case clearance. Instead, cases with physical evidence collected during the initial police response phase were nearly 50% less likely to result in clearance. Ultimately, the authors underscore the pressing need for further investigation into this particular type of crime.

Impact of Technology on Homicide and Aggravated Assault Investigations

Merely a handful of investigations have delved into the influence of innovative police technology on resolving violent crimes. Instead, examinations of measures associated with homicide clearance typically center on the organizational, investigative, and case dimensions, as per a review of 46 studies on violent crime conducted by Wellford et al. (2019). The limited amount of research in police technology may stem from various factors, such as insufficient funds for technology development and implementation, inadequate comprehension of the potential benefits of technology in policing, and the intricacy of integrating technology with preexisting policing practices. For instance, while advanced forensic evidence (such as cellphone records, CCTV, or DNA) is acknowledged as crucial during the prosecutorial stage of case processing, investigators often encounter frustration due to their limited expertise in handling the vast quantities of data produced by

these technologies, as well as the resource constraints that come with them (Gibsb Van Brunschot, et al., 2022).

Research endeavors investigating the role of technology generally focused on its potential to function as a preventive mechanism or crime deterrent and have evaluated its ability to reduce criminal activities in specific areas. For instance, Caplan, Kenndey, and Petrossian (2011) conducted an assessment of the impact of police-monitored CCTV cameras on shootings, auto thefts, and thefts in Newark, NJ, and found that the benefits were minimal. Furthermore, they noted that the benefit was more significant for auto thefts than for shooting incidents and that the cameras were not as effective in discouraging street crimes.

Similarly, Piza and colleagues (2014) discovered that the integration of proactive CCTV monitoring with direct police patrol leads to greater crime control benefits than solely relying on CCTV deployment. In their study, they evaluated the effects of this combination and determined that it surpasses the efficacy of using CCTV as a stand-alone measure. This intervention was particularly beneficial for street and disorder crimes. More recently, Gerell (2021) investigated whether the implementation of police operator-controlled CCTV cameras in a nightlife area would result in a decrease in assaults. Contrary to their initial hypothesis, the outcome revealed a noticeable increase in reported assaults in the area where actively monitored cameras were employed to assist law enforcement in their crime-fighting efforts.

In a quasi-experimental evaluation study conducted by Wheeler and Phillips (2018), the efficacy of roadblocks and automated license plate readers (ALPRs) in reducing crime rates in high-crime areas of Buffalo, NY was investigated. The Buffalo Police

Department (BPD) implemented high-visibility roadblocks in various locations throughout the city during a two-month period in 2013, known as Operation Strikeforce, and used ALPR to identify vehicles associated with warrants, revoked licenses, or stolen status. Only those vehicles flagged as problematic were stopped. Overall, the evaluation showed mixed results in deterring part one violent crimes and reducing traffic accidents, where part one violent crimes (which included homicides and aggravated assaults) showed a statistically significant decrease, while accidents showed a statistically significant increase. This inquiry added to the findings of two preceding empirical investigations that employed ALPRs at crime hot spots. Notably, Lum and colleagues (2011) study revealed no noteworthy effects, whereas Taylor et al.'s (2012) research documented a deterrent impact for longer-term ALPR interventions.

In an attempt to determine impact on case clearance, Doucette and colleagues (2021) conducted a rigorous investigation to evaluate the impact of ShotSpotter technology on case clearance rates. Their study utilized a pooled, cross-sectional time-series analysis to thoroughly examine the influence of this technology on murder arrests, firearm homicides, and weapons arrests across sixty-eight metropolitan areas across the United States between 1999 and 2016. The authors posited that the broad adoption of ShotSpotter technology in diverse jurisdictions would lead to a decline in homicide rates while concurrently yielding a rise in firearm crime arrests. This expectation was rooted in the capacity of the technology and the information offered by the vendor. However, the findings indicated no significant differences in homicide rates, murder arrests, and weapons arrests between counties with ShotSpotter technology and those without it, even after controlling for multiple demographic variables and state firearm laws. Nevertheless, the

authors acknowledged a limitation of their study, namely the inability to attribute the results to the exact census tract covered by ShotSpotter, which may have introduced some bias and limited the accuracy of the evaluation of the technology's impact on crime clearance rates.

Despite the insufficient research on the use of ShotSpotter and case clearance, it is important to acknowledge that, as with other forms of technology, law enforcement personnel widely acknowledge the effectiveness of gunshot detection software in investigations. This technology has demonstrated considerable positive outcomes such as reducing response times of officers to the scene of an incident (Mazerolle et al., 1998), facilitating the collection of crucial evidence such as identifying the location of gunfire (Choi, Librett, & Collins., 2014), and generating comprehensive reports on gun-related crimes (Carr & Doleac, 2016).

Impact of Real-Time Crime Centers

Currently, assessments of RTCCs are restricted to the Strategic Decision Support Centers implemented by the Chicago Police Department and the centralized Real-Time Crime Center employed by the Miami Police Department. [9] The former evaluation revealed some indications of success in reducing homicides and aggravated assaults in districts where SDSCs were deployed (Hollywood et al., 2019), whereas the latter study did not focus on homicides and aggravated assaults exclusively but instead found support for improved crime clearance rates across all categories of violent crime which also included sexual assaults, robberies, and domestic violence incidents (Guerette et al., 2022).

[9] For more information regarding the evaluations refer to Chapter 2.

Current Study

Considering that this data is derived from a broader empirical investigation on the influence of real-time crime centers on case processing and outcomes of violent crime incidents (Guerette, et al., 2022) there is a basis to argue that these violent crimes, when assessed individually, could also profit from the assistance of these investigative entities. Additionally, informed by the clearance rates of this category of crime on a national scale, it is anticipated that there is a higher probability of clearing homicide and aggravated assault cases when compared to all other forms of violent crime (Li & Lartey, 2022). Thus, the following hypotheses are advanced.

[H1] Compared to all other forms of violent crime, homicide and aggravated assault incidents will have an improved likelihood of case clearance, controlling for the support of RTCC technologies.

[H2] RTCC-technologies will be firmly integrated in the case processing of homicides and aggravated assaults.

[H3] RTCC-assisted homicide and aggravated assault incidents will experience longer time to case clearance rates as a result of increased investigative effort.

For the purpose of this study homicides and aggravated assaults will be classified as commensurate offenses due to their shared characteristic of involving intentional violence against another person.[10] While homicide refers to the act of killing another individual, aggravated assault pertains to a severe physical attack that inflicts serious harm

[10] The sample size for homicides was too small to run a regression analysis with all relevant variables.

or injury to another. This classification may arise when an individual, who intends to commit homicide, instead inflicts severe harm on the victim but does not succeed in ending their life. Similarly, when an individual commits aggravated assault, the severity of the injury inflicted may lead to the victim's death, resulting in a homicide charge. Furthermore, both these offenses may entail similar factors, such as the use of weapons, the intent to harm, and the presence of anger or aggression. The gravity of the wounds incurred by the target can also ascertain whether the transgression is classified as an aggravated assault or a homicide. Thus, for some of these reasons, scholars have progressively gravitated toward viewing these classifications of crimes as interconnected, particularly with the incorporation of firearms in the perpetration of the offense (Hipple, 2022).

Analytical Approach

This study employed four distinct analytical techniques and two forms of analytical software programs to examine the influence of RTCC technologies on the case clearance of homicides and aggravated assaults. First, a set of nested multivariate logistic regression models were employed to determine the likelihood of case clearance for aggravated assault and homicide incidents compared to all violent crime incidents and controlling for RTCC support. The multivariate model was adjusted for various solvability factors or case characteristics that could potentially impact the clearance probability of cases outside the scope of the technologies employed within the unit. Second, module network analysis was used to empirically examine the extent to which RTCC technologies influence the investigative process of these unique crime types. Network analysis involves a set of integrated techniques that aim to depict the relational properties of individual elements and analyze the structures that emerge from the recurrence of these relations (Chiesi, 2001;

Marsden, 2016). Kaplan Meier survival analysis was used to determine differences in the time to case clearance for RTCC-assisted homicide and aggravated assault cases compared to the control sample and finally, due a suboptimal model fit, Cox regression with hazard model outputs was used to determine differences in the time to case clearance for RTCC-assisted homicide and aggravated assault incidents compared all other forms of violent crime assisted by RTCC technologies. To conduct the logistic regression and survival analyses, Statistical Package for the Social Science (SPSS) was utilized, whereas an open-source online program, Jeffrey's Amazing Statistics Program (JASP) was used to perform modular network analysis.

Source of Information

The dataset utilized for this study was acquired through a collaborative Smart Policing Initiative with the City of Miami Police Department, which concluded in October of 2022. This section will detail the process involved in gathering this information, with emphasis on the measures included for analyses that follow.

RTCC Internal Incident Log

The MPD RTCC kept an internal log within their unit as a method of performance measurement. The significance of the internal log lies in its dual role: firstly, as an in-unit record that enabled RTCC personnel to document their daily undertakings and their efforts toward providing requested incident support, and secondly, as a basis for the development of an extended log that facilitated the eventual evaluation of the unit's performance. The incidents recorded within the internal RTCC log served as a foundation for the extended log, which in turn provided essential data for the initial outcome evaluation and ultimately served as a valuable resource for the present dissertation.

The RTCC unit personnel (e.g., detectives, sworn personnel) were tasked with recording all activity that occurred during operating hours within this established internal log. This included any occurrences that were brought to the attention of the RTCC staff through either dispatch or officers in the field, and whether material assistance was provided by the RTCC. This also included incidents that were identified by RTCC detectives through active monitoring and cases in which the RTCC was asked to assist in the post-incident investigative process.

When an incident was brought to the attention of RTCC personnel during the time of initial response, information sequencing occurred in one of two ways. RTCC personnel were either informed of the incident through dispatch or solicited by officers in the field to provide support. When a call was dispatch-initiated, the incident number was generated through the computer-aided dispatch system and was entered into the log as an identifier by RTCC personnel. When a call was raised by patrol, either the same process occurred, or patrol advised the RTCC detectives of a generated case number. Any immediate information obtained from dispatch or officers in the field was also entered into the log at that time (e.g., location, vehicle, weapon used, etc.) along with an indication of the information sources utilized and any additional notes pertaining to the incident.

For post-investigative support purposes there were two procedures for obtaining support from the RTCC. For requests related to video retrieval, an online video request form was submitted by the lead detective which was routed to the RTCC. This form was reviewed by the RTCC Sergeant who either approved or denied the request based on whether the RTCC had resources that covered the area of the incident in question. If the request was denied, the incident was not recorded in the log. For those that were approved

and processed, the case number was recorded in the log. The second procedure related to all other requests that were not video related. In those instances, detectives sent an email request to the RTCC unit for specific support. For occurrences that were identified through active monitoring, RTCC personnel simply recorded the activity in the log. For these instances, there were no formal case numbers that were developed and required no added data validation.

Time Period of Internal Log Data

The assembly of the internal RTCC log began in January of 2020. The internal RTCC consisted of 3,636 entries, of which 2,228 involved an incident resulting in an assigned case number. Once duplicate case numbers were removed, 1,871 unique incidents remained, and select violent crime and property crime types were extracted for the purpose of eventual evaluation. Despite the date of assembly, the oldest incident that the RTCC provided post-incident investigative support to transpired on January 16th, 2019. [11]

Validity Check

On a weekly basis, the internal RTCC log was reviewed and validated. Any missing information was also added at this time. For instances in which the RTCC personnel are unable to obtain an identifier, MPD's crime analyst tool was utilized to locate the appropriate case number. This tool can extract all incidents that occurred in a specified area and timeframe. Once an incident or case number was identified, the Department's records management system, *PremierOne* was utilized. The record management system stores

[11] Note that recorded incidents may precede the development of the log as new investigative leads develop and additional assistance is needed.

information pertaining to the type of incident (i.e., signal number), the location of the incident, whether an arrest was made and any additional case details. This information was extracted and added in the log.

Three other sources were used to validate the information in the log. First, the RTCC personnel indicated the information technologies used to assist the individual or unit in the notes section of the database when initially logging the incident. It was then ensured that those details were properly coded within the relevant measures. Second, case files within *PremierOne* were reviewed to identify instances when the RTCC provided support and the types of information received. Each case entry was then compared to the record management system information to ensure proper coding. Third, request emails sent directly to the unit detectives were reviewed to ensure that the log properly documented all the applicable information technologies used to support the investigation.

Data and Measures

The internal log included sixteen distinct measures that were determined in collaboration with the MPD command staff to capture the RTCC's activities while responding to or assisting with incidents. These measures aided in identifying incidents and recording relevant numerical and dichotomous measures. For example, the log recorded the case number, date, time, type of incident signal, location, NET area, unit assisted, and type of service provided (real-time, post-incident, active monitoring). Numerical measures such as arrests and charges (felony or misdemeanor) were also included, as were dichotomous measures such as the recovery of weapons or narcotics and involvement of a vehicle. The RTCC resources accessed were coded categorically, and each resource used for each incident (e.g., CCTV, Clearview, ShotSpotter) was likewise

recorded. Lastly, personnel provided their personal identification number and described their assistance in a column designated for additional notes.

Treatment Group

The purpose of the expanded database was to capture additional measures for all violent and property crime incidents which received support from the RTCC. The incidents recorded within the expanded RTCC database had all received real-time or post-hoc deployment of information from the RTCC to responding officers or detectives. The extended database allowed for more in-depth analysis in the assessment of selected outcomes for incidents treated by the RTCC compared to non-RTCC incidents. The procedure for incident collection and recording within the extended RTCC database was as follows. With each identified case number, exhaustive steps were taken to acquire the additional information needed to build the extended database and to ensure that measures within the database were coded with certainty. These steps involved accessing the police department's record management system, PremierOne, reviewing reports of investigations (301 reports), querying the department-wide email system, attending bi-weekly violent crime meetings, and reading through OneNote documents. The following illustrates the systematic process developed in the expansion of the log.

Each incident number within the extended database was first queried within PremierOne to access the case folder devoted to the incident. The initial report uploaded within the system allowed the research partner to obtain information on the time of occurrence, the amount of manpower assigned to the case, and the overall description of the incident (i.e., what transpired, number of victims, suspects, and witnesses). The initial report also provided some detail on the traditional investigative methods utilized (e.g.,

whether CSI was requested). This information was often submitted into the system by dispatchers or patrol officers who first arrived on the scene.

Next, reports of investigation (or 301 reports) were accessed. These reports were often found attached to the PremierOne case files and provided a more detailed account of the incident through subsequent investigation. These reports were typically submitted by detectives and approved by their superiors. In the instance that a 301 report was not attached to the case file, a secondary search was conducted within the department-wide email system. Email queries also produced additional information that was not always readily available within the RMS system. These queries verified items such as: whether information flyers were produced on the subject, victim, or vehicles involved, whether felony apprehension team packets were submitted, whether arrest forms were uploaded, and whether the National Integrated Ballistic Information Network (NIBIN) was utilized. Finally, the email queries also drew out RTCC personnel end-of-shift reports which documented their involvement in note-worthy incidents.

Additional steps were taken for classified cases such as homicides. To capture all the resources employed within the investigation of these incidents, the research partner attended bi- weekly violent crime meetings and read through the subsequent OneNote documents that were assembled as part of the information-sharing process developed within the agency. These documents captured information regarding new investigative leads, any additional witnesses or victims that may have surfaced, the use of cellphone and search warrants, and any additional intelligence information or activities that may have developed over the course of the investigation.

Time Period of Treatment Group

The data collection for the extended RTCC analytical database occurred simultaneously with the internal RTCC incident log. Data collection was concluded on April 30th, 2022, to allow for the assembly of the non-RTCC treated control sample. The violent crime incidents recorded within the extended RTCC database are comprehensive. That is, all homicides, aggravated assaults, robberies, sexual assaults, and domestic violence incidents that were included and validated within the internal RTCC log, were also captured within the extended database. Selected property crimes (e.g., burglaries, larceny thefts, vandalism) were likewise included within the extended database The extended analytical database consisted of a total number of 762 incidents, of which 648 were violent crimes. The time period of cases within the expanded RTCC database was naturally the same as that in the RTCC incident log, from January 16th, 2019, to April 30th, 2022.

Measures for Treatment Group

The extended RTCC analytical database included 41 additional measures beyond those contained in the RTCC data log. Of these measures, roughly 80 percent were coded dichotomously. These measures consisted of all unique technology and information sources accessed (e.g., CCTV, LPR, facial recognition), as well as traditional investigative methods utilized (witness statements, information flyers, line-ups). These were dichotomously coded in one of two ways: 1) in a manner that detailed the access and use of the selected technology/information source or method (1 = yes it was utilized, 0 = no it was not utilized); or 2) in a manner that depicted is evidence was recovered (e.g., 1 = yes there was evidence from the CCTV, or 0 = no, CCTV was accessed and reviewed but there

was no evidence). Numeric measures detailed the number of officers, victims, witnesses, and suspects involved in the processing of the case, as well as the hours and days it took to clear all applicable cases. The detailed coding arrangement for these measures, and all others, can be found in Appendix A.

Control Group

A control sample of similar cases not receiving RTCC support was randomly drawn to compare and isolate the impact of RTCC assistance on case outcomes. These non-treated calls for service were those dispatched to officers who responded to incidents through traditional operating procedures and without the assistance of the centralized RTCC unit. The procedure for incident collection and recording within this control sample entailed the following. To ensure equivalency between the RTCC-treated sample and the control sample, a proportionate stratified random sampling procedure was used. First, the frequency of cases in the RTCC sample within neighborhood enforcement areas (NETs) and for each violent crime type was calculated.

These two criteria, crime type and NET area, were used to stratify the groups from within which the control sample would be drawn so that the proportion of cases in the control sample would be the same for these two strata. Thus, the RTCC sample and the control sample had the same number of cases for each NET area and for each crime type. This ensured equivalency on these two dimensions and effectively controlled for the influence of neighborhood effects and crime-type effects on case outcomes. The distribution of frequencies and tests for equivalence between the RTCC and control sample are reported in Appendix B. To generate a sampling frame from which the control sample could be randomly drawn from within each stratum, an automated pull was queried through

the RMS system for each unique crime type and specified NET area. Once the sampling frame was generated a three-step process was followed. First, the incidents from the expanded RTCC analytical data log were extracted from the automated pull to ensure that no RTCC-assisted cases were drawn in the process of random selection.

Second, a random integer set generator was utilized to identify the incidents selected for the control log (Randomness and Integrity Services, n.d.). One set was generated with a specified number of unique random integers. The integers were assigned a value between 1 and the total number of incidents generated from the automated pull for the specific crime signal and NET area. The randomness from the selected generator stems from atmospheric noise, which for many purposes is better than pseudo-random number algorithms traditionally used in computer programs (Marangon, Vallone, & Villoresi, 2014). Finally, the number produced by the generator was matched with the number assigned to each incident within the specified automated pull. The matched incident was extracted and placed into the control log. This process was repeated for each signal type and NET area in the log.

Time Period of Control Sample

Data collection for the control analytical sample began in May 2022 and concluded in July 2022. To support the proposed analytical design, the control log consisted of 762 non-RTCC treated cases. These incidents were equivalent to the RTCC-treated group as it concerns the crime type/signal and associated NET areas (see

Appendix B).[12] The sampling period from which the control cases were randomly drawn was the same as that of the RTCC log and the RTCC expanded analytical sample. That is, January 16th, 2019, to April 30th, 2022.

Measures for Control Group

The exact same measures included within the RTCC analytical sample were replicated within the control group to meet the conditions of the experimental design (see Appendix A).

Findings

Table 3 presents a comparative overview of the measures employed in the analyses to evaluate the influence of RTCC technologies on the case processing and outcomes of homicide and aggravated assault incidents, as compared to other forms of violent crimes. Here the other forms of violent crime included within the model consist of misdemeanor assaults (n=74) robberies (n=466), domestic violence (n=90), and sexual assaults incidents (n=56). For all continuous measures in the table, independent samples t-tests were used, equal variances not assumed, to ascertain differences between groups. In contrast, for dichotomous measures, Fisher's Exact Test was implemented. This test is used when the sample size is small, and the assumptions of the chi-squared test are not met. The complete model of violent crimes (N =1296) will be presented first, followed by separate analyses for homicides (N =123) and aggravated assaults (N = 496).

[12] For crime types, Chi-Square = .000, 7 df, p = 1.000, indicating there no significant differences between groups. For neighborhood areas Chi-Square = 0.462, 12 df, p = 1.000, indicating there was no significant differences between groups.

Full Violent Crime Model

For the two outcome measures, a significantly greater proportion of cases were cleared within the RTCC-assisted sample (N = 234, 36 percent) compared to the control sample, with only 198 cleared cases (31%). Alternatively, RTCC-assisted cases had a significantly longer time to clearance, with an average of 43 days to case clearance (SD = 112.4) compared to the control sample average of just 19 days (SD = 54.5). Among the measures used in the analysis to adjust for victim and crime investigation characteristics, there were several significant differences between the two samples. Among victim characteristics, both victim age and whether the victim was black were significantly different, although, in frequency, the differences were small. The RTCC-assisted sample had an average victim age of 36 (SD = 14.4), while the control sample was slightly older, with an average victim age of 38 (SD = 15.8). Among the RTCC sample, roughly half of the victims were black (n = 322), while 46 percent of the control sample victims were black (n = 290).

Within the measures comprising human evidence, three of the five were significantly different between the two groups. There were statistically significant greater proportions of lineups (n = 145 vs. n = 109) and flyers (n = 213 vs. n = 63) used, and the presence of external CCTV evidence (n = 314 vs. n = 79) within RTCC-assisted cases compared to the control sample. There were no significant differences in the presence of victim and witness interviews. For crime scene-based evidence, five of the six measures were significantly different between the samples. There was significantly more firearm present (n = 337 vs. n = 218) and vehicles used (n = 377 vs. n = 210) in the commission of a crime, and accordingly, greater use of NIBIN ballistics tracing (n = 193 vs. n = 57),

processing of gunshot residue (n = 80 vs. n = 31), and greater use of the CSI unit (n = 350 vs. n = 246). Lastly, there were also significantly greater numbers of search warrants (n = 51 vs. n = 15) and cell warrants (n = 39 vs. n = 12) used, and more officers utilized (mean = 4.06 vs. mean = 3.82) on RTCC-assisted cases compared to the control sample.

Homicides

As it concerns homicide incidents, for the two outcome measures, 34% of cases were cleared within the RTCC-assisted sample and 27% of control sample cases were cleared, with no significant difference between the two. However, RTCC-assisted homicide cases had a significantly longer time to clearance, with an average of 40 days to case clearance (SD = 40.5) compared to the control sample average of just 9 days (SD = 15.3). Further, only four additional measures proved to be significantly different between the two samples. Three of these measures were present within the human evidence category, including the use of an information flyer (n = 24 vs. n = 7) and suspect lineup (n = 19 vs. n = 7), and also the use of external CCTV footage (n = 33 vs n = 14), and one within the category that comprises crimes scene evidence, with that being latent/DNA (n = 12 vs. n = 24).

Aggravated Assaults

The outcomes measures for aggravated assault incidents did not show any notable disparities between the two samples. This suggests that both samples had a similar quantity of incidents cleared by arrest, and that time to clear these incidents did not take significantly longer in one sample compared to the other. All remaining categories produced at least one measure that proved to be significantly different between the samples examined. In terms

of victim characteristics, there were more black victims in the RTCC-assisted sample (n = 132) compared to the control sample (n = 99).

Table 3. *Comparative overview of measures used in the analysis of homicide and aggravated assault incidents*

	Full Violent Crime Model N = 1296		Homicide n = 123		Aggravated Assaults n = 496	
Measures	**RTCC-assisted cases n (%)**	**Control cases n (%)**	**RTCC-assisted cases n (%)**	**Control cases n (%)**	**RTCC-assisted cases n (%)**	**Control cases n (%)**
Cleared[2]	234(36%)	198(31%)*	21(34%)	17(27%)	79(32%)	81(33%)
Time to Clearance	43(112.4)	18(54.5%)**	30(40.5)	9(15.3)**	27(76.7)	17(45.9)
RTCC technologies						
CCTV	564(87%)	--	54(87%)	--	222(90%)	--
Social media	120(19%)	--	8(13%)	--	46(19%)	--
Facial recognition	165(26%)	--	18(29%)	--	60(24%)	--
License plate reader (LPR)	82(13%)	--	6(10%)	--	45(18%)	--
ShotSpotter	125(19%)	--	11(18%)	--	72(29%)	--
Victim characteristics						
Victim age[1]	36(14.1)	38 (15.8)**	40(15.2)	41(16.1)	34(13.2)	39(16.3)
Victim male[2]	428(69%)	410(65%)	48(81%)	43(71%)	160(68%)	168(69%)
Victim black[2]	322(52%)	290(46%)*	34(58%)	44(73%)	132(56%)	99(41%)**
Human evidence						
Victim interview[2]	585(90%)	591(91%)	56(90%)	47(76%)	232(94%)	223(90%)
Witness interview[2]	234(36%)	200(31%)	20(32%)	25(40%)	88(36%)	78(32%)
Flyer[2]	213(33%)	63(10%)***	24(39%)	7(11%)***	73(29%)	21(9%)***
Lineup[2]	145(22%)	109(17%)*	19(31%)	7(11%)*	50(20%)	36(15%)
External CCTV[2]	314(49%)	79(12%)***	33(53%)	14(23%)***	131(53%)	18(7%)***
Crime scene evidence						
Firearm used[1]	337(52%)	218(34%)***	32(52%)	27(44%)	151(61%)	60(24%)***
Vehicle involved[2]	377(58%)	210(32%)***	33(53%)	24(39%)	166(67%)	68(27%)***
Ballistics tracing (NIBIN)[2]	193(30%)	57(9%)***	19(31%)	12(19%)	100(40%)	16(7%)***
Gunshot residue[2]	80(12%)	31(5%)***	4(7%)	5(8%)	41(17%)	8(3%)***
Latent prints/DNA[2]	149(23%)	175(27%)	12(19%)	24(39%)*	65(26%)	65(26%)
Crime Scene Investigation[2]	350(54%)	246(38%)***	37(60%)	28(45%)	154(62%)	94(38%)***
Warrant evidence & manpower						
Search warrant[2]	51(8%)	15(2%)***	3(5%)	3(5%)	21(9%)	2(1%)***
Cell warrant[2]	39(6%)	12(2%)***	3(5%)	2(3%)	13(5%)	1(.4%)**
Number of officers[1]	4.06(1.9)	3.82(1.9)**	4.19(2.0)	4.05(1.4)	4.25(1.9)	3.67(1.3)

* $p \leq .05$, ** $p \leq .01$, *** $p \leq .001$; [1] For these measures, independent samples t-tests were used to determine differences, equal variances not assumed. [2] Fisher's Exact Test was used to determine differences for these dichotomous measures.

The RTCC-assisted sample also deployed suspect information flyers (n = 73) and obtained external CCTV video more frequently (n = 131) than the control sample (n = 21 and n =18, respectively). In the crime scene evidence category, five of the six measures showed significant differences between the two samples, with the RTCC sample using them more frequently. These measures include firearm usage (n = 151 vs. n = 60), involvement of a vehicle (n = 166 vs. n = 68), NIBN utilization (n = 100 vs. n = 16), GSR testing (n = 41 vs. n = 8) and calling the crime scene investigation unit to the incident location (n = 154 vs. n = 94). In terms of warrant evidence and manpower, the RTCC sample was more likely to obtain a search warrant (n = 21 vs. n = 2) and cell warrant (n = 13 vs. n = 1) compared to the control sample.

Case Processing

Module network analysis helped to determine the influence of RTCC technologies on the case processing of homicide and aggravated assault incidents. The network modules here were constructed with an EBICglasso estimation procedure with a nonparanormal transformation. EBICglasso is a powerful statistical approach for conducting network analysis on high-dimensional datasets and this method excels at accurately estimating the conditional dependence relationships between variables, which is a crucial aspect of network analysis (Foygel & Drton, 2010). A notable advantage of this approach is its ability to efficiently determine the sparsity pattern of the graphical model, allowing for identification of the most relevant connections while discarding insignificant ones. Additionally, EBICglasso is capable of handling scenarios where the number of variables greatly exceeds the number of observations, a common situation in genomics and other fields that rely on high-throughput data (Williams, Rhemtulla, Wysocki, & Rast, 2019).

Here, the analyzed data was drawn from the expanded data set derived from the RTCC log and the stratified randomly drawn control sample as discussed in the previous "Data and Measures" section. Here the entire sample was used (N=1296).The analysis utilized measures from the data set which coded whether various evidential sources were used in the investigation of the violent crime incident regardless of whether usable evidence was obtained from the evidence sources. This is because the purpose of the analysis here was to determine how various technology or information-based evidence sources were being used by officers and detectives, exclusive of whether they yielded usable evidence for prosecutorial purposes. These evidence sources were grouped into four types and included RTCC technologies (CCTV, LPR, social media, facial recognition, and ShotSpotter), crime scene evidence (NIBIN, GSR, CSI, Latents/DNA), human-based evidence (victim interviews, witness interviews, media flyers, lineups, and external CCTV) and warrant based evidence (search warrants, cell warrants).[13]An additional crime type variable was intentionally coded to represent investigations of homicides and aggravated assaults, with the aim of examining the role of police technology in investigating these crimes. Here, victim characteristics and other solvability factors were not included as they were outside of the scope of the analysis.

Beyond these distributions presented in Table 4, the findings from the network analysis indicate several important things. First, the network of RTCC-supported cases and the randomly drawn control cases had similar levels of sparsity within the network.

[13] For an overview of the measures used in this analysis, refer back to the comparative overview of both samples presented in the previous table.

Table 4. *Weight matrix of variables in homicide and aggravated assault network*

Variable **(N=1296)**	**RTCC Sample (Weight)**	**Control Sample (Weight)**
Crime of Interest		
Homicides and Agg. Assaults	--	--
RTCC Evidence		
CCTV	.000	--
Social media	.000	--
Facial recognition	.000	--
LPR	.028*	--
ShotSpotter	.049*	--
Human evidence		
Victim interview	.000	.000
Witness interview	.000	.000
Flyer	-.008*	-.015*
Lineup	.000	-.006*
External CCTV	.000	.000
Crime scene evidence		
NIBIN	.015*	.053*
Gunshot residue	.000	.000
Latent prints/DNA	.000	.000
CSI	.025*	.034*
Warrant Evidence		
Search warrant	.025*	.000
Cell warrant	.000	-.017*
Number of Nodes	17	12
Non-zero edges	76/136	46/60
Sparsity	0.441	0.303

* Represents an existent connection to the crime of interest node.

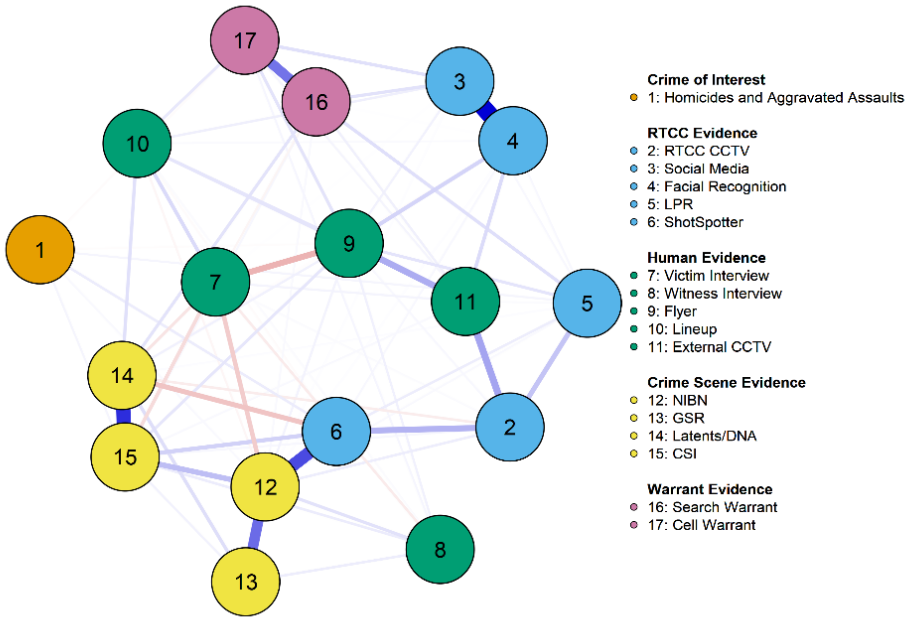
Where the RTCC-assisted sample had a sparsity measure of .441 compared to that of .303 of the control sample. This measure quantifies the difference between a fully connected network, where all nodes have a relationship with one another, and the actual number of relationships observed in the analyzed network. A lower sparsity metric suggests greater interconnectedness among nodes, while a higher metric indicates weaker connectivity. In

this case, the network is moderately dense, as supported by the number of non-zero edges present in the matrix.

Despite the greater number of evidential nodes present in the RTCC network (n = 18), the level of connectedness with other evidential nodes remains about the same as in the control sample (n = 13). This implies that the integration of RTCC technologies into case processing has been proportionate to other evidential nodes, indicating a successful implementation despite the Miami RTCC being a new system as of 2019. This contradicts previous research suggesting that police agencies are resistant to incorporating innovative technologies into traditional practices and structures (Ioimo & Aronson, 2004; Manning, 2003; Manning, 2008). In this case, technology is seen as beneficial to officers' work rather than a hindrance, leading to expected positive outcomes and outputs (Brunschot, Abela, Witt & Hak, 2022; Lum, Koper, & Willis, 2016).

Second, RTCC technologies appear to occupy a substantial role in the processing of homicide and aggravated assaults to a greater extent than traditional evidential sources. Presented in Figure 3, all RTCC evidence nodes appear to have many connections with other traditional evidence sources. Two of the RTCC evidence sources, however, appear to have greater influence on the investigation of the particular crimes of interest. According to the weight matrix (Table 4), the gunshot detection system, ShotSpotter (influence of .049) and the LPR (.028) appear to have stronger connections to the node representing homicides and aggravated assaults as compared to other traditional nodes, such as the CSI unit (.025) and NIBIN (.015).

Figure 3. *RTCC-assisted module network analysis for homicide and aggravated assaults incidents*

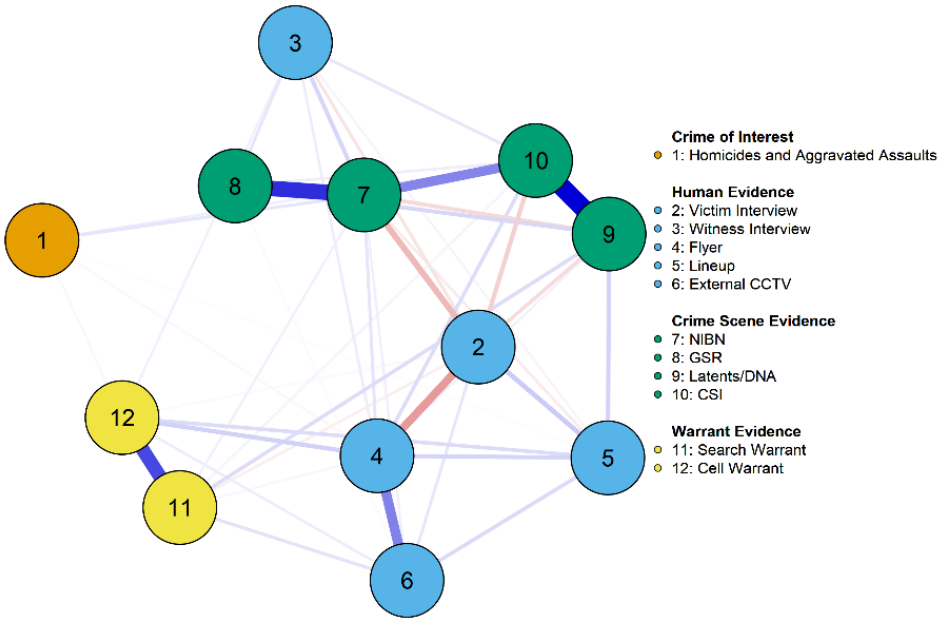


These findings are perhaps as expected, as more than half of the incidents within the sample either utilized a firearm or had a firearm present during the commission of the crime. Two of the measures within this network, suspect flyers (-.008) and cell warrants (-.010), showcased a negative relationship with the crimes of interest, implying that 1) their connection to the node relied on the presence of other connections, and 2) the presence or absence of these measures would not affect the strength of the other connections.

Within the control sample (Figure 4.) RTCC technologies were removed from the network plot to determine which sources outside the scope of the RTCC maintained strength within the investigative process regardless of the integration of the unit. Here, we see that crime scene evidence sources played the greatest role in the investigation. As seen in the weight matrix (Table 4) the ballistics tracing NIBIN measures had the greatest

influence on the investigation of these crimes (.053), followed by the deployment of the CSI unit to the scene of the crime (.034).

Figure 4. *Non-RTCC-assisted module network analysis for homicide and aggravated assaults incidents*



Once again, given the high incidence of firearm use, these findings are expected. Moreover, within the MPD standard operating procedures, the CSI unit is tasked with collecting firearm casings when there are ten or more present at the scene of the incident (whereas patrol collects the casings if there is less than ten). These casings then get sent to the property unit before being transferred to the internal gun crime processing unit which houses the ballistics tracing mechanisms. Additionally, three of the measures within this network, suspect flyers (-.015), suspect lineups (-.006) and cell warrants (-.017), exhibited a negative relationship with the crimes of interest. This indicates that these measures' association with the node was reliant on the presence of other connections, and their absence would not weaken the other connections' strength in the network.

Case Clearance

Table 5 presents the analysis results aimed at examining whether RTCC technologies influenced the case clearance of homicides and aggravated assault incidents. Although Table 3. shows that RTCC-assisted homicide and aggravated assault incidents were not cleared more frequently than the control sample, these numbers alone do not consider the differences in the case processing of these crimes. Therefore, to accurately determine the impact of RTCC technologies on clearance rates for these incidents, the multivariate logistic regression analysis controlled for such differences.

Here, missing data was identified in 52 total cases and deleted using the listwise deletion method. Specifically, 3.5% (n = 46) of all cases had missing victim gender information and 3.9% (n = 50) had missing victim race information. Moreover, two cases or .2% of the sample had missing data regarding whether the CSI unit was deployed to the scene of a crime. As for victim age, fifty-five incidents were missing data, but this information was transformed and replaced with the series mean. The Chi-square and -2 log-likelihood coefficients were statistically significant in all models, with the exception of *Model One*, indicating good model fit. Within the significant models Nagelkerke pseudo R-square indicated an increasing ability of the models to predict the model variance ranging from a low of .8% in *Model Two*, to a high of twenty-three percent in *Model Six.* All independent measures used in these models were examined for multicollinearity using the variance inflation factor (VIF) with no multicollinearity revealed. All VIF factors ranged from a low of 1.034 to the highest at 2.067.[14]

[14] These scores are far below the area of concern for VIF where convention dictates that scores of 5 or greater suggest the presence of multi-collinearity among independent measures.

Model One only included homicide and aggravated assault incidents. Here, we see that without the presence of any other covariates, the odds of these crimes being cleared, as compared to all other crimes included in the model, were not significant. In *Model Two*, all violent crimes that received RTCC-support, were included within the model to determine the influence of technology on case clearance. These crimes established significance within the model, indicating approximately 36% increased odds of case clearance (≤ .05). Here, we see that the crimes of interest improved slightly in likelihood of case clearance but did not reach levels of significance.

Homicide and aggravated assault incidents remained insignificant in *Model Three,* in which victim characteristics were subsequently introduced. Of all covariates included within the model, RTCC-assisted violent crimes and whether the victim was male also proved to be significantly related to whether a case was cleared. Here, violent crime cases receiving RTCC technology support being 38% more likely to be cleared than those without RTCC support and male victims resulted in 34% lower odds of case clearance. Human evidence sources were added to *Model Four*. Within this model, all violent crime RTCC-assisted cases maintained significance (47% more likely to be cleared), as did whether the victim was a male, which still resulted in roughly 33% lower odds of case clearance. All human evidence sources proved to be of statistical significance, with suspect lineups being the most impactful. In fact, when a lineup was present, the odds of the case being cleared were five times greater than when there was no lineup, while controlling for other factors (≤ .001). External CCTV, or private surveillance CCTV did not result in increased odds, but rather the presence of this form of evidence was associated with 58 percent decreased odds in case clearance.

Model Five incorporated evidence from the crime scene, revealing that incidents of homicide and aggravated assaults remained insignificant. Further, cases involving RTCC-assistance had a 47% greater likelihood of being cleared (≤ .05). Of the victim characteristics, a male victim resulted in 26% lower odds of case clearance (≤ .05). Among the five human evidence elements, all five remained significant, with suspect lineups having the greatest impact on case clearance (OR = 5.679; ≤ 001). External CCTV, once again, depicted lesser odds of case clearance (OR = .423; ≤ .001). As for crime scene variables, only the presence of a firearm during the commission of the crime was significant, revealing 48% lower odds of case clearance.

In *Model Six*, all variables, including those related to warrant evidence and manpower, were included in the final analysis. Here, the crimes of interest remained insignificant. A male victim sustained statistical significance, revealing an unchanged 26% lower odds of case clearance (≤ .05). Among all the variables, the support of RTCC technology, witness interviews, lineups, suspect flyers, and search warrants significantly increased the odds of case clearance. RTCC-assisted violent crime cases had 64% better odds of case clearance, and the use of a suspect lineup remained the strongest indicator in the model (OR = 5.416; ≤ .001). External CCTV presence (OR = .403; ≤ .001) and the use of a firearm (OR = .511; ≤ .001) significantly decreased the likelihood of the case being cleared, controlling for all other solvability factors.

Table 5. *Nested multivariate logistic regression of predictors of homicide and aggravated assault case clearance and influence of RTCC technology*

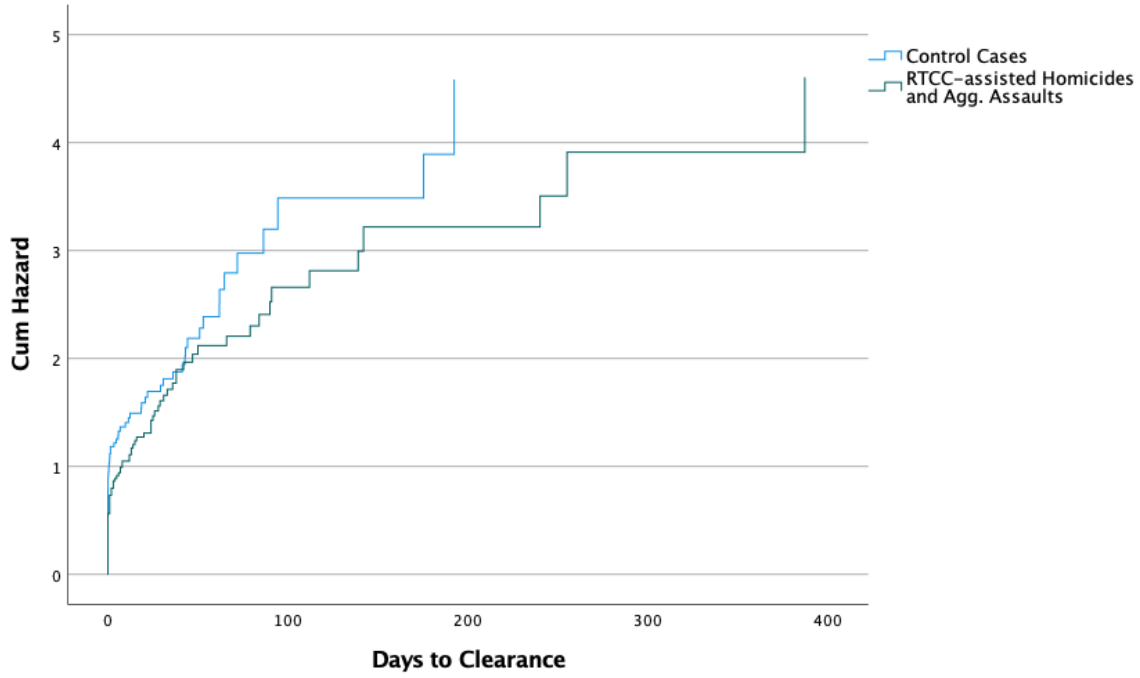
***Measures (*N=1244)**	***Model One*** **B (Odds ratio)**	***Model Two*** **B (Odds ratio)**	***Model Three*** **B (Odds ratio)**	***Model Four*** **B (Odds ratio)**	***Model Five*** **B (Odds ratio)**	***Model Six*** **B (Odds ratio)**
Crime of Interest						
Homicide and Agg. Assaults	-.103(.902)	-.104(.901)	-.078(.925)	-.002(.998)	.022(1.022)	.049(1.051)
Impact of Technology						
RTCC-Assisted Violent		.307(1.359)*	.324(1.383)**	.384(1.468)**	.506(1.659)**	.495(1.640)**
Victim characteristics						
Victim age			-.002(.998)	.002(1.002)	-.001(.999)	-.001(.999)
Victim male			-.414(.661)**	-.405(.667)**	-.296(.744)*	-.306(.736)*
Victim black			-.006(.994)	.038(1.039)	.118(1.125)	.132(1.141)
Human evidence						
Victim interview				.533(1.703)*	.367(1.443)	.388(1.475)
Witness interview				.456(1.578)***	.475(1.609)***	.450(1.569)**
Flyer				.609(1.838)***	.702(2.018)***	.645(1.907)***
Lineup				1.676(5.342)***	1.712(5.539)***	1.689(5.416)***
External CCTV				-.861(.423)***	-.811(.445)***	-.908(.403)***
Crime scene evidence						
Firearm used					-.650(.522)*	-.671(.511)***
Vehicle used					-.297(743)	-.376(.686)*
Ballistics tracing (NIBIN)					.148(1.160)	.137(1.147)
Gunshot residue (GSR)					.105(1.111)	.053(1.054)
Latent prints/DNA					-.009(.991)	-.113(.893)
Crime Scene Investigation					-.152(.859)	-.174(.841)
Warrant evidence & manpower						
Search warrant						.911(2.488)**
Cell warrant						.848(2.330)*
Number of officers						.049(1.050)
Model Chi-Square	.732	7.219*	18.215**	179.449***	208.982***	230.047***
-2 Log likelihood	1586.130	1579.643	1568.647	1407.414	1377.880	1356.815
Nagelkerke Pseudo R-square	.001	.008	.020	.186	.215	.234
Constant				-.685(.504)***		

*$p \leq .05$, **$p \leq .01$, ***$p \leq .001$

Time to Case Clearance

The analyses to determine whether RTCC-assisted cases improved the time that it took to clear cases are reported in Figures 5 and 6, and Table 6. The simple frequency of days to case clearance reported in Table 3 suggests that RTCC assistance did not improve the speed at which arrests were made in homicide and aggravated assault incidents. However, this does not account for the various victim characteristics and case processing characteristics that could influence the speed at which cases are cleared which are outside the scope of RTCC influence.

Figure 5. *Cumulative rate of case clearance RTCC-assisted homicides and aggravated assaults vs. control sample*



It may be possible that when accounting for fundamental that exist within different crime incidents, the difference in the time that it takes to clear cases is adjusted. To determine this, first, a Kaplan-Meier survival analysis was performed to identify the

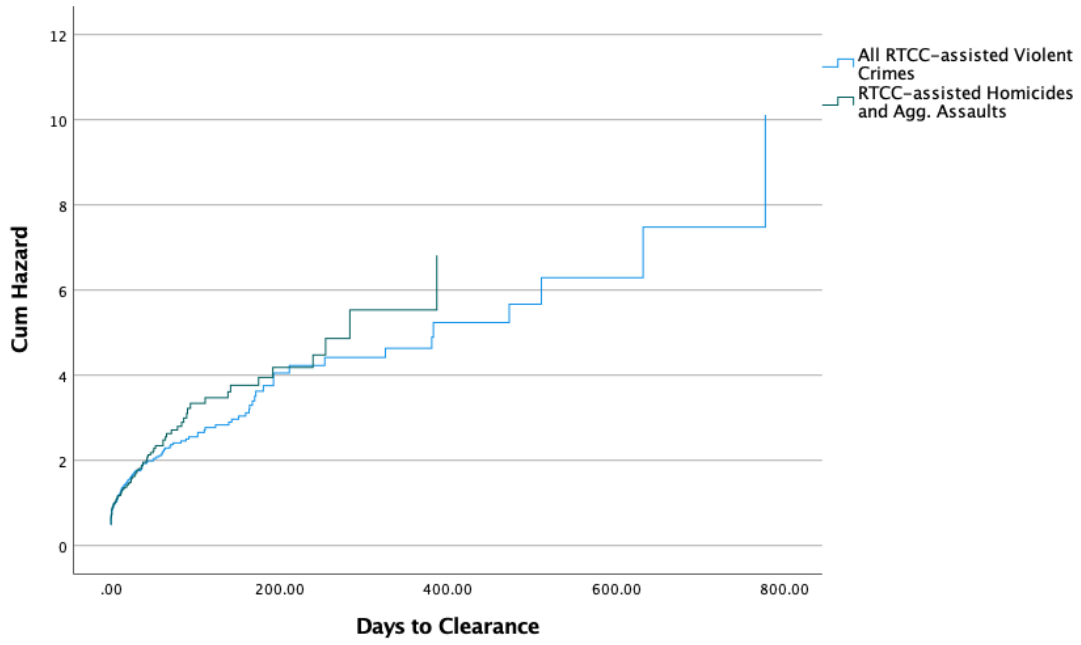
trajectories in the time- to-case clearance within the two samples. Here, only the treatment and control groups for aggravated assaults and homicides were included within the analysis. The total number of cleared cases between the two samples was 198, of which 100 were cleared in the RTCC-assisted sample, and 98 in the control. The outcome of that analysis is reported in Figure 5. This analysis does not adjust for case characteristics and only plots the proportion of cases cleared within each sample across the period of days present within the observation period. Here we do not see a clear difference in the trajectory of RTCC-assisted cases and the control sample, with control sample cases being cleared at a similar rate as those which were assisted by RTCC technologies. The Mantel-Cox Log Rank comparison indicates that these trends were not significantly different ($\chi 2 = 2.736$, 1 df, $p = .098$).

A Cox Proportional Hazard regression model was utilized to address the dissimilarities in victim and case processing characteristics between two samples. This analysis was initially conducted solely on homicide and aggravated assault cases (N = 198) due to their distinct characteristics. However, the sample size was limited, and the resulting Chi-Square coefficient did not attain statistical significance, indicating a suboptimal model fit. Consequently, all cleared violent crimes were incorporated in the analysis and further stratified by technology support to examine whether the time to clearance varied for RTCC-assisted homicides and aggravated assaults in comparison to other forms of RTCC-assisted violent crimes.

The comparison of the trend line of that model is presented in Figure 6, and the measures included in the model and coefficients are reported in Table 6. Each of the victim characteristics, human evidence, crime scene evidence, warrants, and manpower measures

were included to consider their impact on the time taken to clear cases. With this adjustment and holding constant the effect of all other measures, no significant difference was observed in the days to case clearance between the RTCC-assisted homicide and aggravated cases and all other RTCC-assisted violent crimes (B = -.056, SE = .104, *p* = .590, n.s.).

Figure 6. *Cumulative rate of case clearance for RTCC-assisted homicides and aggravated assaults vs. other forms of violent crime adjusted at mean for case characteristics*



Among the remaining measures in the analysis, four were significantly related to the time that it took to clear cases. Of the victim characteristics, the only significant predictor was whether the victim was male. When the victim was male the case had significantly greater odds (OR = 1.530) of being cleared before female victims. Among human evidence measures, whether a public information flyer was disseminated was

associated with 39 percent (OR = .612) lower odds of being cleared before those cases without a flyer. Within the crime scene group, whether a firearm was present during the commission of the crime lowered the odds of being cleared before incidents that did not utilize a firearm, with 29% lower odds ($p = \leq .05$). Finally, within the warrant and manpower measures, the presence of a cell phone warrant lowered the odds of a case being cleared as compared to incidents that did not require the obtaining of a cell phone warrant (OR = .600; $p = \leq .05$). The model Chi-Square coefficient was statistically significant indicating a good model fit ($\chi^2 = 73.357$, 17 df, $p = <.001$).

Table 6. *Cox proportional hazard model coefficients predicting days to clearance (homicides and aggravated assaults).*

Measures (N=413)	B (SE)	Odds Ratio	*p* - value
Crime of Interest			
RTCC-assisted Homicide and Agg. Assault	.056(.104)	1.058	.590
Victim Characteristics			
Victim age	.000(.004)	1.000	.950
Victim male***	.425 (.116)	1.530	< .001
Victim black	-.071(.104)	.932	.496
Human Evidence			
Victim interview	.328(.234)	1.389	.160
Witness interview	.053(.105)	1.054	.615
Lineup	-.214(.115)	.808	.063
Flyer***	-.419(.143)	.612	< .001
External CCTV	-.196(.138)	.822	.156
Crime Scene Evidence			
Firearm used*	-.347(.141)	.707	.014
Vehicle used	-.210(.116)	.811	.070
Ballistics tracing (NIBIN)	.085(.184)	1.089	.642
Gunshot residue (GSR)	.314(.218)	1.369	.149
Latent prints/DNA	.122(1.130)	1.130	.421
Crime Scene Investigation (CSI)	.025(.142)	1.025	.860
Warrant Evidence & Manpower			
Search warrant	.309(.222)	1.363	.181
Cell warrant*	-.511(.234)	.600	.029
Number of officers	.038(.037)	1.039	.310
Model Chi-Square		73.357***	
-2 Log likelihood		4215.245	

*p ≤ .05, **p ≤ .01, ***p ≤ .001

Supplemental Analysis

A supplemental analysis was conducted to further augment the results of the former nested regression model (see Table 7). The research questions posed here was the following: "Do RTCC-assisted homicide and aggravated assault crime incidents have

greater rates of case clearance compared to homicide and aggravated assault cases that did not receive RTCC support?"

[H4] RTCC-assisted homicide and aggravated assault crime incidents will have greater rates of case clearance as compared to the control cases that did not receive RTCC support.

Here, only RTCC-assisted homicides and aggravated assaults were used in the model and compared to those that did not receive RTCC-support. Within this model, missing value analysis resulted in the following missing variables: victim sex 20 incidents (3.2%); victim race 24 incidents (3.9%), and CSI 2 incidents (.3%). Data was deleted using the listwise deletion method, resulting in 594 total incidents remaining in the analysis, or 26 removed incidents. As for victim age, twenty-seven incidents were missing data, but this information was transformed and replaced with the series mean.

The Chi-square and -2 log-likelihood coefficients suggested models 3, 4 and 5 were statistically significant indicating good model fit, and within those models Nagelkerke pseudo R-square indicated an increasing ability of the models to predict the model variance ranging from a low of 21% in *Model Three,* to a high of twenty-five percent in *Model Five*. All independent measures used in these models were examined for multicollinearity using the variance inflation factor (VIF) with no multicollinearity revealed. All VIF factors ranged from a low of 1.083 to the highest at 2.184.

Despite the exclusion of all violent crimes, RTCC-assisted homicides and aggravated assaults did not reach statistical significance in any of the five models, indicating that the use of technologies housed within the investigative unit do not increase the odds of case clearance for these two crime types. Rather, as seen in the final model,

whether a victim or witness interview took place, whether an information flyer was deployed, and whether a suspect lineup was conducted, all significantly improved the odds of a case being cleared. Within this analysis, external CCTV and the use of a firearm continued to depict lower odds of case clearance, 64% and 59% respectively.

Table 7. *Multivariate logistic regression of predictors of case clearance among RTCC-assisted homicide and aggravated assaults and control sample*

***Measures (*N=594)**	***Model One*** **B (Odds ratio)**	***Model Two*** **B (Odds ratio)**	***Model Three*** **B (Odds ratio)**	***Model Four*** **B (Odds ratio)**	***Model Five*** **B (Odds ratio)**
RTCC-Assisted Homicides and Agg. Assaults	.051(1.052)	.032(1.032)	-.027(.974)	.064(1.066)	.036(1.036)
Victim characteristics					
Victim age		-.006(.994)	-.002(.998)	-.006(.994)	-.006(.994)
Victim male		-.365(.694)	-.284(.753)	-.175(.839)	-.174(.840)
Victim black		-.026(.975)	.116(1.123)	.124(1.133)	.115(1.122)
Human evidence					
Victim interview			1.173(3.230)**	.996(2.706)*	.975(2.651)*
Witness interview			.519(1.680)*	.546(1.727)**	.524(1.689)*
Flyer			.777(2.174)**	.866(2.378)**	.837(2.309)**
Lineup			1.895(6.650)***	1.921(6.827)***	1.892(6.632)***
External CCTV			-.954(.385)***	-.930(.394)***	-1.029(.357)***
Crime scene evidence					
Firearm used				-.861(.423)**	-.892(.410)**
Vehicle used				.079(1.082)	.037(1.038)
Ballistics tracing (NIBIN)				.223(1.250)	.206(1.229)
Gunshot residue (GSR)				.309(1.362)	.217(1.243)
Latent prints/DNA				-.051(.951)	-.127(.880)
Crime Scene Investigation				.019(1.019)	-.033(.968)
Warrant evidence & manpower					
Search warrant					.601(1.824)
Cell warrant					.946(2.575)
Number of officers					.096(1.101)
Model Chi-Square	.083	5.231	97.509***	108.543***	116.473***
-2 Log likelihood	747.503	742.356	650.078	639.044	631.114
Nagelkerke Pseudo R-square	.000	.012	.211	.233	.249
Constant			-.739(.478)***		

*p ≤ .05, **p ≤ .01, ***p ≤ .001

Chapter Five: The Influence of RTCC Technologies on the Case Outcomes of Robbery Incidents

Introduction

Whereas the former chapter assessed homicides and aggravated assaults, the primary objective of this chapter was to examine the influence of police technologies on the outcome measures of cases that involve robbery incidents. In particular, the study sought to evaluate the impact of RTCC technologies on case clearance rates, ascertain their efficacy in facilitating investigative procedures, and investigate whether cases receiving RTCC assistance demonstrates accelerated clearance rates. The research questions guiding this study were therefore three-fold: (1) "What is the impact of RTCC technologies on the clearance of robbery incidents?" (2) "To what extent have RTCC technologies been integrated into the case processing of this particular crime type?" and (3) "Does RTCC support improve the time that it takes to clear these cases?"

The opening section of this chapter presents a review of the relevant literature pertaining to robberies and the role of technological advancements in their investigation. A subsequent section will detail the analytical approach taken and provide a brief overview of data collection process and the measures involved within the analysis. Once all analytical approaches are exhausted, a supplemental analysis will be presented to augment the findings and provide additional insight into the impact of the technologies assessed. Limitations and policy implications for this and all subsequent studies will be located in chapter 8.

Review of Literature

Although research on the investigation of robbery incidents is comparatively limited when contrasted with other types of violent crimes, there exist two pertinent and distinctive bodies of scholarship that warrant discussion. The first pertains to the factors that impact the solvability of robbery cases, while the second concerns the empirical literature on the use of technology in the investigation of such crimes. In the following sections, a comprehensive review of both is presented.

Factors Known to Impact Case Clearance

Despite its frequent occurrence and grave nature as a violent interpersonal crime, scholarly inquiry on robbery has been comparatively sparse. National data reveals that the clearance rate for robberies is merely 27%, which is significantly lower than the clearance rates for homicides and assaults, standing at 50% and 47%, respectively (Li & Lartey, 2022). Although the average robbery clearance rate in the United States has remained stagnant over the past four decades, there has been variation among law enforcement agencies, with some consistently surpassing the national average in long-term robbery clearance rates, while others falling below it. Scholars have put forth diverse explanations for these disparities (Scott, et al.,2019).

The low clearance rates of robberies have been attributed to the distinctive features of this crime and the variables that impact its solvability. Previous research has emphasized the difficulty in identifying suspects in robbery cases, given that they are often committed by individuals leading a high-risk lifestyle (Cook, 1987; Wright & Decker, 1997). Additionally, scholars have pointed out that insufficient resources, outdated technologies,

and limited time and motivation on the part of law enforcement officers to investigate these incidents have all contributed to the low clearance rates (Horvath, Meesig, & Lee, 2001).

The understanding of case clearance in the domain of robberies has remained limited, prompting scholars to adopt a broad approach to comprehensively present legal and extralegal factors known to affect the case clearance of homicides and aggravated assaults. However, the application of the same variables to their models has yielded inconsistent results. While some studies indicate that extralegal variables have no effect on case clearance (Roberts, 2008), others suggest that such variables may have an impact, particularly when the victim is female or white (Snyder, 1999). Additionally, it has been observed that the seriousness of the offense is a critical factor in determining case clearance likelihood (Stolzenberg & D'Alessio, 2004), as well as the existence of a victim-offender relationship (Eitle, Stolzenberg & D'Alessio, 2005).

The impact of forensic evidence on the successful resolution of robbery cases is often recognized as a crucial factor. Nevertheless, scientific evidence is utilized in only a small fraction of violent crime cases and even when it is present, its contribution to case outcomes is typically negligible. Contrarily, non-scientific evidence like eyewitness identification has been found to be a stronger predictor of conviction. Nonetheless, Baskin & Sommers (2010) study revealed that the presence of guns and crime scene evidence, as well as the victim's and witness's statements, increased the likelihood of clearance in robbery cases. The collection of physical evidence at the scene resulted in an arrest in 48% of cases compared to only 14% of cases where such evidence was absent. However, the examination of physical evidence before arrest happened in just 16% of cases, and about 25% of arrests resulted from direct arrest techniques. Moreover, a systematic review by

Higginson, Eggins, and Mazerolle in 2017 showed that the presence of crime scene technicians at robbery crime scenes significantly enhanced the likelihood of case clearance by three times and the probability of an arrest by almost double.

In a more recent study, Scott and colleagues (2018) compared the aforementioned factors and other relevant characteristics to determine the difference in case clearance amongst high and low performing agencies. The findings revealed that high performing agencies prioritize investigating robberies and assign all cases to a detective who acts as a first responder to the crime scene. They also employed centralized organizational structures, required specialized training for detectives, and provided adequate resources, such as support from forensic units, intelligence, crime analysis, and fugitive task forces. Conversely, low performing agencies reported higher caseloads per detective and complaints about resources. It should be noted that robbery units typically experience higher case or workloads as compared to other violent crime units, such as those comprising homicide investigators (LoFaso, 2020).

In the same study, the authors additionally collected data on over 350 robbery incidents across seven of the eight agencies included in the preliminary analyses. Utilizing logistic regression analysis, the researchers sought to identify factors linked to successful robbery clearance, discovering that the early collection of physical evidence and the presence of an investigative supervisor during initial response significantly augmented the likelihood of resolving the crime. Moreover, collecting physical evidence later in the investigation was associated with substantially enhanced case clearance rates. In contrast, investigators following up with the victim decreased the likelihood of clearance, potentially due to their preference for tackling more complex cases.

Impact of Technology on Robbery Investigations

A discussion on the role of technology in robbery investigations can be quite complex. Intuitively, technology has the potential to greatly benefit the investigation of robberies by providing police with new tools and methods for collecting and analyzing evidence, identifying suspects, and tracking their movements. For example, forensic technologies such as, ballistics testing, and digital forensics can help police connect suspects to crime scenes and build stronger cases against them. Moreover, as a form of street crime, the commission of robbery incidents in external environments and in the purview of CCTV cameras, could further assist investigators in corroborating victim/witness testimonies and help investigators piece together events leading up to the event. Social media analysis and LPR can further help assist in identifying the whereabouts of the perpetrator and property stolen at the scene of the robbery. However, despite these potential benefits, the use of police technology has not necessarily led to an improvement in case clearance rates for robberies.

There are several reasons for this. First, the sheer amount of data that police must sift through in order to identify suspects can be overwhelming, particularly in the age of big data (Gibs Van Brunschot et al., 2022). While technology has made it easier to collect and store data on potential suspects and crime scenes, it has not necessarily made it easier to analyze and interpret this data in a timely and effective manner. Second, technology can sometimes create false leads and waste valuable investigative resources. For example, Hill and Mac (2023) have pointed out that facial recognition technology has been subjected to criticism for its ability to potentially misidentify innocent individuals as suspects, ultimately leading law enforcement agencies to pursue incorrect investigative leads.

Finally, the effectiveness of police technology in solving robberies can be limited by broader systemic issues such as underfunding, understaffing, and lack of training. Even the best technology cannot compensate for these fundamental issues.

Perhaps, the only study that assessed the impact of police technology on the case clearance of robbery incidents in specific, found that license plate readers are a promising tool in law enforcement for investigating this crime type. In their study, involving the large-scale deployment of 100 fixed LPRs throughout Charlotte-Mecklenburg Police Department's jurisdiction, Koper & Lum (2019) found that the technology was associated with an increase in the number of robbery investigations that resulted in an arrest or prosecution. The LPR technology significantly increased the number of investigative leads, which in turn increased the likelihood of a successful outcome for robbery investigations.

The study also found that the effectiveness of LPR technology was influenced by other factors, such as the level of training and experience of the officers using the technology, the quality of the LPR system, and the availability of other resources for investigations. Overall, the study suggests that large-scale deployment of LPR technology can be a valuable tool for law enforcement agencies in investigating robbery cases but caution that it should not be viewed as a 'cure-all' for all criminal investigations.

The employment of CCTV surveillance in the investigation and case clearance of violent crime incidents has also attracted considerable attention in recent years. However, it is important to acknowledge that most studies examining the impact of CCTV tend to view it as a tool for deterrence and primarily focus on its potential for reducing crime, which is not the intention of the present study. Of those trying to achieve the same objective, Morgan and Dowling (2018) conducted a study in the New South Wales region

of Australia, which revealed that requesting surveillance footage from rail network authorities resulted in an astounding 18% increase in the clearance rate of police investigations into robberies. Furthermore, Ashby's (2018) analysis of the British rail network showed a marked difference in the clearance rates of robbery incidents with and without access to relevant CCTV footage. The findings demonstrated that CCTV surveillance significantly increased the clearance rate of robberies by more than two-fold, underscoring the invaluable role of this technology in criminal investigations.

Impact of Real-Time Crime Centers

Thus far, only two evaluations of RTCCs have been conducted. As it pertains to the crime reduction potential of Chicago SDSC's on the robbery incidents throughout select districts, the evaluation found clear support. In all evaluated models, a statistically significant decrease in crime was observed. The SDSC systems were identified as "critical enablers" for SDSC operations due to their ability to provide real-time situational awareness of incidents and support a framework for long-term decision-making (Hollywood et al., 2019). The second evaluation of MPD's RTCC did not provide insight into the impact of technology on robbery but rather, the researchers found support for improved crime clearance rates across all categories of violent crime which also included homicides, assaults, sexual assaults, and domestic violence incidents (Guerette et al., 2022). The same data and quasi-experimental design utilized in the latter evaluation will be replicated here, but with a particular emphasis on the robberies.

Current Study

Considering that this data is derived from a broader empirical investigation on the influence of real-time crime centers on case processing and outcomes of violent crime

incidents (Guerette, et al., 2022), it can be inferred that robbery cases receiving RTCC support are more likely to be cleared, involve higher levels of technological integration in their case processing, and not experience any delays in case processing time. (i.e., time to arrest) as a result of investigative effort exerted and the subsequent actions taken in that process. Nonetheless, in contrast to other forms of violent crime such as homicides, assaults, and sexual assaults, robbery incidents may have lower clearance rates according to national data, even without taking into account any other factors. As a result, the following hypotheses are proposed to address the original research questions.

[H1] Compared to all other forms of violent crime, robbery incidents will have a lower likelihood of case clearance, controlling for the support of RTCC technologies.

[H2] RTCC-technologies will be firmly integrated in the case processing of robbery incidents.

[H3] RTCC-assisted robbery incidents will experience longer time to case clearance rates as a result of increased investigative effort.

Analytical Approach

This study employed four distinct analytical techniques and two forms of analytical software programs to examine the influence of RTCC technologies on the case clearance and processing of robbery incidents. First, a set of nested multivariate logistic regression models were employed to determine the likelihood of case clearance for robbery compared to all other forms of violent crime incidents and controlling for RTCC support. The multivariate model was adjusted for various solvability factors or case characteristics that could potentially impact the clearance probability of cases outside the scope of the

technologies employed within the unit. Second, module network analysis was used to empirically examine the extent to which RTCC technologies influence the investigative process of these unique crime types. Finally, Kaplan Meier survival analysis and Cox regression with hazard model outputs was used to determine differences in the time to case clearance for RTCC-assisted robbery incidents compared to the control sample, holding all other case characteristics constant. To conduct the logistic regression and survival analyses, SPSS was utilized, whereas an open-source online program, JASP was used to perform modular network analysis.

Data and Measures

The analysis relied on data from the expanded RTCC incident log analytical database which contained additional measures of violent crime incident characteristics and sources of evidence used by responding officers and detectives. For a detailed overview of the data collection process, please refer back to the "Data and Measures" section in chapter four. Here, the violent crime of interest is robbery. Within the following analyses, all but two of measures were coded dichotomously, with 1 indicating it was present within the investigation of the case and 0 indicating its absence. Two measures, victim age and the number of officers responding, were coded as continuous measures. There were two dependent measures in the outcome analysis, whether the case was cleared by arrest (coded as 1 indicating an arrest was made and 0 one was not) and the number of days from the time the incident occurred to the time the arrest was made. Days to arrest was coded as a continuous measure.

Findings

Table 8 presents a comparative overview of the measures employed in the analyses to evaluate the influence of RTCC technologies on the case processing and outcomes of robbery incidents, as compared to other forms of violent crimes. Here the other forms of violent crime included within the model consist of homicides (n = 123), aggravated assaults (n = 496), misdemeanor assaults (n = 74), domestic violence (n = 90), and sexual assaults incidents (n = 56). For all continuous measures in the table, independent samples t-tests were used, equal variances not assumed, to ascertain differences between groups. In contrast, for dichotomous measures, Fisher's Exact Test was implemented. This test is used when the sample size is small, and the assumptions of the chi-squared test are not met. The full violent crime model (N=1296) will be presented first, followed by separate analysis for robbery incidents.

Full Violent Crime Model

For the two outcome measures, a significantly greater proportion of cases were cleared within the RTCC-assisted sample (N = 234, 36 percent) compared to the control sample, with only 198 cleared cases (31%). Alternatively, RTCC-assisted cases had a significantly longer time to clearance, with an average of 43 days to case clearance (SD = 112.4) compared to the control sample average of just 19 days (SD = 54.5). Among the measures used in the analysis to adjust for victim and crime investigation characteristics, there were several significant differences between the two samples. Among victim characteristics, both victim age and whether the victim was black were significantly different, although, in frequency, the differences were small. The RTCC-assisted sample had an average victim age of 36 (SD = 14.4), while the control sample was slightly older,

with an average victim age of 38 (SD = 15.8). Among the RTCC sample, roughly half of the victims were black (n = 322), while 46 percent of the control sample victims were black (n = 290).

Within the measures comprising human evidence, three of the five were significantly different between the two groups. There were statistically significant greater proportions of lineups (n = 145 vs. n = 109) and flyers (n = 213 vs. n = 63) used, and the presence of external CCTV evidence (n = 314 vs. n = 79) within RTCC-assisted cases compared to the control sample. There were no significant differences in the presence of victim and witness interviews. For crime scene-based evidence, five of the six measures were significantly different between the samples. There was significantly more firearm present (n = 337 vs. n = 218) and vehicles used (n = 377 vs. n = 210) in the commission of a crime, and accordingly, greater use of NIBIN ballistics tracing (n = 193 vs. n = 57), processing of gunshot residue (n = 80 vs. n = 31), and greater use of the CSI unit (n = 350 vs. n = 246). Lastly, there were also significantly greater numbers of search warrants (n = 51 vs. n = 15) and cell warrants (n = 39 vs. n = 12) acquired, and more officers utilized (mean = 4.06 vs. mean = 3.82) on RTCC-assisted cases compared to the control sample.

Robberies

Once isolated from the full violent crime model, a significantly greater proportion of robbery cases were cleared within the RTCC-assisted sample (n = 89, 38 percent) compared to the control sample, with only 64 cleared cases (28%). Alternatively, RTCC-assisted cases had a significantly longer time to clearance, with an average of 51 days to case clearance (SD = 126.4) compared to the control sample average of just 22 days (SD = 49.1).

Table 8. *Comparative overview of measures used in the analysis of robbery incidents*

	Full Violent Crime Model N = 1296		Robberies Only n = 466	
Measures	**RTCC-assisted cases n (%)**	**Control cases n (%)**	**RTCC-assisted cases n (%)**	**Control cases n (%)**
Cleared[2]	234 (36%)	198 (31%)*	89(38%)	64(28%)*
Time to Clearance	43 (112.4)	18(54.5%)**	51(126.4)	22(49.1)*
MRTCC technologies				
CCTV	564(87%)	--	197(85%)	--
Social media	120(19%)	--	49(21%)	--
Facial recognition	165(26%)	--	60(26%)	--
License plate reader (LPR)	82(13%)	--	21(9%)	--
ShotSpotter	125(19%)	--	27(12%)	--
Victim characteristics				
Victim age[1]	36(14.1)	38(15.8)**	35(14.5)	36(14.8)
Victim male[2]	428 (69%)	410 (65%)	157(70%)	133(59%)*
Victim black[2]	322 (52%)	290 (46%)*	91(41%)	102(45.3%)
Human evidence				
Victim interview[2]	585 (90%)	591 (91%)	206(88%)	223(96%)**
Witness interview[2]	234 (36%)	200 (31%)	92(40%)	64(28%)**
Flyer[2]	213 (33%)	63 (10%)***	84(36%)	25(11%)***
Lineup[2]	145 (22%)	109 (17%)*	40(17%)	56(24%)
External CCTV[2]	314 (49%)	79 (12%)***	35(15%)	109(47%)***
Crime scene evidence				
Firearm used[1]	337 (52%)	218 (34%)***	103(44%)	81(35%)*
Vehicle involved[2]	377 (58%)	210 (32%)***	119(51%)	84(36%)**
Ballistics tracing (NIBIN)[2]	193 (30%)	57 (9%)***	49(21%)	15(6%)***
Gunshot residue[2]	80 (12%)	31 (5%)***	25(11%)	9(4%)**
Latent prints/DNA[2]	149 (23%)	175 (27%)	51(30%)	53(23%)
Crime Scene Investigation[2]	350 (54%)	246 (38%)***	114(49%)	79(34%)**
Warrant evidence & manpower				
Search warrant[2]	51 (8%)	15 (2%)***	16(7%)	6(3%)*
Cell warrant[2]	39 (6%)	12 (2%)***	15(6%)	6(3%)
Number of officers[1]	4.06(1.9)	3.82(1.9)**	3.85(1.8)	3.73(1.4)

*$p \leq .05$, **$p \leq .01$, ***$p \leq .001$; [1] For these measures, independent samples t-tests were used to determine differences, equal variances not assumed. [2] Fisher's Exact Test was used to determine differences for these dichotomous measures.

Among the measures used in the analysis to adjust for victim and crime solvability characteristics, there were several significant differences between the two samples. Among victim characteristics, the only significant difference between the two samples consisted of victim sex. The RTCC-assisted sample had a greater proportion of male victims (n=157, 70 percent) as compared to the control sample (n = 133, 59 percent).

Four of the five measures comprising human evidence were significantly different between the two groups. There were significantly greater proportions of victim interviews (n = 223 vs. n = 206) in the control sample, as well as more presence of external CCTV evidence (n = 109 vs. n = 35). Within the sample that received RTCC support, however, there more witness interviews present (n = 92 vs. n = 64), and more suspect information flyers (n = 84 vs. n = 25) deployed. For crime scene-based evidence, five of the six measures were significantly different between the samples. There were significantly more firearms present in the RTCC sample (n = 103 vs. n = 81) and vehicles used (n = 119 vs. n = 84) in the commission of a crime, and accordingly, greater use of NIBIN ballistics tracing (n = 49 vs. n = 15), processing of gunshot residue (n = 25 vs. n = 9), and greater use of the CSI unit (n = 114 vs. n = 79). Lastly, there were also significantly greater numbers of search warrants (n = 16 vs. n = 3) utilized within RTCC-assisted cases compared to the control sample.

Case Processing

Module network analysis helped to determine the influence of RTCC technologies on the case processing of robbery incidents. As in the previous chapter, the network modules here were constructed with an EBICglasso estimation procedure with a nonparanormal transformation The analyzed data was drawn from the expanded data set

derived from the RTCC log and the stratified randomly drawn control sample as discussed in the previous "data" section, with a total sample size of 1296 included within the analysis. The data was coded to represent whether various evidential sources were used in the investigation of the violent crime incident regardless of whether usable evidence was obtained from the evidence sources.

These evidence sources were grouped into four types and included MRTCC technologies (CCTV, LPR, social media, facial recognition, and ShotSpotter), crime scene evidence (NIBIN, GSR, CSI, Latents/DNA), human-based evidence (victim interviews, witness interviews, media flyers, lineups, and external CCTV) and warrant based evidence (search warrants, cell warrants).[15] An additional crime type variable was intentionally coded to represent the investigation of robbery incidents, with the aim of examining the role of police technology in the processing of these crimes. Victim characteristics and other investigative solvability factors were not included as they were outside of the scope of the analysis.

The findings from this network indicate a few things. First, the network of RTCC-supported cases and the randomly drawn control cases had similar levels of sparsity within the network, where the RTCC-assisted sample had a sparsity measure of .449 compared to that of .333 in the control sample (see Table 9). Despite the greater number of evidential nodes present in the MRTCC network (n = 17), the level of connectedness with other evidential nodes remains about the same as in the control sample (n = 12).

[15] For an overview of the measures used in this analysis, refer back to the comparative overview of both samples presented in the previous table.

Table 9. *Weight matrix of variables included in robbery network*

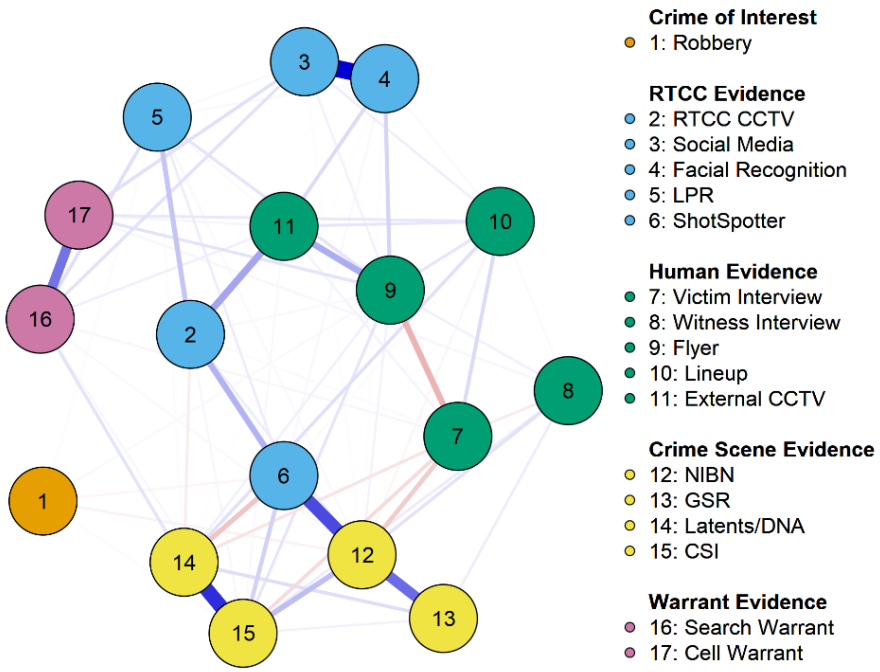
Variable **(N=1296)**	**RTCC Sample (Weight)**	**Control Sample (Weight)**
Crime of Interest		
Robbery	--	--
RTCC Evidence		
CCTV	.000	--
Social media	.000	--
Facial recognition	.000	--
LPR	-.009*	--
ShotSpotter	-.024*	--
Human evidence		
Victim interview	.000	.000
Witness interview	.000	.000
Flyer	.015*	.026*
Lineup	.000	-.006*
External CCTV	.000	.000
Crime scene evidence		
NIBIN	-.034*	-.059*
Gunshot residue	.000	.000
Latent prints/DNA	.000	.000
CSI	-.010*	-.017*
Warrant Evidence		
Search warrant	.000	.000
Cell warrant	.000	-.017
Number of Nodes	17	12
Non-zero edges	75/136	44/60
Sparsity	0.449	0.333

* Represents an existent connection to the crime of interest node.

Notably, these sparsity measures are not an improvement over the homicide and aggravated assault model (.441 and .303, respectively), however they still imply that, overall, the integration of RTCC technologies into case processing has been proportionate to other evidential nodes and again this contradicts previous research that suggest police agencies are resistant to incorporating innovative technologies into traditional practices and structures (Ioimo & Aronson, 2004; Manning, 2003; Manning, 2008). Instead, it shows

that robbery units, which tend to have a heavier case load than homicide units (Cook et al., 2019) have also become acquainted with RTCC technologies as a mechanism for improving case processing efficiency.

Figure 7. *RTCC-assisted module network analysis for robbery incidents*

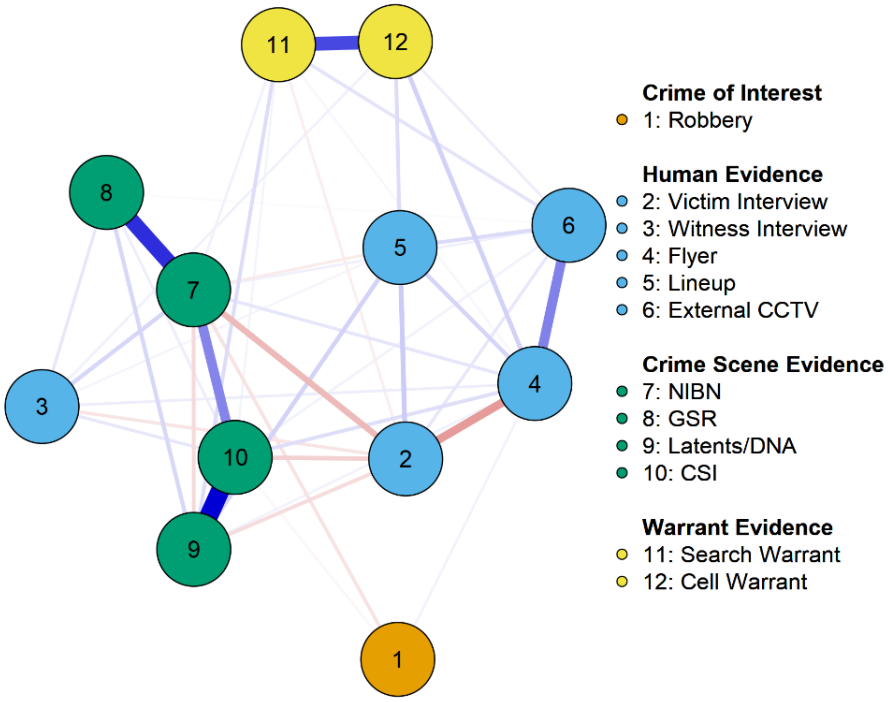


Further, as evidenced in Figure 7. it is apparent that RTCC technologies fulfill a comparable function to conventional evidential sources in the context of robbery investigations. This observation is corroborated by the fact that each of the RTCC evidence nodes depicted in Figure 7 demonstrates a significant number of associations with other customary sources of evidence throughout the network as a whole. Two of the RTCC evidence sources, however, appear to have greater influence on the investigation of the

particular crime of interest, as compared to other violent crime incidents. According to the weight matrix (Table 9), the gunshot detection system, ShotSpotter (influence of .049) and the LPR (.028) appear to have stronger connections to the node representing robbery incidents. Here, the more traditional nodes that work alongside the RTCC technologies and influence the processing of robbery incidents include the following: suspect information flyers (.015), CSI (-.010) and NIBIN (-.034). These findings are perhaps as expected, as approximately 40% of the incidents within the isolated robbery sample either utilized a firearm or had a firearm present during the commission of the crime. Moreover, the increased reliance on suspect information flyers in this model may be due to the tendency of robberies to have more investigative information than other crime types.

Robbery incidents often involve face-to-face interactions between the perpetrator and the victim, which can provide more potential evidence for investigators to work with. For example, eyewitnesses may be able to provide descriptions of the perpetrator, the vehicle they used, or the direction they fled after the crime, all of which could subsequently involve images or sketches to be produced. The negative direction of the nodes does not represent a negative relationship as typically reported and observed within a regression model, but rather it depicts that 1) their connection to the node relied on the presence of other connections, and 2) the presence or absence of these measures would not affect the strength of the other connections.

Figure 8. *Non-RTCC-assisted module network analysis for robbery incidents*



Within the control sample, RTCC technologies were removed from the network plot to determine which sources outside the scope of the RTCC maintained strength within the investigative process regardless of the integration of the unit (see Figure 8). Here, we see that human and crime scene evidence sources played the greatest role in the investigation. As seen in the weight matrix (Table 9) the dissemination of a public information flyer (.026) had the greatest utility within the investigation of robberies. Three other nodes had a notable connection with the crime of interest: that is, NIBIN (-.059), suspect lineup (-.006), and CSI (-.017). This indicates that these measures' association with the node was reliant on the presence of other connections, and their absence would not weaken the other connections' strength.

Case Clearance

Table 10 presents the analysis results aimed at examining whether RTCC technologies influenced the case clearance of robbery incidents. Although Table 8. shows that RTCC-assisted robbery incidents were cleared more frequently than the control sample, these numbers alone do not consider the differences in the case processing of these crimes. Therefore, to accurately determine the impact of RTCC technologies on clearance rates for these incidents, the multivariate logistic regression analysis controlled for such differences.

Here, missing data was identified in 52 total cases and deleted using the listwise deletion method. Specifically, 3.5% (n = 46) of all cases had missing victim gender information and 3.9% (n = 50) had missing victim race information. Moreover, 2 cases or .2% of the sample had missing data regarding whether the CSI unit was deployed to the scene of a crime. As for victim age, fifty-five incidents were missing data, but this information was transformed and replaced with the series mean.

The Chi-square and -2 log-likelihood coefficients were statistically significant in all models, with the exception of *Model One*, indicating good model fit. Within the significant models Nagelkerke pseudo R-square indicated an increasing ability of the models to predict the model variance ranging from a low of .7% in *Model Two* to a high of twenty-four percent in *Model Six*. All independent measures used in these models were examined for multicollinearity using the variance inflation factor (VIF) with no

multicollinearity revealed. All VIF factors ranged from a low of 1.032 to the highest at 2.062.[16]

[16] These scores are far below the area of concern for VIF where convention dictates that scores of 5 or greater suggest the presence of multi-collinearity among independent measures.

Table 10. *Nested multivariate logistic regression of predictors of robbery case clearance and influence of RTCC technology*

***Measures (*N=1244)**	***Model One*** **B (Odds ratio)**	***Model Two*** **B (Odds ratio)**	***Model Three*** **B (Odds ratio)**	***Model Four*** **B (Odds ratio)**	***Model Five*** **B (Odds ratio)**	***Model Six*** **B (Odds ratio)**
Crime of Interest						
Robbery	-.050(.951)	-.051(.950)	-.072(.931)	-.127(.881)	-.177(.837)	-.180(.835)
Impact of Technology						
RTCC-Assisted Violent Crime		.306(1.359)*	.324(1.383)**	.384(1.468)**	.510(1.666)**	.497(1.644)***
Victim characteristics						
Victim age			-.002(.998)	.002(1.002)	-.001(.999)	-.001(.999)
Victim male			-.422(.656)***	-.411(.663)**	-.300(.741)*	-.309(.734)*
Victim black			-.017(.983)	.027(1.028)	.107(1.113)	.122(1.130)
Human evidence						
Victim interview				.536(1.709)*	.360(1,433)	.384(1.468)
Witness interview				.455(1.577)***	.479(1.615)***	.455(1.576)**
Flyer				.616(1.852)***	.713(2.040)***	.656(1.927)***
Lineup				1.678(5.352)***	1.715(5.555)***	1.692(5.429)***
External CCTV				-.864(.421)***	-.813(.444)***	-.907(.404)***
Crime scene evidence						
Firearm used					-.654(.520)***	-.675(.509)***
Vehicle used					-.297(.743)*	-.375(.688)*
Ballistics tracing (NIBIN)					.128(1.137)	.120(1.127)
Gunshot residue (GSR)					.114(1.121)	.060(1.062)
Latent prints/DNA					-.015(.985)	-.118(.888)
Crime Scene Investigation					-.160(.852)	-.178(.837)
Warrant evidence & manpower						
Search warrant						.916(2.498)**
Cell warrant						.852(2.345)*
Number of officers						.046(1.047)
Model Chi-Square	.158	6.484*	18.117**	180.313***	210.569***	231.545***
-2 Log likelihood	1586.704	1580.221	1568.745	1406.550	1376.294	1355.317
Nagelkerke Pseudo R-square	.000	.007	.020	.187	.216	.236
Constant			-.695(.504)***			

* $p \leq .05$, **$p \leq .01$, ***$p \leq .001$

Model One only included robbery. Here, we see that without the presence of any other covariates, the odds of these crimes being cleared, as compared to all other crimes included in the model, were not significant. In *Model Two*, all violent crimes that received RTCC-support, were included within the model to determine the influence of technology on case clearance. This variable established significance within the model, indicating approximately 36% increased odds of case clearance (≤ .05). Here, we see that robbery incidents improve slightly in likelihood of case clearance, but do not reach levels of significance.

Robbery incidents remained insignificant in *Model Three,* in which victim characteristics were subsequently introduced. Of all covariates included within the model, RTCC-assisted violent crimes and whether the victim was male also proved to be significantly related to whether a case was cleared. Here, violent crime cases receiving RTCC technology support being 38% more likely to be cleared than those without RTCC support and male victims resulted in 34% lower odds of case clearance. Human evidence sources were added to *Model Four*. Within this model, violent crime RTCC-assisted cases maintained significance (47% more likely to be cleared), as did whether the victim was a male, which still resulted in roughly 34% lower odds of case clearance. All human evidence sources proved to be of statistical significance, with suspect lineups being the most impactful. In fact, when a lineup was present, the odds of the case being cleared were five times greater than when there was no lineup, while controlling for other factors ($p = \leq .001$). Here, external, or private CCTV footage resulted in 58% decreased odds of case clearance.

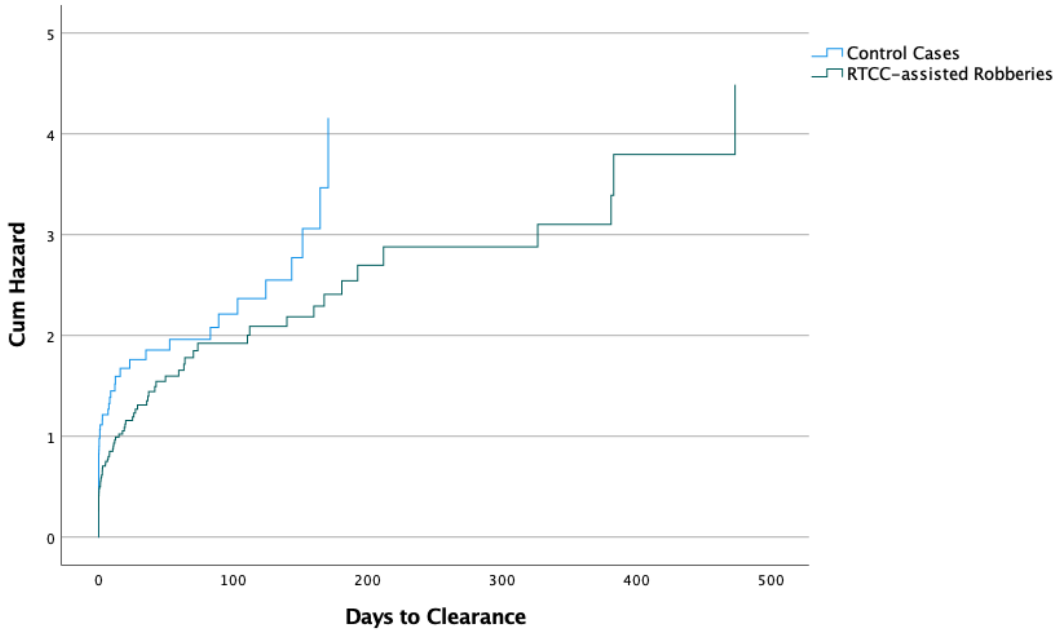
Model Five incorporated evidence from the crime scene, revealing that incidents of robbery remained insignificant. Further, cases involving RTCC-assistance had a 47% greater likelihood of being cleared ($p = \leq .05$). Of the victim characteristics, a male victim resulted in 26% lower odds of case clearance ($p = \leq .05$). Among the five human evidence elements, only four remained significant, with suspect lineups having the greatest impact on case clearance (OR=5.555; $p = \leq .001$). External CCTV, once again, depicted 56% lesser odds of case clearance (OR=.444; $p = \leq .001$). As for crime scene variables, the use of a firearm and vehicle in the commission of the crime and was significant, revealing 48% and 26% lower odds of case clearance, respectively.

In *Model Six*, all variables, including those related to warrant evidence and manpower, were included in the final analysis. Here, the crimes of interest remained insignificant. A male victim sustained statistical significance, revealing 27% lower odds of case clearance ($p = \leq .05$). Among all the variables, the support of RTCC technology, witness interviews, suspect lineups, public information flyers, and search warrants significantly increased the odds of case clearance. RTCC-assisted violent crime cases had 64% better odds of case clearance, and the use of a suspect lineup remained the strongest indicator in the model (OR = 5.429; $p = \leq .001$). External CCTV presence (OR = .404; $p = \leq .001$), the use of a firearm (OR = .509; $p = \leq .001$), and the use of a vehicle (OR = .509; $p = \leq .001$) significantly decreased the likelihood of a case being cleared, controlling for all other solvability factors. Within the warrant and manpower group, both search and cellphone warrants significantly increased the odds of case clearance two-fold (OR = 2.237) and (OR = 2.534).

Time to Case Clearance

The analyses to determine whether RTCC-assisted cases improved the time that it took to clear cases are reported in Figures 9 and 10, and Table 11. The simple frequency of days to case clearance reported in Table 8 suggests that RTCC assistance did not improve the speed at which arrests were made in robbery incidents. However, this does not account for the various victim characteristics and case processing characteristics that could influence the speed at which cases are cleared which are outside the scope of RTCC influence.

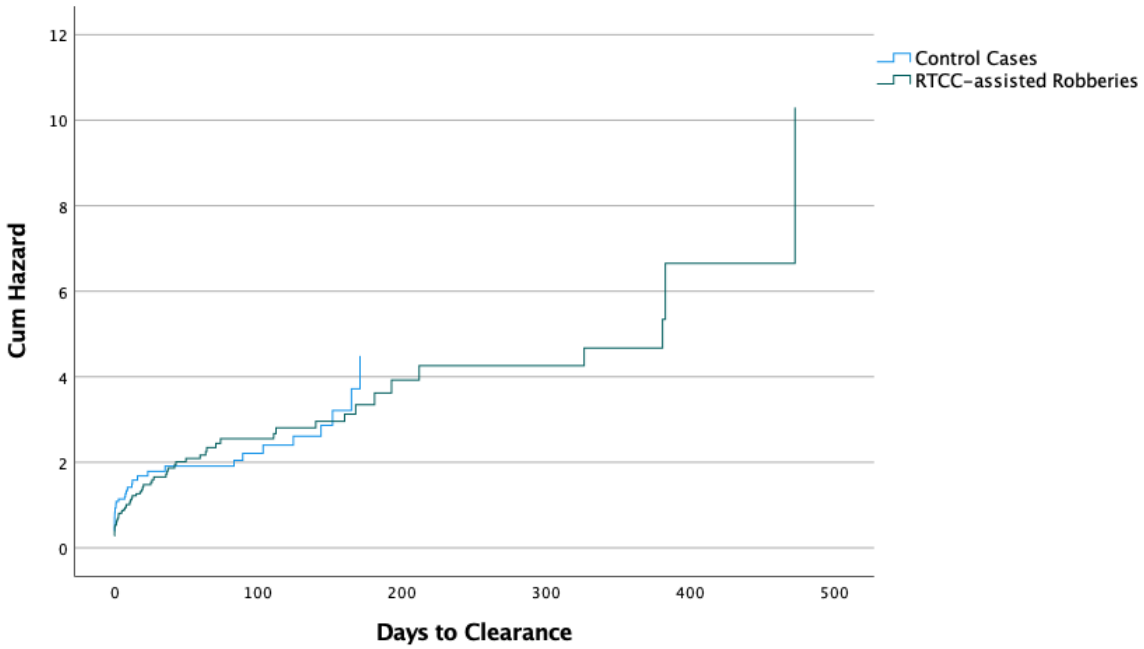
Figure 9. *Cumulative rate of case clearance RTCC-assisted robberies vs. control sample*



It may be possible that when accounting for fundamental differences that exist within different crime cases, the difference in the time that it takes to clear cases is adjusted. To determine this, first, a Kaplan-Meier survival analysis was performed to identify the trajectories in the time- to-case clearance within the two samples. Here, only the treatment

and control groups for aggravated assaults and homicides were included within the analysis. The total number of cleared cases between the two samples was 153, of which 89 were cleared in the RTCC-assisted sample, and 63 in the control. The outcome of that analysis is reported in Figure 10. This analysis does not adjust for case characteristics and only plots the proportion of cases cleared within each sample across the period of days present within the observation period. Here we see a clear difference in the trajectory of RTCC-assisted cases and the control sample, with control sample cases being cleared at a significant faster rate than the sample assisted by RTCC technologies. The Mantel-Cox Log Rank comparison further indicates that these trends were significantly different ($\chi 2 = 6.283$, 1 df, $p = .012$).

Figure 10. *Cumulative rate of case clearance for RTCC-assisted robberies and control sample adjusted at mean for case characteristics.*



To account for differences in victim and case processing characteristics between the two samples a Cox Proportional Hazard regression model was fitted. The trend line

comparison of the model is presented in Figure 10 and the measures included in that model and coefficients are reported in Table 11. Each of the victim characteristics, human evidence, crime scene evidence, warrants, and manpower measures were included to account for their influence on the time taken to clear cases. As visible in Figure 10, after controlling for case characteristics the trend lines indicating the occurrence of case clearance over time overlap and largely exhibit the same trajectory as the previous model.

With this adjustment, there are no apparent differences between the RTCC-assisted cases and the control sample in the time-to-case clearance trajectories. This was also evident in the coefficients (see Table 11). Holding constant the influence of all other measures there was no significant difference in the days to case clearance between the RTCC-assisted robbery cases and the control sample (B = -.176, SE = .192, p = .360, n.s.). The findings here suggest that the implementation of RTCC information technologies in the investigation of robbery cases has not reduced the amount of time that it takes to clear cases, but once case level circumstances are taken into consideration, the use of RTCC technologies does not prolong the time that it takes to clear these cases either.

Among the remaining measures in the analysis, two were significantly related to the time that it took to clear cases. For victim characteristics, the only significant predictor was whether the victim was male. When the victim was male the case had significantly greater odds (OR = 1.931) of being cleared before female victims. Among human evidence measures, whether a public information flyer was used was associated with 41 percent (OR = .593) lower odds of being cleared before those cases without a flyer, and the same finding held true for the use of a suspect lineup in cases (OR = .639) which resulted in longer

clearance rates. The model Chi-Square coefficient was statistically significant indicating a good model fit ($\chi^2 = 44.445$, 18 df, $p = \leq .001$).

Table 11. *Cox proportional hazard model coefficients predicting days to clearance (robberies)*

Measures (N = 153)	**B (SE)**	**Odds Ratio**	***p* - value**
Crime of Interest			
RTCC-assisted Robberies	-.176(.192)	.839	.360
Victim Characteristics			
Victim age	-.002(.007)	.998	.808
Victim male**	.658(.239)	1.931	.006
Victim black	-.320(.202)	.726	.113
Human Evidence			
Victim interview	.655(.475)	1.924	.168
Witness interview	.046(.197)	1.047	.817
Lineup*	-.448(.228)	.639	.049
Flyer*	-.523(.243)	.593	.031
External CCTV	-.159(.233)	.853	.494
Crime Scene Evidence			
Firearm used*	-.412(.261)	.662	.114
Vehicle used	-.382(.215)	.682	.076
Ballistics tracing (NIBIN)	.180(.389)	1.197	.644
Gunshot residue (GSR)	.323(.452)	1.382	.475
Latent prints/DNA	.474(.303)	1.606	.117
Crime Scene Investigation (CSI)	-.182(.252)	.834	.471
Warrant Evidence & Manpower			
Search warrant	.450	1.568	.306
Cell warrant*	-.768	.464	.059
Number of officers	-.031	.970	.660
Model Chi-Square		44.445***	
-2 Log likelihood		1151.630	

*p ≤ .05, **p ≤ .01, ***p ≤ .001

Supplemental Analysis

A supplemental analysis was conducted to further augment the results of the former nested regression model (Table 12). The research question of interest was: "Do RTCC-assisted robbery crime incidents have greater rates of case clearance compared to homicide

and aggravated assault cases that did not receive RTCC support?" Here, only RTCC-assisted robbery incidents were used in the model and compared to those that did not receive RTCC-support. Within this model, missing value analysis resulted in the following missing variables: victim sex 18 incidents (3.9%); victim race 18 incidents (3.9%), and CSI 2 incidents (.2%). Data was deleted using the listwise deletion method, resulting in 448 total incidents remaining in the analysis, or 18 removed incidents. As for victim age, 19 incidents were missing data, but this information was transformed and replaced with the series mean. The mean average is presented in Table 8.

> **[H4]** RTCC-assisted robbery crime incidents will have greater rates of case clearance as compared to the control cases that did not receive RTCC support.

The Chi-square and -2 log likelihood coefficients suggested all five models were statistically significant, indicating good model fit, and Nagelkerke pseudo R-square indicated an increasing ability of the models to predict the model variance ranging from a low of two percent in *Model One* to a high of twenty-four percent in *Model Five*. All independent measures used in these models were examined for multicollinearity using the variance inflation factor (VIF) with no multicollinearity revealed. All VIF factors ranged from a low of 1.098 to the highest at 1.965.

In contrast to the supplemental analysis conducted in the previous chapter, here we see that the influence of technology significantly improves the case clearance of robbery incidents. As reported in the first row of Table 3, the odds ratio for RTCC-assisted robbery cases exhibited increasingly greater clearance odds after adjusting for each component of case characteristics across the four models.

Table 12. *Multivariate logistic regression of predictors of case clearance among RTCC-assisted robberies and control sample*

***Measures (*N=448)**	***Model One*** **B (Odds ratio)**	***Model Two*** **B (Odds ratio)**	***Model Three*** **B (Odds ratio)**	***Model Four*** **B (Odds ratio)**	***Model Five*** **B (Odds ratio)**
Crime of Interest					
RTCC-Assisted Robberies	.565(1.759)**	.571(1.769)**	.540(1.715)*	.630(1.878)**	.625(1.869)*
Victim characteristics					
Victim age		-.009(.991)	-.006(.994)	-.011(.989)	-.011(.989)
Victim male		-.150(.861)	-.109(.896)	.11(1.122)	.178(1.195)
Victim black		-.092(.913)	-.093(.911)	.097(1.102)	.115(1.222)
Human evidence					
Victim interview			.509(1.663)	.204(1.226)	.378(1.459)
Witness interview			.167(1.182)	.204(1.227)	.200(1.221)
Flyer			.627(1.873)*	.729(2.073)*	.701(2.016)*
Lineup			1.344(3.833)***	1.380(3.973)***	1.504(4.499)***
External CCTV			-.642(.526)*	-.538(.584)	-.735(.480)*
Case Characteristics					
Firearm used				-.633(.531)*	-.629(.533)*
Vehicle used				-.368(.692)	-.509(.601)
Ballistics tracing (NIBIN)				-.369(.692)	-.320(.726)
Gunshot residue (GSR)				-.005(.995)	.089(1.093)
Latent prints/DNA				-.123(.884)	-.243(.785)
Crime Scene Investigation				-.065(.937)	-.130(.878)
Warrant evidence & manpower					
Search warrant					1.785(5.957)**
Cell warrant					.898(2.456)
Number of officers					-.049(.952)
Model Chi-Square	7.777**	10.116*	51.496***	68.230***	85.989***
-2 Log likelihood	559.250	556.912	515.531	498.797	481.038
Nagelkerke Pseudo R-square	.024	.031	.151	.197	.243
Constant			-.717(.488)***		

*p ≤ .05, **p ≤ .01, ***p ≤ .001

In *Model One*, RTCC-assisted cases were associated with 76 percent greater odds (OR = 1.759) of being cleared compared to the control cases . In *Model Two*, while holding victim characteristics constant, those odds improved. RTCC-assisted robbery cases exhibited 77 percent greater odds (OR = 1.482) of being cleared compared to control cases while accounting for victim age, gender, and race.

In *Model Three* the odds of case clearance decreased slightly with RTCC-assisted cases exhibiting 72 percent greater odds (OR = 1.715) of being cleared compared to control cases while accounting for the presence of the human-based forms of evidence. Here, the dissemination of a public information flyer was also associated with improved likelihood of case clearance (OR = 1.873) as was the use of a suspect lineup, which resulted in case clearance to be 3 times more likely. External CCTV was associated with a decreased likelihood of case clearance, with cases that involved the extraction of private CCTV footage resulting in 47% lesser odds of being cleared.

In *Model Four* crime scene evidence was accounted for and again we see significantly greater odds of RTCC robbery cases being cleared compared control cases with an improvement of 88 percent (OR = 1.878). The use of public information flyers (2.073) and suspect lineups (OR = 3.973) remained significant. Within this model, the presence of external CCTV and a firearm during the commission of a crime was associated with 42% and 47% lower odds of clearance. Finally, in *Model Five*, after accounting for the influence of warrants and the number of officers used in the case, RTCC-assisted cases continued to exhibit 87 percent greater odds (OR = 1.869) of being cleared compared to cases that did not receive RTCC support. Additionally, where dissemination of a public information flyers (OR=2.016) and use of suspect lineup (OR = 4.499) and search warrants

(OR = 5.957) resulted in increased odds of case clearance, the presence of external/private CCTV footage and firearm resulted in decreased odds (47% and 52%, respectively).

Chapter Six: The Influence of RTCC Technologies on the Case Outcomes of Property Crimes

Introduction

The primary objective of this chapter is to examine the impact of police technologies on outcome measures of cases that involve burglaries and larceny thefts. Specifically, the following study aims to assess whether the use of RTCC technologies influence the case clearance of these cases as compared to violent crimes, what role they play in the investigative processing of these cases, and whether the time to clearance of cases that receive RTCC support are cleared at a faster rate. The research questions guiding this study are therefore three-fold: (1) "What is the influence of RTCC technologies on the clearance of burglary and larceny theft incidents compared to all other violent crimes?" (2) "To what extent have RTCC technologies been integrated into the case processing of this particular crime type?" and (3) "Does RTCC support improve the time that it takes to clear these cases as compared to violent crime RTCC-support cases?"

This chapter will follow a similar outline to the preceding chapter. It will commence by presenting a literature review pertaining to the examination of burglary and larceny theft and the utilization of technology in their investigation. A subsequent section will detail the analytical approach taken and provide an overview of the data collection process and the measures involved within the analysis. A discussion of the findings, limitations and policy implications for this and all subsequent studies will be located in chapter 8.

Review of Literature

There are two pertinent bodies of research related to an examination of property crime case clearance as it concerns this assessment. The first relates to the factors found to

influence the solvability of burglary and larceny incidents, and the second pertains to the role of technology in the investigation of these crimes. Following a review of each, a brief overview of evaluations associated with real-time crime centers will be presented to further orient the study.

Factors Known to Impact Case Clearance

As per the latest data published by the Federal Bureau of Investigation (FBI), the occurrence of property crimes in the United States outweighs that of violent crimes. The agency's report for the year 2019 reveals an alarming rate of 2,110 incidents of property crimes per 100,000 individuals, while the cases of violent crimes were relatively lower, standing at 379 per 100,000 people (Gramlich, 2020). The most prevalent form of property crime in 2019 was larceny or theft, which constituted a significant portion of all property crimes. Furthermore, the occurrence of burglary and motor vehicle theft was also notably high. These figures illustrate that Americans are more vulnerable to property crimes compared to the incidence of violent offenses. With that being said, it is important to acknowledge that these crimes have varying degrees of severity and can have diverse impacts on the victims, such as financial loss, emotional trauma, and property damage (Lockwood, 2014).

Regarding the clearance of property crimes, they remain remarkably low in comparison to all violent crime types. According to Li and Lartey (2022), the national clearance rates for thefts were slightly higher than those for burglary in 2020, at 15% and 14%, respectively. It can be argued that clearance rates for property crimes may be lower than those for violent crimes due to a complex interplay of various factors. For instance, some scholars, including Friedmanl, Grawert, and Cullen (2016), have pointed to case-

level characteristics that may hinder improved clearance rates, such as the availability of forensic evidence and witnesses. Therefore, property crimes tend to be less visible and more challenging to detect and investigate, leading to lower clearance rates. Conversely, violent crimes such as murder or assault may receive more attention and resources from law enforcement agencies, resulting in higher clearance rates due to their overtly violent and easily recognizable nature.

As mentioned earlier, the scarcity of research in the field of property crimes can be traced back to challenges associated with data collection, such as underreporting and inconsistencies in definitions and police practices, which were identified by Avdija and Akgul (2021), Merry and Harsent (2018), and Steffensmeier, Harris, and Painter-Davis (2015). For example, it can be difficult to differentiate between various types of burglaries, such as residential, commercial, non-occupied dwelling, and occupied dwelling burglaries, since they may be classified under the same police signal in a database. Moreover, the distinction between theft and burglary is often a fine line, making it challenging to determine which category an offense falls under. These issues can significantly impact the accuracy of data collection, resulting in limited knowledge and understanding of property crimes.

Society also tends to allocate more resources to studying violent crimes, particularly homicides, as they are often perceived as more severe (Braga and Dusseault, 2018; Cook et al., 2019). This prioritization has resulted in relatively few studies being conducted on property crimes. However, it is important to note that property crimes have a significant economic and social impact, as they can cause financial loss, damage property, and harm individuals' sense of security. Therefore, there is a pressing need for further

research in this field to better understand the underlying factors contributing to property crimes and develop effective prevention strategies.

Recently, scholars have worked diligently to bolster this area of research. Notably, Roth (2017) has contributed to this pursuit by conducting a study that aimed to augment the limited literature available on the subject. The research analyzed a sample of cities drawn from the 110 largest cities in the United States. Roth's approach mirrors that of other scholars, such as Eitle, Stolzenberg, and D'Alessio (2007), Jang et al. (2008), and Roberts (2008). The study revealed significant variations in property crime clearance rates across the cities examined, with some cities having higher clearance rates than others. The city's size, population density, and the police department's budget were identified as critical factors influencing clearance rates. Workload also emerged as a particularly influential predictor of burglary and theft clearance rates, in which a higher workload resulted in less cleared incidents. With regard to demographic and social disorganization variables, only the non-white population was found to be significant for all three offenses in the full models. This finding contrasts with that of Jang et al. (2008), who found a positive association between the proportion of black residents and clearance rates for theft and vehicle theft, but not for burglary.

In a later study, Roth (2018) took a similar approach and explored factors that influence property crime clearance across smaller jurisdictions. The study used multilevel modeling and data from the Uniform Crime Reporting Program to investigate predictors of clearance rates for burglary, larceny, and vehicle theft in agencies serving populations of 50,000 individuals or fewer. The results of the study indicated that police resources, community cohesion, and socioeconomic status all play a significant role in property crime

clearance rates. Specifically, increased police resources, higher community cohesion, and higher socioeconomic status were associated with higher clearance rates. Surprisingly, the study found that case workload was positively associated with larceny clearance rates. Moreover, the only variable that influenced the clearance of burglaries was whether the suspect was a juvenile, which made it more challenging for investigators to solve the case.

Lockwood's (2014) investigation into the determinants of crime clearance focused specifically on burglary incidents, and yielded support for a range of perspectives, including non-discretionary, discretionary, and broken windows approaches. The author estimated the effects of predictors at multiple levels of aggregation by analyzing burglary incident data and measures of crime and community context aggregated to Philadelphia block groups. The results demonstrated that non-discretionary factors, such as the day and time of the crime, discretionary factors, such as the race and gender of the complainant, and broken windows enforcement, specifically low-serious crime enforcement, were significantly associated with increased clearance rates. Specifically, the analysis showed that burglaries reported earlier in the week, at night, with white and male complainants, and in areas with higher levels of broken windows police enforcement are more likely to be cleared by arrest.

With respect to gender, Roth and King (2019) conducted an investigation into the determinants of larceny and burglary arrest rates for both males and females, building upon prior research on the relationship between gender and property offending (Steffensmeier & Haynie, 2000; Steffensmeier & Streifel, 1992). Using a macro-level approach and gender-disaggregated predictors, the results indicated that male and female arrest rates

were comparably influenced by factors associated with disadvantage, such as unemployment, but these factors had a stronger influence on female arrest rates.

Although underutilized in the case of property crimes, forensic DNA analysis has been acknowledged as a valuable tool in the detection and resolution of burglary cases. The process of residential burglary typically follows a defined sequence of steps, beginning with the formation of a motive, followed by target selection, entry into the premises, a comprehensive search of the property, procurement of desired items, exit from the location, and disposal of any unwanted items (Swanson, Chamelin, and Territo, 2002; Wright and Decker, 1994). At each stage, DNA evidence can potentially be uncovered, such as at the point of entry, the tools or other items left at the scene, and unclaimed but handled property.

According to Roman et al. (2008), the processing of DNA evidence in property crime cases results in over twice the number of identified suspects, twice the number of arrested suspects, and more than double the number of accepted prosecution cases compared to conventional investigations. Paine and Ariel (2013) have also found that footprints, fingerprints, and DNA were more prevalent in resolved than unresolved cases, which aligns with the findings of other researchers (Coupe and Griffiths, 2000; Bond, 2007), but not others (Burrows et al., 2005).

Moreover, it has been unveiled that crimes associated with the prompt apprehension of suspects and the collection of suspect data are more likely to be classified as cleared, particularly concerning non-residential burglary, in contrast to ongoing offenses (Coupe and Kaur, 2005). Furthermore, Pelfrey and Hanna's (2011) research indicates that the likelihood of resolving business burglary cases is positively correlated with the degree of knowledge gathered about a suspect, with the identification of a name being particularly

advantageous. Various other factors have also been identified as contributing to the likelihood of solving such offenses, including the occurrence of a witnessed offense (Donnellan, 2011; Paine, 2012), the commission of the offense during daylight hours (Coupe and Girling, 2001; Coupe and Blake, 2006), and the availability of investigative resources (Coupe and Griffiths, 2000).

Impact of Technology on Property Crime Case Clearance

As previously discussed in the earlier chapters, there has been an abundance of scholarly research examining the correlation between closed-circuit television (CCTV) systems and the efficacy of crime reduction or resolution. However, these studies have adopted a comprehensive approach by exploring a broad range of criminal activities, rather than limiting their scope exclusively to property-related offenses. For instance, Gerell's (2021) investigation of CCTV deployment in impoverished neighborhoods during a brief period presents compelling findings. The study revealed that the installation of CCTV had a statistically significant impact on reducing violent crime rates in the short term but failed to reduce property crimes, and there was no substantial impact on crime clearance rates for property crimes. This study emphasizes the relevance of contextual factors, such as camera placement, community participation, and camera quality, in determining the effectiveness of CCTV as a crime prevention tool in impoverished neighborhoods.

Coupe and Kaur's (2005) research on the efficacy of alarms and closed-circuit television in detecting non-residential burglaries demonstrated a significant increase in clearance rates for these types of crimes when they occurred within the field of vision of a CCTV-monitored location. Additionally, Morgan and Dowling's (2019) investigation into the impact of CCTV footage on clearance rates for crimes within the South Wales rail

network revealed a noteworthy rise of 7% and 8% in clearance rates for theft, burglary, and property damage, respectively, when investigators obtained CCTV footage. Similarly, Jung and Wheeler's (2023) assessment of CCTV installations and clearance of all reported crimes within the jurisdiction of the Dallas Police Department revealed that the deployment of CCTV resulted in a 2% increase in crime clearance, although only in cases of theft.

The integration of license plate reader technologies with CCTV has also gained significant traction in recent times to bolster crime-fighting efforts, as noted by La Vigne et al. (2011). Nonetheless, only one study, which is extensively detailed in chapter five, has comprehensively evaluated the impact of LPRs on property crime. In this regard, Koper and Lum (2019) investigated the impact of widespread LPR deployment on criminal investigations and determined that it enhanced the clearance rates of robbery cases and vehicular component-related thefts. However, further research is imperative to fully comprehend the utility of LPR in property crime investigations.

Impact of Real-Time Crime Centers

To date, only one assessment has been conducted on real-time crime centers concerning property crime incidents. This particular evaluation did not examine crime clearance rates but was exclusively focused on identifying crime reduction effects. The evaluation of Chicago's Strategic Decision Support Centers found a statistically significant reduction in burglary incidents across all districts and models evaluated, while significant reduction in larceny thefts was observed only in a few districts and models (Hollywood et al., 2019).

Although the data for this dissertation was collected as part of the MRTCC evaluation, the purpose of the initial valuation was to assess the impact of the investigative

unit on the case clearance of violent crime, whereas property crimes did not receive attention (Guerette et al., 2022). Nevertheless, it is important to acknowledge that the technologies implemented within of RTCCs resulted in a statistically improved clearance rate for violent crimes as a whole.

Current Study

This research represents an attempt to investigate the influence of real-time crime on property crime case processing, clearance, and time to clearance. In the context of this study, property crimes are defined as incidents of burglary and larceny theft. Given the comparatively limited sample size, burglary incidents encompass instances of residential, occupied, and commercial burglaries. Larceny thefts exclude theft of motor vehicles but may incorporate theft of vehicle components. It is noteworthy that property crimes are prevalent offenses in the United States, but they are also notoriously difficult to solve. The formulated hypotheses in this study draw mainly on the two prior national evaluations of RTCCs. They are also informed by my experience as an embedded criminologist within the RTCC for several years, which has generated valuable anecdotal evidence that underpins the assumptions of this investigation.

[H1] Compared to all other forms of violent crime, property crime incidents will have a higher probability of case clearance, controlling for the support of RTCC technologies.

[H2] RTCC-technologies will be integrated into the case processing of property crimes as compared to other violent crimes.

[H3] RTCC-assisted property crime incidents will experience longer time to case clearance rates as a result of increased investigative effort.

Analytical Approach

This study employed four distinct analytical techniques and two forms of analytical software programs to examine the influence of RTCC technologies on the case clearance of property crimes. First, a set of nested multivariate logistic regression models were employed to determine the likelihood of case clearance for property crimes compared to all violent crime incidents and controlling for RTCC support. The multivariate model was adjusted for various solvability factors or case characteristics that could potentially impact the clearance probability of cases outside the scope of the technologies employed within the RTCC unit. Second, module network analysis was used to empirically examine the extent to which RTCC technologies influence the investigative process of this understudied category of crime.

Kaplan Meier survival analysis was used to determine differences in the time to case clearance for RTCC-assisted property crime cases compared to the control sample and finally, due to the small sample size associated with property crimes, Cox regression with hazard model outputs was used to determine differences in the time to case clearance for RTCC-assisted property crime cases compared all other forms of crime assisted by RTCC technologies. To conduct the logistic regression and survival analyses, SPSS was utilized, whereas an open-source online program, JASP was used to perform modular network analysis.

Data and Measures

The analysis relied on data from the expanded RTCC incident log analytical database which contained additional measures of violent crime incident characteristics and

sources of evidence used by responding officers and detectives. For a detailed overview of the data collection process, please refer back to the "Data and Measures" section in chapter four. Here, property crimes are of interest (burglaries and larcenies). Within the following analyses, all but two of measures were coded dichotomously, with 1 indicating it was present within the investigation of the case and 0 indicating its absence. Two measures, victim age and the number of officers responding, were coded as continuous measures. There were two dependent measures in the outcome analysis, whether the case was cleared by arrest (coded as 1 indicating an arrest was made and 0 one was not) and the number of days from the time the incident occurred to the time the arrest was made. Days to arrest was coded as a continuous measure.

Findings

Table 13 presents a comparative overview of the measures employed in the analyses to evaluate the influence of RTCC technologies on the case processing and outcomes of property crime incidents, as compared to other forms of violent crimes.[17] Here the other forms of violent crime included within the model consist of homicides (n = 123), aggravated assaults (n = 436), misdemeanor assaults (n = 74), domestic violence (n = 90), and sexual assaults incidents (n = 56). For all continuous measures in the table, independent samples t-tests were used, equal variances not assumed, to ascertain differences between groups. In contrast, for dichotomous measures, Fisher's Exact Test was implemented. This test is used when the sample size is small, and the assumptions of the chi-squared test are

[17] Property crimes were compared to violent crimes to provide broader context. Further, the sample size for burglary and larceny incidents was too small to conduct a regression model with all relevant independent and control variables.

not met. The complete model of all crimes (N = 1534) will be presented first, followed by separate analyses for robbery incidents.

Full Crime Model

For the two outcome measures, a significantly greater proportion of cases were cleared within the RTCC-assisted sample (n = 261, 34 percent) compared to the control sample, with only 210 cleared cases (27%). Alternatively, RTCC-assisted cases had a significantly longer time to clearance, with an average of 45 days to case clearance (SD = 110.6) compared to the control sample average of just 20 days (SD = 58.8). Among the measures used in the analysis to adjust for victim and crime investigation characteristics, there were several significant differences between the two samples. Among victim characteristics, both victim age and whether the victim was male were significantly different, although, in frequency, the differences were small. The RTCC-assisted sample had an average victim age of 36 (SD = 13.6), while the control sample was slightly older, with an average victim age of 38 (SD = 15.5). Among the RTCC sample, sixty-two percent were male (n = 477), while 59 percent of the control sample victims were male (n = 454). This constitutes a change from the violent crime models in the previous chapters, in which whether the victim was black was significantly different, suggesting that many of the property crime victims in this sample were of African American descent.

Within the measures comprising human evidence, four of the five were significantly different between the two groups. There were statistically significant greater proportions of witness interviews (n = 250 vs. n = 210), flyers (n = 259 vs. n = 72) disseminated, lineups (n = 157 vs. n = 114) conducted , as well as increased presence of external CCTV evidence (n = 409 vs. n = 115) within RTCC-assisted cases compared to

the control sample. There were no significant differences in the presence of victim interviews. For crime scene-based evidence, all six measures were significantly different between the samples. There was significantly more firearm present (n = 347 vs. n = 218) and vehicles used (n = 413 vs. n = 219) in the commission of a crime, and accordingly, greater use of NIBIN ballistics tracing (n = 195 vs. n = 57), processing of gunshot residue (n = 80 vs. n = 31), and greater use of the CSI unit (n = 381 vs. n = 285) in the RTCC-assisted sample. Latent/DNA prints were collected more within the control sample (n = 214 vs. n = 178). Lastly, there were also significantly greater numbers of search warrants (n = 52 vs. n = 15) and cell warrants (n = 39 vs. n = 13) acquired, and more officers utilized (M = 3.81 vs. M = 3.58) on RTCC-assisted cases compared to the control sample.

Burglaries

Once isolated from the full violent crime model, a significantly greater proportion of burglary cases were cleared within the RTCC-assisted sample (n = 21, 24 percent) compared to the control sample, with only 11 cleared cases (13%). Alternatively, RTCC-assisted cases had a significantly longer time to clearance, with an average of 69 days to case clearance (SD = 104.5) compared to the control sample average of just 44 days (SD = 114.2). Among the measures used in the analysis to adjust for victim and crime solvability characteristics, there were only a handful of significant differences between the two samples. These differences fell within the human evidence and crime scene evidence categories.

Table 13. *Comparative overview of measures used in the analysis of burglary and larceny incidents*

	Full Crime Model N = 1524		Burglary n = 164		Larceny n = 64	
Measures (N=1524)	**RTCC-assisted cases n (%)**	**Control cases n (%)**	**RTCC-assisted cases n (%)**	**Control cases n (%)**	**RTCC-assisted cases n (%)**	**Control cases n (%)**
Cleared[2]	261 (34%)	210 (27%)**	21(24%)	11(13%)	6(19%)	1(3%)
Time to Clearance	45(110.6)	20(58.8%)**	69(104.5)	44(114.2)	28(19.1)	4(-)
MRTCC technologies						
CCTV	628(82%)	--	45(52%)	--	19(59%)	--
Social media	134(17.5%)	--	10(12%)	--	4(13%)	--
Facial recognition	231(30%)	--	53(61%)	--	13(41%)	--
License plate reader (LPR)	92(12%)	--	7(8%)	--	3(9%)	--
ShotSpotter	125(16%)	--	--	--	--	--
Victim characteristics						
Victim age[1]	36(13.6)	38(15.5)***	42(10.1)	44(13.1)	39(9.0)	38(12.9)
Victim male[2]	477 (62%)	454 (59%)*	33(38%)	32(37%)	16(50%)	12(38%)
Victim black[2]	335(44%)	310(40%)	8(9%)	13(15%)	5(16%)	7(22%)
Human evidence						
Victim interview[2]	699(91%)	710(93%)	82(94%)	(87%)	32(100%)	32(100%)
Witness interview[2]	250 (33%)	210 (27%)*	12(14%)	9(10%)	4(13%)	1(3%)
Flyer[2]	259 (34%)	72(9.4%)***	42(48%)	9(10%)***	4(13%)	--
Lineup[2]	157 (21%)	114 (15%)**	7(8%)	5(6%)	5(16%)	--
External CCTV[2]	409 (53%)	115 (15%)***	75(86%)	28(32%)***	20(63%)	8(25%)**
Crime scene evidence						
Firearm used[1]	347 (45%)	218 (28%)***	6(7%)	--*	4(13%)	--
Vehicle involved[2]	413 (54%)	219 (29%)***	25(29)	7(8%)***	11(34%)	2(6%)*
Ballistics tracing (NIBIN)[2]	195 (25%)	57 (7%)***	2(2%)	--	--	--
Gunshot residue[2]	80 (10%)	31 (4%)***	--	--	--	--
Latent prints/DNA[2]	178 (23%)	214 (28%)*	28(32%)	38(44%)	1(3%)	1(3%)
Crime Scene Investigation[2]	381 (50%)	285 (37%)***	29(33%)	38(44%)	2(6%)	1(3%)
Warrant evidence & manpower						
Search warrant[2]	52 (7%)	15 (2%)***	1(1%)	--	--	--
Cell warrant[2]	39 (5%)	13 (2%)***	--	1(1%)	--	--
Number of officers[1]	3.81(1.9)	3.58(1.5)**	2.54(1.50)	2.62(1.64)	2.06(1.24)	1.34(.94)

* $p \leq .05$, ** $p \leq .01$, *** $p \leq .001$; [1] For these measures, independent samples t-tests were used to determine differences, equal variances not assumed. [2] Fisher's Exact Test was used to determine differences for these dichotomous measures.

Here, there was a greater proportion of public information flyers deployed (n = 42 vs. n = 9), more presence of external/private CCTV (n = 75 vs. n=28), and significantly more firearms present (n = 6 vs. n = 0), as well as vehicles involved in the commission of a crime (n = 27 vs. n = 7) in the RTCC-assisted sample as compared to the control. It should also be noted that while NIBIN was present in only the treatment group, GSR was not present in either the treatment or control group.

Larceny Thefts

For larceny thefts, there were only two significant differences between the treatment and control groups. Those differences were also apparent within the human and crime scene evidence groups, with a significantly greater proportion of external CCTV presence (n = 20 vs. n = 8), and more vehicles used (n =11 vs. n = 2) within the RTCC-assisted group. There were no differences in outcomes measures, with six larceny thefts being cleared within the RTCC-sample, and one within the control group. Moreover, on average it took the RTCC 28 days to clear larceny theft incidents, whereas in the control sample it only took four. However, this difference was not significant as the sample here was relatively small.

Case Processing

Module network analysis helped to determine the influence of RTCC technologies on the case processing of property crimes. Network modules here were constructed with an EBICglasso estimation procedure with a nonparanormal transformation[18] The analyzed data was drawn from the expanded data set derived from the RTCC log and the stratified

[18] Refer to chapter four for a detailed description of this analytical approach.

randomly drawn control sample as discussed in the previous "Data and Measures" section, with a total sample size of 1534 included within the analysis. The data was coded to represent whether various evidential sources were used in the investigation of the violent crime incident regardless of whether usable evidence was obtained from the evidence sources.

These evidence sources were grouped into four types and included MRTCC technologies (CCTV, LPR, social media, facial recognition, and ShotSpotter), crime scene evidence (NIBIN, GSR, CSI, Latents/DNA), human-based evidence (victim interviews, witness interviews, media flyers, lineups, and external CCTV) and warrant based evidence (search warrants, cell warrants).[19] An additional crime type variable was intentionally coded to represent the investigation of property crimes, with the aim of examining the role of police technology in the processing of these crimes. As with all other network analyses performed in this dissertation, victim characteristics and other investigative solvability factors were not included as they were outside of the scope of the analysis.

[19] For an overview of the measures used in this analysis, refer back to the comparative overview of both samples presented in the previous table.

Table 14. *Weight matrix of variables included in property crime network*

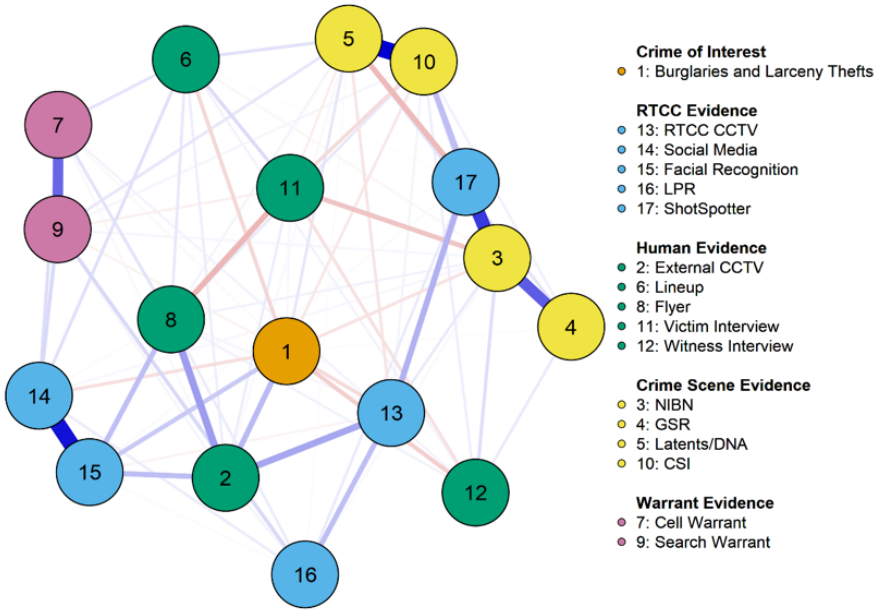
Variable **(N=1534)**	**RTCC Sample (Weight)**	**Control Sample (Weight)**
Crime of Interest		
Burglaries and Larcenies	--	--
RTCC Evidence		
CCTV	.003*	--
Social media	-.064*	--
Facial recognition	.341*	--
LPR	.107*	--
ShotSpotter	-.007*	--
Human evidence		
Victim interview	.219*	.418*
Witness interview	-.127*	-.098*
Flyer	.110*	.367*
Lineup	-.649*	-.006*
External CCTV	.452*	.691*
Crime scene evidence		
NIBIN	-.307*	-.279*
Gunshot residue	-.195*	-.330*
Latent prints/DNA	.342*	.362*
CSI	.003*	.004*
Warrant Evidence		
Search warrant	-.328*	-.350*
Cell warrant	-.328*	-.334*
Number of Nodes	17	12
Non-zero edges	120/136	58/60
Sparsity	0.118	0.121

* Represents an existent connection to the crime of interest node.

The findings from this network indicate a few things. First, the network of RTCC-supported cases and the randomly drawn control cases had similar levels of sparsity within the network, where the RTCC-assisted sample had a sparsity measure of .118 compared to that of .121 in the control sample. Despite the greater number of evidential nodes present

in the MRTCC network (n = 17), the level of connectedness with other evidential nodes remains about the same as in the control sample (n = 12). Notably, these sparsity measures are much lower than those of the violent crimes assessed in chapter four and chapter five, which is an improvement in the model and indicates a much more connected network.

Figure 11. *RTCC-assisted module network analysis for property crime incidents*



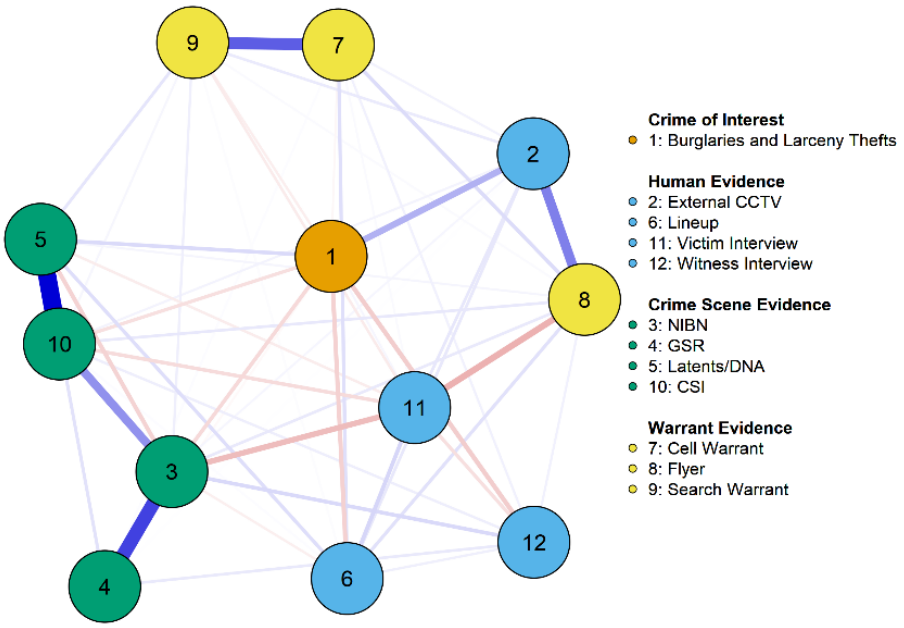
In Figure 11, it is apparent that RTCC technologies play a significant role in investigating property crime incidents[20]. These incidents are more central in the module network than other types of crime that have been examined. Among the RTCC evidence

[20] It is evident that the numbers in the table of contents are not arranged sequentially, which contrasts with all other network analyses performed in the dissertation. However, JASP has discontinued its open-source software as of April 2023, and modifications to the network visualization can no longer be executed.

sources, two of them have a greater impact on investigating the particular crime in question in the model. The weight matrix (Table 14) indicates that facial recognition software technology (with an influence of .341) and the license plate reader system (with an influence of .107) are strongly linked to the property crime node. CCTV (.003), social media searches (-.064), and ShotSpotter (-.007) also maintain some form of connection to the property node, albeit weaker.

Here, as perhaps expected, the more traditional nodes that work alongside the RTCC technologies with the highest influence the processing of property crime incidents include external/private CCTV (.452), latent prints/DNA (.342), and victim interviews (.219). Suspect lineups (-.649) also appear to be connected to the property node; however, the negative relationship indicates that it is highly contingent on the presence of other nodes.

Figure 12. *Non-RTCC-assisted module network analysis for property crime incidents*



To determine which sources outside the scope of RTCC technologies maintained strength in the investigative process, the network plot within the control sample was created by removing the RTCC technologies. The results showed that human and crime scene evidence sources played a significant role in the investigation. According to the weight matrix (Table 14), external CCTV (.691) has the highest connection to the property node, followed by victim interviews (.418), and collection of latent prints or DNA evidence (.362). The negative direction of most of the remaining nodes does not indicate a negative relationship as typically reported and observed within a regression model. Instead, it shows that 1) their connection to the node relied on the presence of other connections, and 2) the presence or absence of these measures would not affect the strength of the other connections.

Case Clearance

Table 15 presents the analysis results aimed at examining whether RTCC technologies influenced the case clearance of property incidents. Although Table 13. shows that RTCC-assisted burglary and larceny theft incidents were cleared more frequently than the control sample, these numbers alone do not consider the differences in the case processing of these crimes. Therefore, to accurately determine the influence of RTCC technologies on clearance rates for these incidents, the multivariate logistic regression analysis controlled for such differences.

Here, missing data was identified in 140 total cases and deleted using the listwise deletion method. Specifically, 8.1% (n = 125) of all cases had missing victim gender information and 9.0% (n = 138) had missing victim race information. Moreover, 2 cases or

.1% of the sample had missing data regarding whether the CSI unit was deployed to the scene of a crime. As for victim age, 136 incidents contained missing data, but this information was transformed and replaced with the series mean.

The Chi-square and -2 log-likelihood coefficients were statistically significant in all models, indicating good model fit. Within the significant models Nagelkerke pseudo R-square indicated an increasing ability of the models to predict the model variance ranging from a low of .7% in *Model Two* to a high of twenty-four percent in *Model Six*. All independent measures used in these models were examined for multicollinearity using the variance inflation factor (VIF) with no multicollinearity revealed. All VIF factors ranged from a low of 1.085 to the highest at 1.979.[21]

Model One only included property crimes. Here, we see that without the presence of any other covariates, the odds of burglary and larceny thefts being cleared, as compared to all other violent crimes included in the model, were statistically significantly lower. Specifically, property crimes had roughly 66 percent lower odds of case clearance. In *Model Two*, all crimes that received RTCC-support, were included within the model to determine the influence of technology on case clearance. This variable established significance within the model, indicating approximately 42% increased odds of case clearance (≤ .01) Here, we see that property crime incidents remain at 66% significantly lowers odds of case clearance, thus RTCC-support was not influential in clearing these crimes.

[21] These scores are far below the area of concern for VIF where convention dictates that scores of 5 or greater suggest the presence of multi-collinearity among independent measures.

Table 15. *Nested multivariate logistic regression of predictors of property case clearance and influence of RTCC technology*

***Measures (*N=1394)**	***Model One*** **B (Odds ratio)**	***Model Two*** **B (Odds ratio)**	***Model Three*** **B (Odds ratio)**	***Model Four*** **B (Odds ratio)**	***Model Five*** **B (Odds ratio)**	***Model Six*** **B (Odds ratio)**
Crime of Interest						
Property Crime	-1.076(.341)***	-1.086(.338)***	-1.132(.322)***	-.768(.464)**	-1.015(.362)***	-.872(.418)**
Impact of Technology						
RTCC-Assisted Crimes		.353(1.423)**	.375(1.455)**	.407(1.502)**	.519(1.680)***	.515(1.673)***
Victim characteristics			-.001(.999)	.003(1.003)	.000(1.000)	.000(1.000)
Victim age			-.413(.662)***	-.392(.767)**	-.285(.752)*	-.290(.748)*
Victim male			-.024(.977)	.029(1.030)	.110(1.116)	.124(1.132)
Victim black						
Human evidence						
Victim interview				.506(1.659)*	.324(1.383)	.335(1.398)
Witness interview				.452(1.572)***	.485(1.624)***	.451(1.570)**
Flyer				.665(1.944)***	.761(2.140)***	.713(2.040)***
Lineup				1.739(5.692)***	1.785(5.958)***	1.754(5.775)***
External CCTV				-.794(.452)***	-.745(.475)***	-.843(.430)***
Crime scene evidence						
Firearm used					-.653(.521)***	-.689(.502)***
Vehicle used					-.256(.774)	-.335(.715)*
Ballistics tracing (NIBIN)					.113(1.120)	.103(1.108)
Gunshot residue (GSR)					.107(1.113)	.043(1.044)
Latent prints/DNA					-.016(.984)	-.123(.885)
Crime Scene Investigation					-.171(.843)	-.201(.818)
Warrant evidence & manpower						
Search warrant						.805(2.237)*
Cell warrant						.930(2.534)*
Number of officers						.075(1.077)
Model Chi-Square	24.946***	34.117***	45.555***	225.672***	255.152***	277.342***
-2 Log likelihood	1711.927	1702.756	1691.318	1511.201	1481.721	1459.531
Nagelkerke Pseudo R-square	.025	.034	.045	.210	.235	.253
Constant			-.777(.460)***			

* $p \leq .05$, ** $p \leq .01$, *** $p \leq .001$

Property crime incidents remained significant in *Model Three,* in which victim characteristics were subsequently introduced. Of all covariates included within the model, RTCC-assisted violent crimes and whether the victim was male also proved to be significantly related to whether a case was cleared. Here, incidents receiving RTCC technology support were 46% more likely to be cleared than those without RTCC support and male victims resulted in 34% lower odds of case clearance. Human evidence sources were added to *Model Four*. Within this model, incidents that received RTCC support maintained significance (50% more likely to be cleared), as did whether the victim was a male, which still resulted in roughly 23% lower odds of case clearance. All human evidence sources proved to be of statistical significance, with suspect lineups being the most impactful. In fact, when a lineup was present, the odds of the case being cleared were five times greater than when there was no lineup, while controlling for other factors ($p = \leq .001$). Here private or external CCTV was associated with 55% decreased odds of case clearance.

Model Five incorporated evidence from the crime scene, revealing that incidents of property crime remained significant. The odds improved slightly, but property crimes were still 64% less likely to be cleared than all other violent crimes included within the model. Further, cases involving RTCC-assistance had a 68% greater likelihood of being cleared ($p = \leq .001$). Of the victim characteristics, a male victim resulted in 25% lower odds of case clearance. Among the five human evidence elements, only four remained significant, with suspect lineups having the greatest impact on case clearance (OR = 5.958; $p = \leq .001$). External CCTV, once again, depicted lesser odds of case clearance (OR = .475; $p = \leq .001$).

As for crime scene variables, the use of a firearm in the commission of the crime and was significant, revealing 48% lower odds of case clearance.

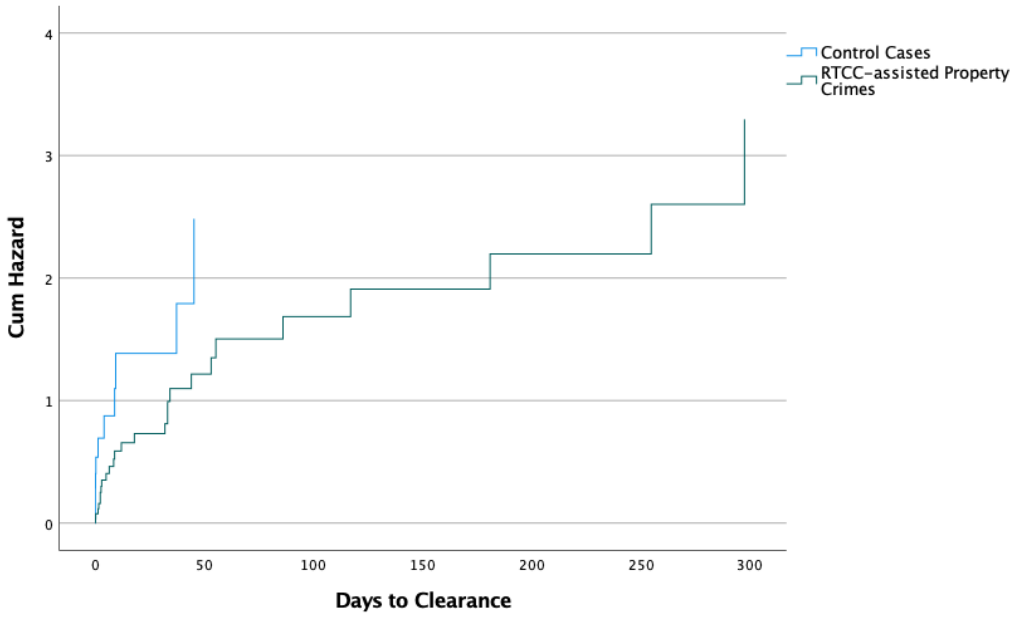
In *Model Six*, all variables, including those related to warrant evidence and manpower, were included in the final analysis. Here, the property crimes also remained insignificant, depicting 58 percent lower odds of case clearance than violent crime. A male victim sustained statistical significance, revealing 25% lower odds of case clearance ($p = \leq .05$). Among all the variables, the support of RTCC technology, witness interviews, suspect lineups, public information flyers, and search warrants significantly increased the odds of case clearance. RTCC-assisted incidents had 67% better odds of case clearance, and the use of a suspect lineup remained the strongest indicator in the model (OR = 5.775; $p = \leq .001$), followed by the dissemination of public information flyers (OR = 2.040; $p = \leq .001$). External CCTV presence (OR=.430; $p = \leq .001$), the use of a firearm (OR =.502; $p = \leq .001$), and the use of a vehicle (OR = .509; $p = \leq .05$) significantly decreased the likelihood of a case being cleared, controlling for all other solvability factors and case characteristics. Within the warrant and manpower group, both search and cellphone warrants significantly increased the odds of case clearance two-fold (OR = 2.237) and (OR = 2.534).

Time to Case Clearance

The analyses to determine whether RTCC-assisted cases improved the time that it took to clear property crime incidents are reported in Figures 13 and 14, and Table 16. The simple frequency of days to case clearance reported in Table 13 suggests that RTCC assistance did not improve the speed at which arrests were made in burglary or larceny theft incidents. However, this does not account for the various victim characteristics and

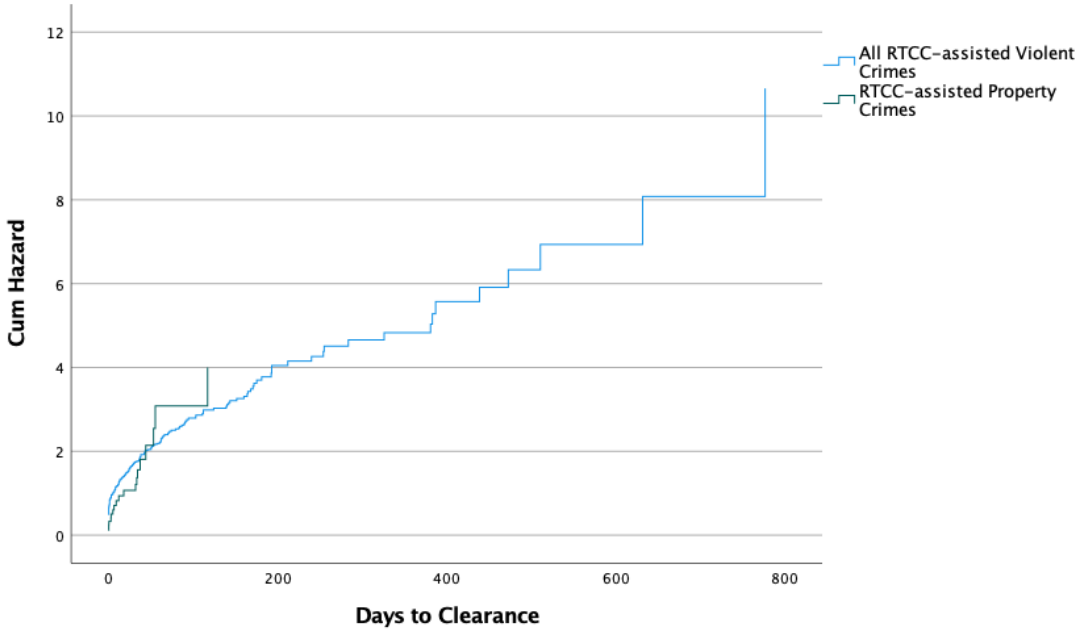
case processing characteristics that could influence the speed at which cases are cleared which are outside the scope of RTCC influence.

Figure 13. *Cumulative rate of case clearance RTCC-assisted property crimes*



First, a Kaplan-Meier survival analysis was performed to identify the trajectories in the time- to-case clearance within the two samples. Here, only the treatment and control groups for property crime incidents were included within the analysis. The total number of cleared cases between the two samples was 39, of which 27 were cleared in the RTCC-assisted sample, and 12 in the control. The outcome of that analysis is reported in Figure 13. This analysis does not adjust for case characteristics and only plots the proportion of cases cleared within each sample across the period of days present within the observation period. Here we do not see a clear difference in the trajectory of RTCC-assisted cases and the control sample, with control sample cases being cleared at a similar compared to the sample assisted by RTCC technologies. The Mantel-Cox Log Rank comparison further indicates that these trends were not significantly different ($\chi 2 = .861$, 1 df, $p = .353$).

Figure 14. *Cumulative rate of case clearance for RTCC-assisted property crimes vs. other forms of violent crime adjusted at mean for case characteristics*



To account for differences in victim and case processing characteristics between the two samples a Cox Proportional Hazard regression model was fitted. Initially, this analysis was conducted solely on property crime incidents due to their distinct characteristics. However, the sample size was too small, and the resulting Chi-Square coefficient did not attain statistical significance, indicating a suboptimal model fit. As a result, all cleared crimes were incorporated in the analysis and further stratified by technology support to examine whether the time to clearance varied for RTCC-assisted property crimes in comparison to other forms of RTCC-assisted violent crimes. The trend line comparison of the model is presented in Figure 14 and the measures included in that model and coefficients are reported in Table 18. Each of the victim characteristics, human evidence, crime scene evidence, warrants, and manpower measures were included to

account for their influence on the time taken to clear cases. As visible in Figure 14, after controlling for case characteristics the trend lines indicating the occurrence of case clearance over time overlap.

With this adjustment, there are no apparent differences between the RTCC-assisted cases and the control sample in the time-to-case clearance trajectories. This was also evident in the coefficients (see Table 16). Holding constant the influence of all other measures there was no significant difference in the days to case clearance between the RTCC-assisted property crime cases and the control sample (B = -.130, SE = .878, p = .591, n.s.). The findings here suggest that the implementation of RTCC information technologies in the investigation of property incidents has not reduced the amount of time that it takes to clear cases as compared to violent crime RTCC-assisted cases, even when considering case level circumstances.

Among the remaining measures in the analysis, five were significantly related to the time that it took to clear cases. For victim characteristics, the only significant predictor was whether the victim was male. When the victim was male the case had significantly higher odds (OR = 1.484) of being cleared before female victims. Among human evidence measures, whether a public information flyer was disseminated was associated with 40 percent (OR = .601) lower odds of being cleared before those cases without a flyer. Within the crime scene evidence characteristics, when a firearm or vehicle was used, the odds of faster case clearance was lower as compared to incidents that did not have these variables (86% and 89%, respectively). Finally, in cases that a cell phone warrant was obtained, it also took significantly longer to clear a case (OR = .589). The -2 log-likelihood and model

Chi-Square coefficient were statistically significant indicating a good model fit (χ^2 = 70.060, 17 df, $p = \leq .001$).

Table 16. *Cox proportional hazard model coefficients predicting days to clearance (property crimes)*

Measures (N = 434)	**B (SE)**	**Odds Ratio**	***p* - value**
Crime of Interest			
RTCC-assisted Property Crimes	-.130(.242)	.878	.591
Victim Characteristics			
Victim age	-.000(.004)	1.000	.910
Victim male***	.395(.111)	1.484	<.001
Victim black	-.059(.103)	.943	.567
Human Evidence			
Victim interview	.340(.231)	1.405	.141
Witness interview	.082(.105)	1.085	.435
Lineup	-.209(.110)	.812	.059
Flyer***	-.508(.132)	.601	.001
External CCTV	-.223(.131)	.800	.088
Crime Scene Evidence			
Firearm used*	-.321(.138)	.727	.020
Vehicle used*	-.225(.113)	.798	.046
Ballistics tracing (NIBIN)	.080(.180)	1.083	.659
Gunshot residue (GSR)	.324(.217)	1.383	.135
Latent prints/DNA	.140(.149)	1.151	.346
Crime Scene Investigation (CSI)	.016(.142)	1.017	.908
Warrant Evidence & Manpower			
Search warrant	.317(.222)	1.373	.153
Cell warrant*	-.530(.233)	.589	.023
Number of officers	.049(.036)	.589	.172
Model Chi-Square		70.060***	
-2 Log likelihood		4413.495	

Note. *$p \leq .05$, **$p \leq .01$, ***$p \leq .001$.

Chapter Seven: Public Perceptions Associated with Police Use and Deployment of Technology

Introduction

The primary aim of this chapter was to investigate how the public perceives police technology through the use of randomized controlled trial research design. The study sought to evaluate public acceptance of technology that is relevant to the operational decision-making processes of law enforcement agencies equipped with real-time crime center units. To this end, survey participants were assigned to differing treatment groups, thereby allowing for a systematic assessment of their attitudes toward such technologies.

The present study pursued two principal objectives: firstly, to ascertain if the tone of narratives and the extent of disseminated public messaging and information pertaining to police technology use and deployment could enhance public acceptance of these technologies, and secondly, to identify the factors that exert the most significant impact on public approval of police technology, such as respondent characteristics and experiences. To attain these objectives, the study was framed by two key research questions: (1) "What is the influence of the messaging strategy employed to communicate information on the use and deployment of police technology in communities on public approval?" and (2) "Which of the three tested concepts - police legitimacy, transparency, or effectiveness - exerts the most substantial influence on obtaining public approval of police technology?"

The chapter begins with a review of the relevant literature concerning the public's attitudes towards police technology. This is followed by a section outlining the analytical methodology employed, including a detailed overview of the data collection procedure and

the analytical measures implemented. In chapter eight, the limitations of this study and policy implications will be presented.

Review of Literature

The dynamic between police officers and the communities they serve has been characterized by a growing sense of tension and fragmentation. This trend is indicative of a broader societal shift, as traditional law enforcement practices are being questioned for their legitimacy and effectiveness. This phenomenon has been fueled in part by high-profile cases of police brutality, particularly those involving the killings of African Americans, which have prompted some of the largest protests in American history, as well as widespread calls for systemic change in law enforcement practices (Buchanan et al., 2020). The root causes of this erosion of trust and sense of alienation are not confined to any one specific region, but rather represent a pervasive phenomenon with significant implications for the future of policing.

Moreover, Gallup undertakes extensive annual surveys encompassing all 50 states of the United States, with the aim of comprehending the degree of confidence that Americans place in various institutions throughout their nation. Over the course of the past few decades, spanning back to 1993, the police have consistently garnered a noteworthy level of trust from the American populace in comparison to other establishments (Gallup, 2023). Throughout the period spanning 1993 to 2018, public confidence in the police exhibited fluctuations, with the lowest recorded levels occurring in 1993 and 2015 (both registering at 52%), while the pinnacle of trust was observed in 2004, boasting an impressive 64%. Overall, the police have commanded considerable respect, maintaining an average level of trust at a commendable 57% over this timespan. However, more recent

findings extracted from Gallup's poll reveal a downward trajectory in the confidence bestowed upon the police. Specifically, in 2020, less than half of the American population, a mere 48%, expressed possessing "a great deal" or "quite a lot" of faith in the police (Gallup, 2023). This occurrence signifies the first instance since 1993 in which confidence in the police has descended below the critical threshold of 50%.

Nonetheless, the integration of technological advancements, such as body-worn cameras, has frequently been proposed as a strategy to harness the digital era and foster positive rapport between law enforcement and society (Lum, Stoltz, Koper &Scherer, 2019). However, it is paradoxical that many instances of police misconduct and aggressive enforcement actions have been documented through video surveillance facilitated by the implicated law enforcement agency or by private citizens utilizing their smartphones or social media platforms (Kochel & Skogan, 2021). Thus, establishing the status of public approval of police technologies has been multifaceted and complex, as the promise of enhanced public safety with limited intrusion on personal privacy is welcomed by civilians, whereas contrasting views on surveillance technology are influenced by their trust in government and police agencies (Pavoe & Esposti, 2012).

However, based on the findings of Saulnier and Thompson's (2016) research, it appears that the degree to which police technology is embraced by the public depends on the intended purpose of the technology or the justification for its implementation. Further, in situations where individuals are not fully informed about the specific goals of the technology or where there is no clear victim, concerns about the underlying motives behind surveillance may be heightened (Sakiyama, 2016). Legitimacy, in conjunction with trust, can also serve as a criterion by which individuals assess whether the police are justified in

expanding the deployment of technological tools. Existing literature suggests that perceiving law enforcement agencies as fair and equitable entities fosters endorsement of behaviors that align with established normative boundaries of policing (Gerber & Jackson, 2017). Whereas accountability and transparency are then seen as levers that promote elements of police legitimacy and trust. Nevertheless, in light of the current scrutiny faced by law enforcement agencies and the recent movement to defund these organizations, gaining public approval has become a crucial priority for police departments nationwide.

Public Perception of Technology

Bragias, Hine, and Fleet (2021) recently conducted a study exploring the public's attitudes towards the utilization of facial recognition technology (FRT) by law enforcement agencies. The researchers focused on analyzing comments posted on YouTube, recognizing both the potential benefits of FRT for enhancing police efficiency and safety, as well as the challenges of striking a balance between security and public support. The study analyzed a diverse dataset of 460 videos pertaining to police use of FRT from November 2017 to March 2021. Employing sentiment analysis, the videos were categorized as positive, neutral, or negative, with only neutral videos being included to minimize biases, resulting in a set of 71 videos created by various individuals, local media, and television recordings.

The study's findings revealed that a substantial majority (75%) of the comments expressed negative sentiments, while only 15% were positive and 9% neutral, indicating a predominantly unfavorable perception of police FRT usage. The public voiced apprehension regarding the accuracy of the technology, given that several private companies are developing FRTs with varying degrees of success. Additionally, concerns

were raised about potential abuses of power and authority by both the state and police in using FRT, as well as the erosion or violation of individual rights and freedoms, indicating discontent with both state governance and private enterprise. To generate support for FRT, the researchers proposed addressing these concerns and highlighting its potential benefits for criminal investigations through public education campaigns outlining the technology's usage, trials, and accompanying legislation.

Even more recently, a national survey of public support for FRT and associated privacy and safety concerns was carried out by Miethe et al., (2023). This study utilized a sample generated through Qualtrics' online panel of survey participants and was conducted in March 2022, with 612 U.S. adults participating. The results of their study showed variability in support across different policing contexts, with 70% of respondents supporting FRT use overall. Support was also found to be higher in high-profile policing activities, than in public-order situations with the highest support for identifying terrorists and the lowest for identifying jaywalkers. Additionally, the impact of privacy concerns on FRT support was found to depend on an individual's level of trust in the police. For example, among those with low trust, higher privacy concerns are associated with lower support for FRT.

On a similar note, but from an international perspective, Bradford et al. (2020) utilized an RCT vignette approach survey data from 1,092 Londoners collected between 23 May and 4 June 2018 to investigate their perspectives on the Metropolitan Police Service's (MPS) gathering and retention of personal data, particularly in relation to Live

Facial Recognition (LFR).[22] The findings are very relevant to the current study. Here the scholars found that there was support for LFR was higher when used to identify potential terrorists or those wanted for serious violent crimes, but lower for minor crimes and nuisance behavior.

Additionally, the study uncovered some variation in views across different socio-demographic categories, with older individuals and white individuals more likely to view police use of LFR as acceptable. Concern about crime and political orientation were also significantly associated with legitimacy, whereas there was little variation in views by gender, country of birth, social class, or crime victimization. The study's models employing different measures of acceptability also revealed a weaker association between privacy concerns and acceptability when participants were prompted to consider specific crimes, indicating that privacy concerns become relatively less important for many individuals in such cases.

Scholars have also recently employed similar analytical strategies to determine public approval of drone usage. For example, Heen, Lieverman & Miethe (2018) investigated the relationship between individual attitudes towards police legitimacy and effectiveness, and the support for police UAV use. The researchers hypothesized that greater legitimacy and effectiveness would lead to increased acceptance of proactive policing measures, while the use of UAVs for reactive policing activities would receive high levels of support regardless of attitudes towards police legitimacy. To test this

[22] It should be noted that the vignette's employed did not achieve a statistical significance between the groups tested, thus additional analyses and research questions were carried out.

hypothesis, a national survey was conducted using Qualtrics and Mechanical Turk, resulting in a sample of 481 adult respondents collected in May 2015.

The results of the study showed that support for UAV use was generally higher in voluntary citizen-initiated contexts, which are reactive in nature, while support was lower for proactive police-initiated interactions. Furthermore, victimization and previous contact with police were found to influence support for UAV use in both proactive and reactive applications. African Americans, who generally had lower confidence and trust in the police, were also less likely to support UAV usage compared to White individuals.

Merola, Lum & Murphy (2018) further implemented a survey-based experiment to understand community perceptions following the adoption of a new policy by an agency, specifically focusing on the use of LPRs by the police. The experiment was conducted in Fairfax County, Virginia, with a sample size of 405 participants. Two versions of the survey were created, with the control group receiving questions about police perceptions first and LPR functions second, while the experimental group encountered the sections in reverse order. The scholars found that roughly 90% of respondents were uncertain if their local police department used LPR, indicating limited exposure to LPR in their communities and a lack of awareness regarding its implementation. Further, the experiment revealed that when participants were given LPR information, their trust in the police declined compared to the control group.

Current Study

The primary aim of the research was to investigate the attitudes of the general populace towards the utilization and deployment of technologies commonly housed within real-time crime center units. To accomplish this objective, a randomized experimental

methodology was employed, where treatment groups were provided with a personalized vignette that portrayed a descriptive scenario of police employment of technology. The present investigation was made feasible thanks to the financial backing of the Association of Doctoral Programs in Criminology and Criminal Justice ($1,000) and the Kimberly and Dorothea Green Fellowship for Advanced Doctoral Students ($1,000). These funds were employed to acquire targeted survey responses via the market research provider, Pollfish.[23] The earlier section of the study highlighted pertinent scholarly works, which offered some empirical backing for the following hypotheses postulated.

[H1] The treatment group depicting police use and deployment of police technology in a positive manner will be associated with improved levels of public approval of technology.

[H2] The treatment group depicting police use and deployment of police technology in a negative manner will be associated with heightened levels of public approval of technology.

[H3] The treatment groups receiving only a neutral description of the technologies commonly used and deployed by police agencies will be associated with reduced levels of public approval.

[H4] The most apparent difference in pre and post stimuli scores will be observed between the control and positive treatment group.

[23] In the forthcoming analytical approach section, a comprehensive rationale for the utilization of Pollfish will be presented.

[H5] Of all three concepts tested, perceptions of police legitimacy will have the most significant impact on the approval of police use and deployment of police technology.

Analytical Approach

The study employed various statistical analyses to address the research questions. Internal consistency or reliability of the survey items was assessed using Cronbach's alpha coefficient. Descriptive statistics were then presented using analysis of variance and Fisher's Exact Test to determine significant differences between the control group and treatment groups. One-way ANOVA, along with a Bonferroni post hoc group comparison, was conducted to compare police legitimacy, transparency, and effectiveness post-test scores. The study also analyzed police technology approval pre and post-test scores and used a difference in differences in test. Finally, ordinal regression analysis was performed to identify predictors of increased public approval for police technology use and deployment.

To conduct this RCT experimental vignette study, Pollfish was selected over other survey marketing tools for a several reasons. Firstly, Pollfish refrains from dispensing cash or airline incentive points, unlike most other platforms, which are prone to generate static and cumulative incentives, ultimately resulting in the emergence of "professional survey-takers" (Andrade, 2022). These individuals possess the skills to circumvent screening questions and are motivated to complete as many surveys as possible without comprehending the underlying rationale questions.

Instead, Pollfish engages directly with app developers, who can formulate a distinct non-cash incentive to encourage genuine consumers to participate in surveys without the

risk of inducing them to become habitual survey-takers. For example, a news app may offer a premium article, while a fitness app may provide a free yoga lesson. The third-party app partners undergo a meticulous vetting process, and their compensation is contingent upon furnishing responses from individuals who fit the targeting criteria specified by the researcher and satisfy the quality controls. This approach diverges from the conventional methodology of impaneling respondents, which incurs a steep cost and results in the elimination of dubious responses.

Additionally, over the past few decades, the realm of market research has undergone a significant transformation with the evolution of techniques utilized, which now spans from traditional random digit dialing to online panels and river sampling. This shift has been driven by a confluence of factors, including technological advancements, shifts in consumer behavior, and the identification of deficiencies in existing methodologies. Hence, Pollfish uses organic sampling built on a new delivery framework called Random Device Engagement, which is considered the future of market research by academics and researchers (Rothschild & Konitzer, 2018).

Pollfish has also developed a proprietary algorithm that uses machine learning technology to detect and remove fraudulent or suspicious responses at the question level. By leveraging artificial intelligence, the platform can immediately eliminate insincere responses, ensuring the integrity and reliability of the collected data. Further, in this study, survey takers were randomly assigned to a treatment or control group. As most survey platforms do not offer the ability to test more than two concepts, the A/B testing feature provided within the Pollfish dashboard was particularly advantageous. This feature enables

a researcher to evaluate multiple concepts simultaneously, making it a valuable tool for experimental research.

Survey Construction

The survey comprised a total of 29 questions. The first question asked for the participant's consent to participate (as per Pollfish protocol) and the second question was presented in a descriptive manner, explaining the purpose of the survey (as per the approved FIU IRB protocol). Participants were informed that they could exit the survey at any point in time (see Appendix C for survey instrument). The remaining questions within the survey were close-ended, single-response questions. Certain demographic questions, including age, education, income, employment, and U.S. census region were provided by the survey marketing tool free of cost. Other respondent characteristics were acquired through the survey instrument itself, such as the gender of the participant and their political affiliation. These questions were later recoded into dichotomous measures.

Respondent experiences were also collected by inquiring as to whether they were a victim of a crime or had an interaction with a police officer in the past three years, whether they have ever been arrested for a crime, and their overall perception of crime in their neighborhood. These questions were included as, in some cases, they are known to influence perception of police practices (Koster, Van der Leun & Kunst, 2020). The responses to these questions were then transformed into dichotomous measures, where "yes" was assigned a value of 1, while "no" was assigned a value of 0, with the exception of the question relating to neighborhood perception, which was recorded on an ordinal scale.

All remaining questions were presented using a Likert scale, where a score of one (1) represented "strongly disagree" and a score of five (5) represented "strongly agree." The pretest included several questions, but one question in particular, which was the dependent variable for the ordinal regression model, was of primary interest. This question asked participants to rate their approval of police technology on a scale of one (1) to ten (10), with one indicating no approval and ten indicating high approval.[24] The same question was also asked as a post-test after the stimuli were administered.

Further, the survey participants were assigned to one of four groups through random assignment to meet the criteria for a randomized controlled trial and stratified by age, gender, and race. The control group was not exposed to any stimuli but was instructed to proceed with answering the post-test survey questions. The three treatment groups consisted of the positive stimuli group, the negative stimuli group, and neutral stimuli group. Overall, the intent was to observe the tone and extent of messaging needed to educate the public and achieve levels of acceptance with respect to the deployment of police technologies within communities. A description of each stimulus is provided in Appendix D.

Positive Stimuli

The principal aim of the positive stimuli was to present a favorable perception to survey respondents concerning the utilization of technology by law enforcement agencies as a means to alleviate public concerns. In this context, the treatment group was introduced to the technical resources available within RTCCs and the measures taken by police

[24] This was later recoded to 1 = 0 and 10 = 9 for subsequent analyses, as with the perception of neighborhood crime scale.

authorities to ensure the ethical and community-oriented use of these investigative tools. Additionally, a specific example was provided to demonstrate how these technological resources enhance crime prevention, victim support, and overall clearance rates.

In terms of policy considerations, the purpose of the positive depiction was to illustrate that community members are willing to adopt law enforcement technology, provided they are informed of the relevant protocols and the manifold ways in which such technologies confer benefits upon them. Consequently, an agency that prioritizes transparency in its utilization and dissemination of technology is likely to achieve greater acceptance of such technology within its jurisdiction.

Negative Stimuli

Conversely, the negative treatment group was established to present survey respondents with an unfavorable outlook on the application of police technology. This was implemented to assess the impact of this information on their overall perception of police practices in relation to the use and implementation of such technologies. While participants were already familiar with the existence of these technologies in RTCCs, the scenario portrayed their usage and implementation in an unfavorable light. Moreover, the description emphasized that these technologies were not advantageous to the community and emphasized the fact that their effectiveness as investigative tools was unproven.

The proposition put forth posits that a pessimistic portrayal of police technology or insufficient disclosure regarding its implementation can impede the general populace's view of the usefulness of these tools, thereby gradually eroding their trust in the validity and efficacy of police procedures. If this supposition were to materialize, it would

underscore the crucial importance of conducting research grounded in empirical evidence and disseminating this knowledge transparently to the public.

Neutral Stimuli

The purpose of the neutral stimuli group was to provide a succinct explanation of the technologies commonly utilized in RTCCs. This group can be viewed as representing "treatment as usual," as this treatment is already present, to some degree, in practice (Freedland et al., 2011). Consequently, participants received a summary of the functions of CCTV cameras, license plate readers, gunshot detection systems, and facial recognition software. While some survey respondents may have had prior knowledge of these technologies, the results of this vignette would reflect the perspectives of these individuals without any positive or negative influence or manipulation.

Intuitively, two outcome scenarios could arise. Firstly, participants may recognize the usefulness of individual technologies commonly employed in an RTCC setup, leading to greater comprehension and higher acceptance of these technologies. Alternatively, some individuals may associate systems such as facial recognition and LPRs with privacy issues, resulting in less positive opinions. Nevertheless, further analysis of the outcomes related to this vignette could be valuable in guiding law enforcement efforts to improve transparency.

Sample Size

A power analysis tool and sample size calculator were used to estimate the minimum sample size required for the experiment, given a desired significance level, effect size, and statistical power. According to the G-power software tool for a small to medium effect size of .20, a sample of 321 participants is recommended to optimally detect an effect

of the independent variables. Moreover, according to Raosoft, a sample size of 377 offers a margin of error of +/- 5% at the 95% confidence level. To ensure the attainment of the desired sample size, accounting for the possibility of invalid responses, a total sample size of 400 was deemed appropriate here.

While increasing the sample size to 500 would improve the reliability by only 0.5%, the cost of obtaining 100 additional responses may not justify the marginal benefit. Industry experts in survey marketing also contend that sample sizes beyond 400 often result in diminishing returns on investment (Kuhn, 2018). Further, to ensure the representativeness of the survey results, the sample was stratified based on age, race, and gender. Stratification is an effective method for minimizing bias in the data and obtaining a more precise and reliable estimate of the opinions and behaviors of different demographic groups, which is essential for informed decision-making and developing targeted interventions (Dunn & Shultis, 2023).

Survey Dissemination

The survey was distributed via Pollfish in April of 2023 and results were published within one day of the dissemination. The survey began with an initial screening question that served as a form of consent, requiring participants to confirm that they were at least eighteen years old and willing to proceed with the survey despite its sensitive nature. Pollfish classified the survey as sensitive due to its questions regarding participants' interactions with law enforcement officials and their arrest history within the past three years. The average completion time for the survey was eight minutes, and participant engagement was remarkable, with a response rate of 98%. Initially, 392 out of 400 participants proceeded to answer the survey questions after passing the screening process.

An additional eight individuals were selected through Pollfish's random sampling procedure to achieve the target sample size of 400 participants.

Of the total responses, 55 were eliminated from subsequent analyses for having "don't know" responses, bringing the final count to 345. Accordingly, seventeen respondents were removed from the control group, thirteen from treatment group one, nine from treatment group two, and sixteen from treatment group three (see Table 17). The decision to incorporate the "don't know" option in the survey was informed by the extant literature, which suggests that when a survey lacks a "don't know" (DK) option, respondents feel compelled to make a decision regarding the concept in question (Denman et al., 2018). Furthermore, individuals lacking sufficient knowledge on a topic may select an answer that does not align with their true beliefs or opt to skip the question altogether.

Reliability Testing

Prior to data analysis, reliability testing was conducted to ensure precision and accuracy in the measurement of the primary concepts of interest (refer to Table 18). The reliability evaluation was carried out using Cronbach's alpha coefficient, a statistical tool employed to determine the internal consistency and dependability of a set of survey items. The primary objective of this technique is to ascertain whether a group of items consistently gauges the same fundamental construct. The resulting alpha score provides a standardized indication of agreement among the items, ranging from 0 to 1, with higher values signifying greater coherence between the items. Consequently, high values of Cronbach's alpha suggest that respondents' answers across multiple questions are consistent, whereas low values indicate that the construct being measured lack reliability (Tavakol & Dennick, 2011).

Table 17. *Randomized controlled trial group stratification*

Group (N=345)	Assignment	Pre-Test	Intervention	Post-Test	Effect
Control Group n=83	Random	**O_1**	No Intervention	**O_2**	**O_2 - O_1**
Treatment Group 1 n=87	Random	**O_1**	**X** Positive Vignette	**O_2**	**O_2 - O_1**
Treatment Group 2 n=91	Random	**O_1**	**X** Negative Vignette	**O_2**	**O_2 - O_1**
Treatment Group 3 n=84	Random	**O_1**	**X** Neutral Description of Technologies	**O_2**	**O_2 - O_1**

This test was used to evaluate the consistency of the measuring instruments employed to quantify the constructs of legitimacy, transparency, and efficacy in the context of police technology implementation (see Table 18)[25]. The goal was to discern any perceptual changes within the treatment and control cohorts. Specifically, three questions were used to assess legitimacy, namely: (1) Police technology is used and deployed in a legitimate manner; (2) Police agencies utilize and deploy technology in a manner that

[25] According to recent research by Trinker et al. (2018) and Huq et al. (2017), legitimacy is a more suitable concept to use than trust, as cognitive aspects of trust are an essential element of legitimacy. Additionally, when individuals perceive law enforcement agencies to be competent, impartial, and actively engaged with their communities, it indicates that they are adhering to expected behavioral norms, thereby promoting a sense of trust. Hence, trust serves as a complete proxy for legitimacy in this context.

serves the best interest of the public, and (3) I trust police officer decision-making as it pertains to the use and deployment of innovative technology. The three items used to form the legitimacy scale yielded a Cronbach's Alpha score of α = .887 for the control group, α = .870 for the positive vignette treatment group, α = .876 for the negative vignette group, and α = .883 for the neutral information group.

In order to evaluate perceptions of police transparency, the survey respondents were asked the following questions: (1) Police technology is used and deployed in a transparent manner; (2) I consider myself informed on the vast array of technology; and (3) Police agencies are transparent about their use of innovative technologies. The three items used to form the transparency scale yielded a Cronbach's Alpha score of α = .714 for the control group, α = .760 for the positive vignette treatment group, α = .775 for the negative vignette group, and α = .725, for the neutral information group.

Finally, as it pertains to effectiveness the following questions were asked to survey respondents: (1) Police technology is used and deployed in an effective manner, (2) The use of innovative technology within police agencies has the potential to improve the efficiency and effectiveness of police practices dramatically, (3) Police technology is effective in reducing crime rates and improving public safety. The three items used to form the effectiveness scale had a Cronbach's Alpha score of α = .804 for the control group, α = .856 for the positive vignette treatment group, α = .749 for the negative vignette group and a value of α = .851 for the neutral information group.

Table 18. *Cronbach alpha's reliability test of legitimacy, transparency, and effectiveness scale*

	Treatment or Control Group of Interest					
Measures	**Items**	**Alpha**	**M(SD)**	**Items**	**Alpha**	**M(SD)**
		Control Group			Positive Treatment	
Legitimacy	3	.887	10.41(2.98)	3	.870	10.61(3.14)
Transparency	3	.714	9.42(2.59)	3	.760	10.36(2.77)
Effectiveness	3	.804	11.11(2.26)	3	.856	11.26(2.84)
		Negative Treatment			Neutral	
Legitimacy	3	.876	10.33(3.02)	3	.883	11.12(2.65)
Transparency	3	.775	9.66(2.96)	3	.725	10.30(2.55)
Effectiveness	3	.749	10.56(2.63)	3	.851	11.37(2.61)

Findings

The following section presents the findings of the study. However, prior to identifying any differences between the treatment groups and the control group, a one-way ANOVA was utilized, along with Fisher's Exact Test to determine any significant differences between multiple groups of data. Ensuring that the demographic characteristics of all treatment and control groups are comparable is paramount, as it assists in guaranteeing that any observed variations between the groups are exclusively attributable to the studied treatment and not any extraneous factors such as age, gender, ethnicity, or socioeconomic status.

When groups are not matched for demographic characteristics, it becomes challenging to differentiate between the effects of the treatment and other factors, and observed differences may be due to chance or biases (Stuart, 2010). Matching groups based on demographic characteristics can minimize the impact of these confounding factors and enhance the study's validity and reliability. Furthermore, having comparable groups is vital for generalizing the study results to the broader population. If groups are not matched for

demographic characteristics, the findings may not be applicable to the larger population, or they may only be relevant to a specific subset of the population.

However, as depicted in Table 19, there were no significant differences between the groups in terms of demographic characteristics. This means that the groups were comparable in terms of factors such as age, gender, income level, victim experiences and crime perception of their neighborhood. Here, eleven of the thirteen measures were coded dichotomously, meaning that they were either present (coded as 1) or absent (coded as 0). The age variable was measured as a continuous variable, while crime perception of neighborhoods was measured on a scale from 0 to 9, with higher scores indicating higher perceptions of crime in the neighborhood.

Impact of Stimuli

The present study aimed to investigate the divergent effects of various treatment types using analyses of variance with post hoc Bonferroni corrections (See Appendix E). Generally speaking, statistical analyses rely on two techniques to manage the family-wise error rate in multiple comparisons, namely Bonferroni corrections and Tukey's HSD (honest significant difference) test. The selection of either approach hinges upon the research inquiry and the attributes of the data. Bonferroni correction displays greater efficacy when the number of comparisons is minimal, while Tukey's HSD test exhibits more strength when assessing numerous means (Lee & Lee, 2018). Thus, for the purpose of this analysis, Bonferroni correction was employed. All four groups were subjected to between-group comparisons. Homogeneity of variance was tested using Levene's test, and significant results were observed for all ANOVAs. Accordingly, Welch's F was repor

Table 19. *Respondent characteristics, experiences, and perceptions of crime within neighborhoods used in ordinal regression analysis to determine impact on approval of technology.*

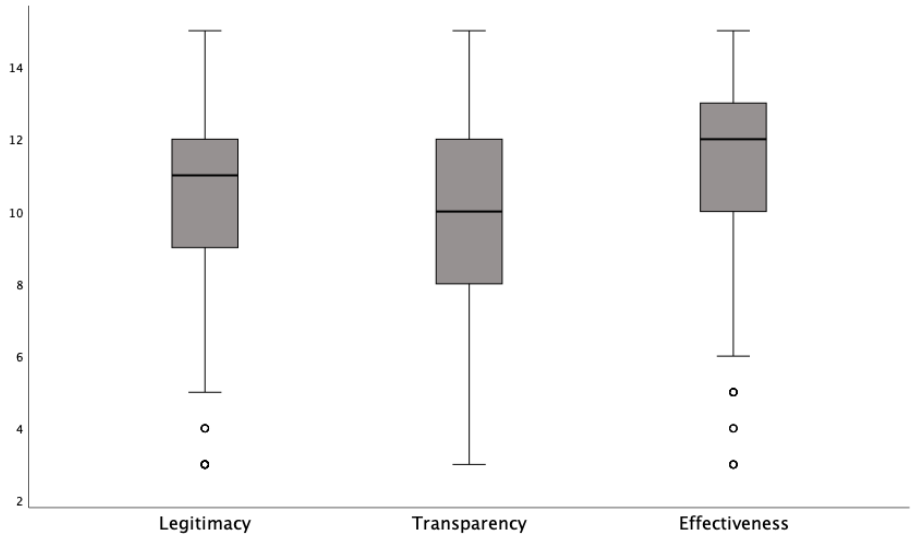
Measures (N=345)	**Control Group (n=83)** **Mean(%)/(SD)**	**Positive Treatment (n=87)** **Mean(%)/(SD)**	**Negative Treatment (n=91)** **Mean(%)/(SD)**	**Neutral Treatment (n=84)** **Mean(%)/(SD)**	***Sig. (Between Groups)*** ***p***
Victim Characteristics					
Age[1]	43(13.5)	31(13.7)	41(13.2)	44(14.1)	.474
Male[2]	43(52%)	41(47%)	46(51%)	39(46%)	.880
Female[2]	40(48%)	46(53%)	45(49%)	45(54%)	.880
White[2]	28(34%)	29(34%)	32(35%)	30(36%)	.987
Black[2]	28(34%)	29(33%)	30(33%)	27(32%)	.997
Hispanic[2]	27(33%)	29(33%)	29(32%)	27(32%)	.997
Income					
Lower Income[2]	56(68%)	52(60%)	55(60%)	51(61%)	.710
Middle Income[2]	19(23%)	25(29%)	23(25%)	23(27%)	.832
Higher Income[2]	8(10%)	10(12%)	13(14%)	10(12%)	.834
Victim Factors					
Crime Victim[2]	18(22%)	19(22%)	26(29%)	22(26%)	.669
Prior Arrest[2]	19(23%)	20(23%)	20(22%)	29(35%)	.210
Prior Interaction[2]	38(46%)	48(55%)	47(52%)	46(55%)	.597
Neighborhood					
Crime Perception[1]	3.92(2.34)	4.38(2.28)	3.99(2.163)	4.21(2.19)	.512

Note. $*p \leq .05$, $**p \leq .01$, $***p \leq .001$. May not equal 100 due to rounding. [1]One-way ANOVA employed to determine significant differences between groups. Fisher's Exact Test was used to determine differences for these dichotomous measures.

The Welch test is commonly used when the assumption of equal variances between two groups and equal sample sizes is violated, a common occurrence in real-world applications (Lakens, 2019). The test is designed to be more robust than the standard t-test when the variances of the two groups are different. The null hypothesis in the Welch test is that there is no significant difference between the means of the two groups, while the alternative hypothesis is that there is a significant difference. The test statistic is calculated as the difference between the means of the two groups divided by the standard error of the difference.

Here it is imperative to acknowledge that the Likert scale, due to its inherent design, follows a pattern whereby a lower mean score is indicative of a stronger negative sentiment towards the variable under investigation. In this particular context, the mean is calculated by summing up the responses to the three questions that pertain to the variable of interest. The specific questions used in each analysis are listed in the reliability testing section. It is noteworthy that the highest achievable score on this scale is 15, which signifies an overall view that the police force is deemed legitimate, transparent, and effective. Conversely, a lower score suggests that the participants perceive the police force as being illegitimate, not transparent, and ineffective in their operational practices. As evidenced in Figure 15., the highest mean scores were achieved for the effectiveness measure, followed closely by legitimacy and transparency.

Figure 15. *Box plot of mean scores across legitimacy, transparency, and effectiveness*



Legitimacy

The police legitimacy mean score of the control group, which did not undergo any intervention, was recorded at 10.41 (SD = 2.98). Treatment group one, which was exposed to a positive vignette, had a higher mean score of 10.92 (SD = 2.71) in comparison to treatment group two, which received a negative vignette, and had a mean score of 10.25 (SD = 3.12). The group that received neutral information achieved the highest mean score of 11.12 (SD = 2.65). Despite Levene's test of homogeneity failing to reveal any significant differences in variances (p =.267), Welch's test was still conducted (p = .156), and a post hoc analysis was performed to visualize the distinctions in mean scores among the groups.

Transparency

The mean score associated with technology transparency for the control group, without an intervention was 9.42 (SD = 2.59). The group that received the positive vignette had a higher mean response score (M = 10.69, SD = 2.33) compared to both the negative

treatment group (M = 9.59, SD = 2.99) and the neutral treatment group (M = 10.30, SD = 2.55). Levene's test of homogeneity suggested significant differences between the variances (p = .005), leading to the reporting of Welch's test (p =.003) that indicated significant differences between the means of the compared groups. This suggests that the difference in means is likely due to the different treatment groups.

To identify where the differences lie, a Bonferroni test was conducted. Results indicated a significant difference between the control group and the positive treatment group with a mean difference of 1.278, revealing that the control group perceived police as significantly less transparent. Furthermore, a significant difference was observed between the positive treatment group and negative treatment group (M =1.096), suggesting that those receiving a positive vignette perceived police as more transparent compared to those who received less favorable information.

Effectiveness

The technology effectiveness mean score for the control group, without intervention, was 11.11 (SD = 2.26). Among the treatment groups, the positive vignette group obtained the highest response mean (M = 11.47, SD = 2.50), surpassing the negative treatment group (M = 10.46, SD = 2.73) and the neutral information group (M = 11.37, SD = 2.61). Levene's test of homogeneity also revealed significant differences between the variances (p = <.041). As a result, Welch's test was reported (p = <.050), indicating that the means of the groups were significantly different from each other. The post hoc analysis indicated that the positive treatment group differed significantly from the negative treatment group in terms of technology effectiveness, with a mean difference of 1.1010.

This finding suggests that the negative group perceived the police to be significantly less effective in their use and. No other significant differences were observed.

Approval of Police Technology Pre-Post Test

Table 20 displays the mean pre-test and post-test scores for police technology approval in each group. To determine the significance of the differences, a paired samples t-test was conducted. The control group had a pre-test mean score of 5.40 (SD = 2.07) and a slightly lower post-test mean score of 5.29 (SD = 1.98). In contrast, the positive treatment group had a higher pre-test mean score of 5.44 (SD = 2.50) and a further increased post-test mean score of 5.64 (SD = 2.38). The negative treatment group had a pre-test mean score of 5.30 (SD = 2.36) and a slightly lower post-test mean score of 5.26 (SD = 2.36). The paired samples t-test showed no significant difference between the pre-test and post-test scores in these groups. However, the neutral treatment group had a significant increase in post-test scores compared to the pre-test means, with scores of 5.61 and 5.93, respectively. The effect size for the difference between the groups was calculated using Cohen's d, resulting in a small effect size or no improvement in the post-test results.

A follow-up analysis was conducted to evaluate whether the differences in pre-post scores among all the treatment groups were significantly different from the pre- and post-test difference in the control sample. The difference-in-difference approach was employed to identify significant variations in outcomes between the treatment and control groups during the pre- and post-treatment phases. This was accomplished by computing the differences in the pre-test and post-test scores of all treatment groups and control group and comparing the differences using an independent samples t-test. The findings revealed significant differences between the mean score differences of the pre-test and post-test for

the neutral treatment group and the control sample, indicating that participants who received an explanation of the technologies achieved significantly higher scores in terms of public approval of police technology.

Table 20. *Paired samples t-test and difference in differences analysis for pre-and post-stimuli responses of approval of police technology*

				Paired Samples T-Test				***Difference in Differences***				
Measures (N=345)	**M**	**Std. Dev**	**S.E. Mean**	**T-value**	***df***	***p***	***Cohen's D***	**Group**	***M***	***SE***	***p***	***Cohen's D***
Control Sample (n=83)								--	--	--	--	--
Pre-test	5.40	2.07	.227	.716	82	.476	-.079					
Post-test	5.29	1.98	.217									
Positive (n=87)								0 vs. 1	.315	.255	.218	.188
Pre-test	5.44	2.50	.269	-1.010	86	.316	.108					
Post-test	5.64	2.38	.255									
Negative (n=91)								0 vs. 2	.075	.264	.776	.043
Pre-test	5.30	2.31	.242	.152	90	.879	-.016					
Post-test	5.26	2.36	.247									
Neutral (n=84)								0 vs. 3*	.525	.240	.030	.339
Pre-test	5.61	2.266	.247	-1.651	83	.103	.180					
Post-test	5.93	2.011	.219									

*p ≤ .05, **p ≤ .01, ***p ≤ .001

Predictors of Increased Public Approval of Technology

Table 21 displays the results of the nested ordinal regression model used to analyze the predictors of increased public approval for the deployment and use of police technology. All independent measures used in these models were examined for multicollinearity using the variance inflation factor (VIF) with no multicollinearity revealed. All VIF factors ranged from a low of 1.082 to the highest at 3.162.[26]

Model one included perceived legitimacy, transparency, and effectiveness of technology, as well as responses from survey participants in the positive, negative, and neutral treatment groups, with the control group as the reference category for the treatment group variables. The analysis revealed that greater levels of police legitimacy were positively associated with higher levels of police technology approval (b =.370, S.E. = .067, p = <.001). Similarly, the same finding held true for effectiveness, where individuals who perceived the police as more effective in their use and deployment of police technology were also more likely to have higher scores on approval of police technology (b =.477, S.E. = .070, p = <.001). Police transparency did not achieve significance in model one, nor did any of the individual treatment group included within the model, although the neutral treatment group was close to achieving statistical significance. Here, the Chi-square and -2 log-likelihood coefficients were statistically significant, indicating good model fit. Moreover, the model accounted for roughly 55% of the variation in perceptions of public approval of technology.

[26] These scores are far below the area of concern for VIF where convention dictates that scores of 5 or greater suggest the presence of multi-collinearity among independent measures.

Model two incorporated respondent characteristics and experiences. Here we see that the Chi-square and -2 log-likelihood coefficients were also statistically significant, indicating a good model fit, and an improvement over the previous model. Police legitimacy and effectiveness remained significant indicating that higher scores were associated with heightened approval of police use and deployment of technology. Transparency scores and treatment groups remained insignificant in this analysis, while only respondent age showed significance among all other variables included in the model. The results indicated that for every one unit increase in age, there was a predicted increase in the log odds of falling at a higher level of police technology approval. The Chi-square and -2 log-likelihood coefficients remained statistically significant, indicating a good model fit. Furthermore, the model explained an improved 56% of the variation in the perceptions of public approval.

Table 21. *Ordinal regression analysis of predictors of increased public approval of the use and deployment of police technology*

Measures **(N=345)**	**Model One** ***b*** **(S.E.)**	**Model Two** ***b*** **(S.E.)**
Legitimacy	.370(.067)***	.339(.068)***
Transparency	-.049(.059)	-.010(.062)
Effectiveness	.477(.070)***	.457(.071)***
Treatment Group		
Positive[1] (ref: control)	-.197(.280)	-.242(.283)
Negative[1]	-.334(.275)	-.376(.276)
Neutral[1]	-.487(.281)	-.483(.285)
Respondent Characteristics		
Age		.021(.008)**
Male		-.246(.280)
Black[2] (ref: white)		-.146(.248)
Hispanic[2]		-.375(.191)
Middle income level[3] (ref:		.139(.229)
low)		.112(.314)
Higher income level[3]		
Respondent Experiences		
Crime victim		.154(.254)
Prior interaction		.151(.213)
Prior arrest		.123(.231)
Neighborhood crime		.038(.046)
perceptions		
Model Fit		
-2 Log Likelihood	1118.082	1173.422
Chi-Square	268.776***	280.190***
Pseudo R-Square		
Nagelkerke	.549	.564

*p ≤ .05, **p ≤ .01, ***p ≤ .001

Chapter Eight: Discussion, Limitations and Conclusion

Introduction

The primary aim of this dissertation was to address a significant gap in the current literature regarding the implementation and use of real-time crime centers. The research concentrated on investigating the disparities in the adoption and implementation of RTCCs on a national scale, their effects on the investigative procedure, including case processing and outcomes within the context of a prominent law enforcement organization in southern Florida, and their impact on public sentiment. This dissertation did not attempt to challenge the findings of the former evaluations that have been published, such as the CPD assessment of the crime reduction impact of district-level SDSC's, but rather aimed to contribute to contribute to the limited body of knowledge that currently exists.

To accomplish the objectives of this study, a comprehensive approach was employed, involving an extensive review of existing literature to develop a typology capable of capturing the nuances of all RTCCs across the nation. Moreover, a quasi-experimental design and network analysis were implemented to evaluate the impact of these centers on both violent and property crimes within the jurisdiction of the City of Miami. Subsequently, a randomized controlled trial was conducted to assess public attitudes and opinions concerning the utilization and implementation of police technologies. The following sections provide a thorough discussion of the findings and limitations of each analytical chapter, followed by an in-depth examination of the policy implications that emerge from the research. Ultimately, the research concludes by proposing further investigations in this area of study.

Discussion of Multi-Perspective Theory Application and Typology

Chapter three of the dissertation utilized three different organizational perspectives, namely the institutional, contingency, and rational perspectives, to provide a framework for understanding the process of adopting and implementing real-time crime centers at a national level. The principal objective of the chapter was to ascertain which of the perspectives was most applicable in comprehending the factors that motivated agencies to adopt and implement an RTCC and how this decision impacted the procurement of technologies and establishment of best practices.

The chapter drew heavily from the research conducted by Willis, Mastrofski, and Weisburd (2007), who used a comparable approach to investigate the organizational transformations associated with the implementation of COMPSTAT in three different police agencies. The authors maintained that the adoption of specific structures is an expected reaction for organizations that operate in a highly developed institutional environment. They also concluded that institutional theory could provide a better explanation for COMPSTAT's internal contradictions than the rational model alone.

The findings here slightly departed from the approach taken by Willis, Mastrofski, and Weisburd (2007) by proposing that each perspective - institutional, contingency, and rational - has inherent value during various stages of RTCC adoption and implementation. Therefore, a multi-perspective approach was formulated to capture the nuances across different RTCC units and the inherent differences their operational structures.

For instance, it was conceptualized that during the initial stage of adoption, external institutional influences may act as compelling catalysts for a neighboring agency to adopt an RTCC unit. In contrast, during the second stage of the RTCC implementation process, the fundamental assumptions of the contingency perspective aligned better with the

structural and operational arrangements of RTCCs that are dependent on factors like funding, stakeholder investments, or community needs. The ultimate stage of the implementation process was deemed ongoing, where the decision-making process concerning the optimal operation of the unit and the procurement of technology can evolve to meet the goals set by the department, as well as adapt to changing societal circumstances over time, aligning with the rational perspective.

Collectively, these perspectives informed the development of the RTCC typology. This resulted in the identification of three primary models - the traditional, community, and virtual models - that exist in the law enforcement domain. The traditional model is characterized by a centralized command and control center responsible for all operations, whereas the community model is decentralized, with multiple smaller centers distributed across a geographic area to cater to specific needs. The virtual model, on the other hand, is fully digital and cloud-based, functioning as a hub for gathering, analyzing, and disseminating critical data and intelligence throughout the established chains of command.

It is important to note that each RTCC model has unique benefits and challenges, and law enforcement agencies should carefully consider their specific needs and circumstances when deciding which model to adopt. For example, while traditional RTCCs may have a more established command structure, they may face difficulties in coordinating with fulfilling the demands of the agency. As indicated in the final MRTCC evaluation report (Guerette et al., 2022), there were some officers that felt that the access to technologies were too restricted and limited to the RTCC unit only. Detectives working within specialized, undercover units, such as the narcotics or gangs' division, stated that they would benefit more from the technologies if they had their own personal access, and

that they often did not want to share classified information with the detectives in the RTCC. This ties into some research that discusses the benefits and drawbacks of centralized investigative units overall (Feeley, 1973; Horvath et al., 2001; Simon, 1947)

While community based RTCCs may possess a more profound comprehension of issues specific to their locale, coupled with the ability to allocate ample time and resources towards predictive analytics, there are still evident limitations that must be acknowledged. For instance, these RTCCs may encounter inefficiencies when attempting to coordinate responses during large-scale incidents or threats, owing to their relatively smaller size and scope of operation. Furthermore, they may lack the centralized resources and sophisticated technological infrastructure of traditional RTCCs, which could inhibit their ability to respond to emergencies in a timely and effective manner. In contrast, virtual RTCCs may provide cutting-edge technological features; nevertheless, their efficacy is contingent upon the existence of a dependable and secure communication infrastructure. Furthermore, the potential for the misapplication of technological resources may exacerbate the chain of command issues. Lastly, monitoring individual accountability and assessing the effectiveness of the technologies within a larger jurisdiction could pose greater challenges in terms of documentation and evaluation.

Nevertheless, an understanding of the inherent benefits and limitations of each real-time crime center model is imperative for law enforcement agencies in making informed decisions regarding the establishment and evaluation of their units. This is vital in order to effectively combat crime in real-time and produce rigorous evaluation reports. Moreover, a deeper understanding of existing RTCC frameworks can prevent an agency from

implementing a unit that is incompatible with its long-term vision of organizational success, thus contributing to sustained success in crime prevention efforts.

Case Processing and Outcomes

This dissertation has highlighted the significance of assessing crimes on an individual basis rather than within a generalized framework. Depending on the distinct attributes associated with each category of crime, specific technological approaches may be more advantageous than others. Therefore, for crimes that involve personal motives or crimes of passion, the availability of solvability factors may inherently vary from those that do not result in harm to a victim. For example, in instances of larceny-theft, the utilization of CCTV coverage or facial recognition may be more beneficial, whereas conventional forms of evidence such as DNA or latent fingerprints may prove more valuable in cases involving homicide or aggravated assault incidents.

Further, upon considering the theoretical frameworks expounded in chapter two, it becomes evident that the findings of this dissertation offer stronger backing for the tenets of the solvability perspective than for those of the devaluation thesis. Although only gender demonstrated statistical significance as a demographic factor, it conflicted with the devaluation thesis by indicating that incidents involving male victims led to poorer case outcomes. Consequently, the findings fail to substantiate the assertion of the devaluation theory, which posits that police discretion in resolving incidents of violent and property crime is influenced by extralegal factors like the victim's demographics and socioeconomic status, resulting in a lower probability of cases being cleared when victims belong to non-white, female, or young categories (Litwin, 2004; Riedel, 2008). This was not observed in the current study.

Discussion of Homicide and Aggravated Assault Incidents

Given the severity of homicides and aggravated assaults and their significant impact on society, these crimes are considered high-profile. As a result, certain factors can affect the clearance rates of these crimes in ways that differ from other types of crimes. Puckett and Lundman's (2003) investigation revealed that law enforcement agencies devote more resources, time, and effort to homicide cases because they are more visible. In addition, Corsianos (2003) found that public outrage and media attention surrounding homicide cases contribute to their increased scrutiny. Consequently, there is increased pressure on law enforcement agencies to solve these cases, resulting in more comprehensive investigations that involve extensive forensic analysis, witness interviews, and other demanding procedures (Cook et al., 2019).

It is apparent that RTCC technologies have been fully incorporated into the case processing of violent crime types. The network analysis of the module revealed a robust association between the homicide and aggravated assault node and two specific RTCC technologies, ShotSpotter, and LPR system. The frequency of firearms involved in such violent incidents within the dataset may explain the influence of RTCC technologies that are optimized for firearm detection and tracking in these cases. Furthermore, the CSI unit and NIBIN were found to be crucial outside of the influence of RTCC technologies. This discovery is not surprising since the MPD recently established an in-house gun crime center

that enables the rapid processing of firearm casings and return of investigative leads to officers.[27]

As it concerns the case clearance model, we see that the influence of the RTCC technologies on the clearance of these crimes is limited. Rather, the case characteristics within the model prove to be better indicators of case clearance in both the initial output and the supplementary analysis conducted, thus supporting the solvability perspective covered in chapter two. Nonetheless, there are certain findings that warrant acknowledgement. For instance, this study found that consistent with prior research, male victims and the involvement of firearms were linked to decreased likelihood of case clearance.

In terms of gender, there are various explanations that could account for these findings. Firstly, biases and stereotypes may propagate the notion that men are less vulnerable to abuse than women, resulting in cases of male victimization receiving less attention (Scott-Storey et al., 2023). Secondly, law enforcement agencies and the public may be less aware of the extent of violence against men compared to women, leading to fewer resources being allocated to identifying perpetrators (Cook, 2009). Additionally, certain areas of Miami, particularly those with high rates of gun violence, are notorious for being heavily influenced by gangs (predominantly male-dominated), where a culture of silence among members is deeply ingrained, and male victims may receive less public support (Bjerregaard & Smith, 1993; Leverso & O'Neill, 2022). Consequently, incidents

[27] For more information regarding the integration of the gun crime center see the following press release: https://mpdc.dc.gov/release/mpd-partners-bureau-alcohol-tobacco-firearms-and-explosives-and-us-attorney's-office-combat

of homicide or aggravated assault may receive minimal cooperation from witnesses, with individuals being less willing to provide their accounts of the events.

When it comes to firearms, the findings are not surprising, as it is widely documented that compared to crimes committed with sharp or blunt instruments that require closer proximity to the victim, firearm-related crimes often lack physical evidence such as latent prints or DNA and often lack the component of victim identification (Braga, Turchan, & Barao, 2019; Cook & Nagin, 1979). Additionally, investigating these crimes can be challenging since many firearms may not be registered or obtained legally, making them challenging to trace back to their owner and challenging for law enforcement agencies attempting to link the firearm to the suspect. For instance, in 2019, the Department of Justice (DOJ) conducted a survey that revealed that around 43 percent of criminals acquired their firearms from the black market, 6 percent obtained them via theft, and 10 percent purchased them from a retail source (McKay, 2020).

Moreover, the logistic regression model revealed a significant decrease in the probability of case clearance with the inclusion of external or private closed-circuit television surveillance. In the investigative process, detectives in the RTCC typically engage in obtaining external CCTV footage through their specialized video retrieval specialists, particularly when their internal cameras are ineffective or do not provide adequate coverage. To access these external surveillance feeds, the police agency often enters into private-public partnerships or a memoranda of understanding with local businesses or establishments. Alternatively, if such partnerships do not exist, cooperative witnesses or individuals may submit privately owned surveillance footage. However,

cooperation may depend on the prevailing street culture, and uncooperative individuals may require detectives to obtain a warrant.

As per Cook and Berglund (2021), the widespread availability of video and other digital resources has brought about a distinctive set of difficulties for investigators. Although law enforcement officers can readily access video footage from city cameras, it is generally not the case for recordings from private cameras. The process of identifying the sources of potentially valuable footage and obtaining it from proprietors can be time-consuming and frequently unfruitful. As a result, the temporal dimension and promptness of video surveillance acquisition are frequently cited as barriers to the successful procurement of video evidence (Morgan & Dowling, 2019).

Suspect lineups and public information flyers are perhaps the best predictor of case clearance in the investigation of homicides and aggravated assaults. The utility of suspect lineups as a valuable tool in the investigative process stems from several factors (Davis, Gibson & Solomon, 2014; Higginson, Eggins & Mazerolle, 2017). Primarily, visual identification of suspects through such lineups can greatly aid investigators in the strenuous task of sifting through potential suspects and apprehending the true culprit. Moreover, properly executed lineups can heighten the possibility of witness identification of the culprit, thereby providing significant evidentiary value in building a compelling case against the suspect. The corroborative role of suspect lineups, particularly when used in conjunction with other forms of evidence, such as physical or forensic evidence, can significantly bolster the overall strength of the case.[28]

[28] It is acknowledged that suspect lineups have undergone some controversy as it pertains to reliability issues (Wixted & Wells, 2017).

Similarly, providing the general public with information flyers that either offer monetary compensation for the reporting of relevant information to law enforcement authorities or provide details about the identity of the suspect(s) can be an effective means of enhancing case clearance rates. These flyers can offer a promise of anonymity to the informant, which can encourage them to come forward with vital information that could be crucial in solving the case. The dissemination of such flyers can also serve as a deterrent to potential suspects and encourage them to surrender or seek legal assistance before being apprehended by law enforcement agencies. Nonetheless, there is no empirical research to support this claim, beyond this study.

Lastly, in relation to the time required to clear homicide and aggravated assault cases, it is apparent that when not accounting for case-level characteristics, the duration for cases receiving RTCC-support does not significantly differ from cases without support. The same holds true when examining RTCC-assisted homicide and aggravated assaults compared to all other RTCC-assisted violent crimes, while controlling for case-level characteristics. Notably, when a firearm is involved, it takes investigators longer to clear a crime due to additional stages in the investigative proceedings. Additionally, the lengthier time to clearance may also be attributed to the increased likelihood of obtaining search or cell phone warrants, which may further prolong the time required to acquire sufficient evidence for a conviction.

Discussion of Robbery Incidents

Robbery, unlike other forms of violent crime such as homicide or assault, is generally categorized as an instrumental crime. This type of criminal behavior is primarily motivated by achieving a specific and tangible goal, such as acquiring property, rather than

being an expression of emotion or impulse (Cohn & Rotton, 2003). Despite its significance in criminology, the literature on robbery has received considerably less attention compared to other forms of violent crime. Additionally, the available literature on robbery outcomes has mostly focused on crime prevention, neglecting the aspect of crime clearance. To bridge this gap, chapter five aimed to investigate not only the established factors that influence case clearance in robbery cases but also the impact of real-time crime center technologies on these outcomes.

As demonstrated by the module network analysis, the ShotSpotter gunshot detection system become a critical factor in the investigation of robbery incidents within the MPD RTCC setting. When examining individual cases, it was found that the use of firearms as a means of intimidation, either before or after the commission of the incident, was prevalent. It is also possible that during the process of data collection, a case initially reported to dispatch as a robbery may have been subsequently reclassified as an aggravated assault over the course of the investigation. Nonetheless, it is crucial to recognize that if the perpetrator did not cause any bodily harm and only stole property, the crime would still be classified as a robbery according to the established procedures of the MPD.

The LPR system was also influential in the case processing of robbery incidents, as it was in homicide and aggravated assault incidents. Interestingly, the utilization of vehicles during the commission of robberies was more prevalent than in the combined instances of homicides and aggravated assaults, although it reduced the chances of case clearance. Willis, Koper, and Lum (2018) suggested that law enforcement officials could enhance their use of the LPR system beyond merely identifying stolen vehicles by exploiting it as a valuable tool for collecting a diverse range of evidence that would bolster witness

accounts. Furthermore, detectives in Miami have noticed a surge in robbery incidents reported fraudulently to facilitate insurance claims. Consequently, the LPRs may have been employed more extensively in these cases to mitigate the effect of such fraudulent claims.

Once more, the findings suggest that the RTCC treatment group did not experience a notable improvement in the time to clearance, as previously hypothesized. It could be argued that this result is due to the RTCC's acquisition of additional investigative leads and the implementation of further steps to gather evidence, which may have prolonged the successful resolution of cases. Additionally, compared to homicides and aggravated assaults, robbery incidents tended to require a longer clearance time on average. Beyond the significant variables of the victim's gender and the dissemination of a public information flyer, it is evident that the implementation of a lineup also resulted in longer clearance times for robbery cases. This observation aligns with the nature of a robbery, as victims typically have more direct contact with their perpetrator during the crime, resulting in the theft of items in their presence.

Finally, the supplementary analysis, which only accounts for only the robbery treatment control sample, puts forth a number of propositions. Firstly, it indicates that RTCC technologies have a disproportionately positive effect on incidents of robbery in comparison to cases of aggravated assault and homicide. Given, a recent assessment of sexual assault incidents and homicides which a markedly negligible impact of Miami RTCC technologies (Przeszlowski, Guerette, & Sudderth, 2023*)*, this may suggest that the presence of robberies may have significantly influenced the broader outcomes of the Miami RTCC evaluation which was concluded in the preceding year (Guerette, et al., 2022).

Here, the supplemental analysis also revealed that the clearance of robbery incidents was more likely to occur through the use of public information flyers or lineups rather than through victim or witness interviews. Although this finding deviates from previous research, it is worth noting that prior studies have often failed to consider additional variables that could impact these results due to over reliance on NIBRS data (Roberts & Lyons, 2009; Overstreet, McNeeley, & Lapsey, 2021).

Discussion of Property Crime Incidents

The findings of property incident study are multi-fold. Notably, it was discovered that police personnel and investigators are significantly less likely to clear property crimes compared to violent crimes, even with the influence of RTCC technologies. This result is not unexpected, as the clearance of crimes with a discernible victim is often prioritized over property crimes in law enforcement practices (Braga and Dusseault, 2018; Cook et al., 2019). Moreover, while the current study did not specifically examine the workload of investigators handling burglary and larceny theft incidents, it is assumed, based on precedent research, that these investigators are often overburdened compared to those assigned to homicide or assault cases (Roberts & Roberts, 2016; Worrall, et al., 2004).

Nonetheless, the highly connected network analysis model demonstrated that RTCC technologies have become a nearly standard practice in property crime investigations, as compared to all other forms of violent crime. However, this raises the question as to why the impact of technology is not more pronounced in the case clearance rates of property crime incidents. Some plausible explanations are the following. Particularly in scenarios of burglary or home invasion, investigators frequently depend on the surveillance recordings obtained from private security cameras mounted outside

residential dwellings or commercial establishments. Victims of such crimes may be more inclined to assist law enforcement personnel and provide such footage, especially in comparison to incidents that are of a more personal nature. However, as noted by Brookman and Jones (2021), despite being the most frequently used forensic tool in identifying offenders of these types of crimes, the utilization of CCTV in investigations presents various challenges, such as inadequate video quality for suspect identification. Thus, any footage received from external or private CCTV surveillance may not be useful, which leads to the subsequent point of consideration.

At the time of data collection, the command staff responsible for overseeing the property crime detectives at MPD mandated that all CCTV images be forwarded to the RTCC, irrespective of their resolution or potential usage facial recognition software. Accordingly, The RTCC detectives, as per their established procedural norms, were duty-bound to subject these images to the software, even if their quality fell short of optimal standards. Consequently, the RTCC team occasionally found itself flooded with image requests from the property unit detectives. So, while this approach could potentially facilitate the identification of more leads and contribute towards the investigation of property crimes, its efficacy in producing a significant impact thus remains uncertain.

Furthermore, in regard to the timeframe required for clearing property crimes, it is apparent that no substantial distinction exists in the duration between the control and treatment cohorts, nor when contrasted with other violent crimes handled by the RTCC. This observation has consistently emerged throughout the dissertation's findings, leading to the proposition that the use of multiple sources of evidence in an investigation neither abbreviates nor elongates the time required for case resolution. In this context, it is apparent

that incidents with public information flyers, which frequently necessitate the use of evidence acquired from RTCC CCTV or external CCTV, are associated with longer clearance times. Although it may seem logical that access to tools that facilitate the identification of suspects would expedite the resolution of crimes, the retrieval of video footage and the subsequent need for additional assistance from the RTCC can sometimes be impeded by "red tape" or bureaucratic procedures (Cook and Berglund, 2021). This tendency is particularly noteworthy for property crimes, which may receive lower priority in the triage of crime reports and requests at the RTCC unit.

Discussion of Public Perception Study

The diminishing confidence in the police, as revealed by the Gallup survey, highlights a significant shift in public perception and underscores the evolving dynamics between law enforcement agencies and the communities they serve (refer to chapter seven). This decline may indicate a growing concern or dissatisfaction among Americans with regards to certain aspects of policing, potentially stemming from high-profile incidents, inappropriate use of technology, or systemic issues that have garnered widespread attention in recent times. As a result, it becomes crucial for policymakers, law enforcement authorities, and society at large to address these concerns and work towards rebuilding trust and ensuring effective and accountable policing practices.

Taken together, the findings of chapter seven present a compelling picture. First, it seems that the stimuli utilized in the study had a discernible impact on perceptions of police technology use and deployment. In comparison to the positive treatment group, the treatment group exposed to negative stimuli exhibited significantly lower scores on their perception of police transparency and effectiveness. This suggests that when individuals

are presented with less favorable information regarding technology use and deployment, such as excessive officer discretion, they may perceive agencies as failing to meet their personal standards for acceptable behavior.

To delve deeper into the impact of the stimuli, additional analyses were conducted. These analyses sought to examine not only the differences in pre- and post-test scores of public approval between all treatment groups and the control group, but also to evaluate the mean differences in these scores compared to the control group that did not receive any stimuli. Surprisingly, the findings revealed that individuals who received only a neutral description of police technologies scored significantly higher on the post-test assessing their approval of police technology than the control group. Furthermore, the mean scores were notably higher than those of the groups that were exposed to positive or negative manipulation.

The findings suggest several significant implications for law enforcement. One crucial aspect is the need for agencies to educate the public about the technologies they use in order to increase public acceptance and approval. It is essential to tailor communication to the specific jurisdiction that an agency serves because each agency operates differently, as documented in chapter three. For instance, consider facial recognition and biometric fingerprint data mechanisms. CVPD is more advanced and progressive than MPD in their deployment of these technologies in their encounters with the public. For example, while MPD only uses facial recognition within the RTCC setting, CVPD allows personnel to use the tool in patrol settings via an application installed on a cellphone or tablet (Strain & Garske, 2013). Additionally, while MPD documents and analyzes latent/DNA data through their CSI unit, other agencies, such as the Anaheim Police Department, have portable

fingerprint scanners for patrol officers (Peralta, 2016). It is important to note that national news or press releases may convey that these practices are standard throughout all jurisdictions, which is not *yet* the case.

Furthermore, the findings of the ordinal regression analysis suggest that the critical factors influencing the acceptance or rejection of police technology are heavily influenced by individuals' perceptions of legitimacy and effectiveness. Law enforcement agencies can utilize this valuable insight by carefully considering their technology acquisitions. For instance, Strom's (2017) research demonstrates that agencies often adopt new technological innovations based on the decisions of executive staff and funding availability, rather than a comprehensive policing strategy. Frequently, an agency member witnesses a technology demonstration elsewhere and perceives a pressing need for the same product within their agency, which aligns with the institutional perspective discussed in chapter three.

Consequently, the agency seeks to justify the technology's purchase to meet its perceived needs, even though the purchase itself is essentially a solution in search of a problem to solve. Such thinking is counterproductive and may lead agencies to procure unsuitable technology that fails to meet their actual needs, resulting in unfulfilled objectives in the long run and increased concerns from stakeholders and the public (Scism & Gallegos, 2020). As a result, the agency's effectiveness and legitimacy may suffer, ultimately resulting in reduced public confidence in police practices.

Limitations

With respect to chapter three, the theories employed in formulating a comprehensive framework for understanding the variations in RTCC models at the national level were not empirically tested, but rather explored to determine their intrinsic value in

the context of these units. Moreover, the typology presented herein ought to be considered preliminary but well-informed, given some of the following reasons. First and foremost, the typology was developed utilizing data obtained from a diffusion of innovation study and survey that was disseminated to sixty-one police agencies across the country. The objective was to attain a more profound understanding of their operational and organizational frameworks. Additionally, leveraging my expertise as a criminologist and having worked as a research assistant for multiple years with the Miami Police Department, I accumulated substantial knowledge not only from that centralized unit but also from external agencies, including the Chicago Police Department, who collaborated with the Miami RTCC on several projects and working groups.

Regarding the chapters pertaining to case processing and outcomes, a significant limitation arose regarding the ability to determine the admissibility of evidence obtained by RTCC detectives and external unit investigators. Specifically, the coding of variables was based solely on whether these personnel utilized sources of evidence or technologies, without considering whether such evidence ultimately proved to be valuable in the investigation process. For instance, although the dataset included CCTV footage, the actual evidentiary value of the footage was not taken into account. Additionally, insufficient sample sizes did not allow for the independent consideration of homicide incidents, aggravated assaults, and the crimes assessed under the property crime model.

Moreover, questions may be raised as to the selection of the control group throughout the studies that consisted of a quasi-experimental design. Despite being selected at random, there may be apparent potential selection effects as there are significant differences between the treatment and control groups. For example, within the violent

crime models, the treatment group had many more incidents involving firearms and vehicles routed through the RTCC, as compared to the control group which may have, to some extent contributed to the clearances depicted, specifically as it pertained to robbery incidents.

Finally, as it concerns the public perception study, the results of the assessment may have proved different if the survey was administered in person, as compared to through an online survey marketing tool. Despite the online marketing tool taking all the precautionary steps to avoid "professional survey takers," the true extent to which this is avoidable is not documented (Andrade, 2022). Nonetheless, the stratification utilized and the RCT design, did alleviate some of the concerns traditionally encountered with this type of research.

Future Research

In order to advance the field, future research must strive to address the aforementioned limitations inherent in the current study. Furthermore, although the current investigation did not examine the nuances of the individual investigative units responsible for case processing, such as those handling assaults versus robberies, it is imperative to conduct further investigation in this area to achieve a more complete understanding of the RTCC's utilization. This is particularly relevant given the notable disparities in case processing observed between property crimes and violent offenses within this dissertation.

There may also be certain police agency educational components that warrant further investigation or insight. Specifically, as it pertains to questions about how the agencies that house real-time crime centers or comparable innovative technological advancements share their information with the populace. For example, perhaps, certain

agencies have a better educational or community-oriented community embedded within their framework and are viewed more favorable by the public. Empirical assessments of this sort, with emphasis on the community-relation unit within police agencies has not garnered much attention, despite playing a significant role.

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APPENDIX

APPENDIX A

Code Book for RTCC Treatment and Control Datasets

Column	*Label*	*Code*
Unique identifier*	Case number	Unique numeric value.
Date/Time Incident	Date and time of incident.	Date in mm/dd/yyyy and time in hh:mm format.
Incident Day	Day of the week that the incident occurred.	Categorical
Type*	Whether the MRTCC response was real-time, post-incident, or proactive monitoring.	Real-time, post-incident, proactive monitoring.
Date Logged*	The date the MRTCC personnel logged the incident.	Date in mm/dd/yyyy format
Time Logged*	The time the incident was logged by MRTCC personnel.	Time in hh:mm format.
Signal*	The type of crime that occurred.	The type of crime that occurred.
Unit Assisted*	The individual unit that the MRTCC assisted (e.g., homicide, burglary).	The individual unit that the MRTCC assisted (e.g., homicide, burglary).
Location*	The address of the incident location.	Descriptive variable
Net Area*	The net area of the incident location.	Categorical
Felony*	The cumulative number of felony charges associated with the arrest.	Numeric; discrete variable

Misdemeanor*	The cumulative number of misdemeanor charges associated with the arrest.	Numeric; discrete variable
Number of Arrests*	The cumulative number of arrests associated with the incident.	Numeric; discrete variable
Arrestee Age	The age of the individual at the time of arrest. If more than one individual was arrested, the demographics of the first arrestee were included.	Numeric; discrete variable.
Arrestee Sex	The sex of the individual at the time of arrest. If more than one individual was arrested, the demographics of the first arrestee were included.	Dichotomous; 1 for male, 0 for female.
Arrestee Race	The race of the individual at the time of arrest. If more than one individual was arrested, the demographics of the first arrestee were included.	Dichotomous, 1 for Black, 0 for White.
Date/Time Arrest	Date and time of first arrest.	Date of arrest in mm/dd/yyyy format. Time of incident in hh:mm format.
Hours to Clearance	The number of hours calculated between the incident time and arrest time.	The formula used: =INT(arrest-incident)*24
Days to Clearance	The number of days calculated between the incident time and arrest time.	The formula used: =INT(arrest-incident)*24, followed by x/24.
Victim Age	The age of the victim at the time of arrest. If more than one individual was involved,	Numeric; discrete variable.

	the demographics of the first victim listed were included.	
Victim Sex	The sex of the victim at the time of arrest. If more than one individual was involved, the demographics of the first victim listed were included.	Dichotomous; 1 for male, 0 for female.
Victim Race	The race of the victim at the time of arrest. If more than one individual was involved, the demographics of the first victim listed were included.	Dichotomous, 1 for Black, 0 for White.
Weapon (Other)	Whether a weapon other than a handgun was involved.	Dichotomous; coded 1 for yes, 0 for no.
Firearm*	Whether a firearm was involved.	Dichotomous; coded 1 for yes, 0 for no.
Vehicle Involved*	Whether a vehicle was involved.	Dichotomous; coded 1 for yes, 0 for no.
Narcotics Recovered*	Whether narcotics were recovered.	Dichotomous; coded 1 for yes, 0 for no.
Notes*	MRTCC personnel case notes.	Descriptive; narrative form.
IBM*	MRTCC personnel identification number.	Descriptive; numeric form.
# Officers	The number of officers involved in the incident. This does not include MRTCC personnel.	Numeric; discrete variable.
External CCTV	Whether external CCTV was accessed.	Dichotomous; coded 1 for yes, 0 for no.
External CCTV Evidence	Whether external CCTV provided evidence.	Dichotomous; coded 1 for yes, 0 for no.

No CCTV	Indicates that there were no accessible CCTV cameras in the area (external and MRTCC).	Dichotomous; coded 1 for yes, 0 for no.
MRTCC CCTV	Whether MRTCC CCTV was accessed.	Dichotomous; coded 1 for yes, 0 for no.
CCTV Evidence	Whether there was video evidence obtained.	Dichotomous; coded 1 for yes, 0 for no.
Social Media	Whether a social media search was conducted.	Dichotomous; coded 1 for yes, 0 for no.
Social Media Evidence	Whether there was social media evidence gathered.	Dichotomous; coded 1 for yes, 0 for no.
Facial Recognition	Whether a facial recognition (Clearview or Faces) search was conducted.	Dichotomous; coded 1 for yes, 0 for no.
Facial Recognition Evidence	Whether a facial recognition match was found.	Dichotomous; coded 1 for yes, 0 for no.
LPR	Whether the LPR system was utilized (Clarity or Vigilant).	Dichotomous; coded 1 for yes, 0 for no.
LPR Evidence	Whether LPR system provided evidence.	Dichotomous; coded 1 for yes, 0 for no.
ShotSpotter	Whether a ShotSpotter alert was associated with the case.	Dichotomous; coded 1 for yes, 0 for no.
Rounds	The number of rounds associated with the ShotSpotter alert.	Numeric; discrete variable.
NIBIN	Whether NIBIN was utilized.	Dichotomous; coded 1 for yes, 0 for no.
GSR	Whether there was gunshot residue collected.	Dichotomous; coded 1 for yes, 0 for no.

Latent/DNA	Whether there was latent, or DNA swabs were conducted.	Dichotomous; coded 1 for yes, 0 for no.
Lineup	Whether a photograph or in-person line up was conducted.	Dichotomous; coded 1 for yes, 0 for no.
Lineup Evidence	Whether evidence resulted from a photograph or in-person line up	Dichotomous; coded 1 for yes, 0 for no.
#Suspects	The total number of suspects involved in the case.	Numeric; discrete variable.
#Witnesses	The total number of witnesses involved in the case.	Numeric; discrete variable.
#Victims	The total number of victims involved in the case.	Numeric; discrete variable.
#Victim Interview	Whether a victim interview was conducted.	Dichotomous; coded 1 for yes, 0 for no.
#Witness Interview	Whether a witness interview was conducted.	Dichotomous; coded 1 for yes, 0 for no.
Cell Warrant	Whether a cellphone warrant was utilized.	Dichotomous; coded 1 for yes, 0 for no.
Search Warrant	Whether a search warrant was utilized.	Dichotomous; coded 1 for yes, 0 for no.
Flyer	Whether a "need to identify (NTI) flyer" was disseminated.	Dichotomous; coded 1 for yes, 0 for no.
CSI	Whether the crime scene unit was requested.	Dichotomous; coded 1 for yes, 0 for no.
Apprehension Team	Whether an apprehension team was utilized.	Dichotomous; coded 1 for yes, 0 for no.

*Denotes that the variable was used as part of the initial MRTCC internal log

APPENDIX B

Comparison of RTCC-Assisted Cases and Randomly Drawn Control Sample by Crime Type and Neighborhood Area Stratifications (N = 1,524)

Variable	**RTCC-assisted cases n (%)[1]**	**Control Sample n (%)[1]**
Crime types		
Homicide	62(8%)	62(8%)
Aggravated assault	232(30%)	232(30%)
Assault	53(7%)	53(7%)
Sexual assault	23(3%)	23(3%)
Domestic violence	45(6%)	45(6%)
Robbery	233(31%)	233(31%)
Burglary	82 (11%)	82 (11%)
Larceny Theft	32(4%)	32(4%)
Neighborhood Areas		
Allapattah	90 (12%)	94 (12%)
Brickell	29 (4%)	29 (4%)
Coconut Grove	21 (3%)	20 (3%)
Coral way	8 (1%)	8 (1%)
Downtown	84 (11%)	84 (11%)
Edgewater	12 (2%)	13 (2%)
Flagami	24 (3%)	24 (3%)
Little Haiti	116 (15%)	112 (15%)
Little Havana	47 (6%)	50 (7%)
Model city	109 (14%)	109 (14%)
Overtown	133 (17%)	133 (17%)
Upper East Side	15 (2%)	15 (2%)
Wynwood	74 (10%)	70 (9%)
Total	N = 762	N = 762

Notes: 1. May not equal 100 due to rounding. 2. For crime types, Chi-Square = .000, 7 df, $p = 1.000$, no significant differences. For neighborhood areas Chi-Square = 0.462, 12 df, $p = 1.000$, no significant differences.

APPENDIX C

RTCC Perception Survey

[Consent for Pollfish]

SQ1.The following survey contains questions regarding police technology. Are you willing to take part in the following survey? Your responses are anonymous.

[Consent for FIU]

Please take a few minutes to complete the following survey. The survey consists of 29 questions and should take no more than 15 minutes of your time. The purpose of the survey is to better understand the nature and use of real-time crime center (RTCC) surveillance technologies and their impact on public perceptions in the United States. The survey responses will be used solely for research, and all information obtained herein will be kept completely confidential. Results of all surveys will only be reported in the aggregate, meaning responses will not be attributable to any single individual. Your participation is completely voluntary, and you may choose not to participate at any time throughout the course of the survey.

Q1. What is your gender?

Q2. What best describes your political affiliation?

Q3. Science and technology are making our lives, healthier, easier, and more comfortable.

Q4. Have you been a victim of crime in the past three years?

Q5. Have you ever been arrested for a crime?

Q6. Have you had a personal interaction with a police officer in the past three years?

Q7. If yes, how would you best describe that interaction with the police officer?

Q8. On a scale of 1-10 how common is crime in the neighborhood that you live in?

[Pre-Test]

Q9.Overall, on a scale of 1 to 10, indicate your approval of police use and deployment of technology.

Q10.I trust that police agencies deploy police technologies in accordance with data-driven decision-making.

Q11.I support the use of police technology as it is deployed in a manner that best serves the community.

Q12. Police agencies are transparent as it concerns the use and deployment of police technology.

Q13.I consider myself familiar with the technologies used by police agencies.

Q14. Police use of technology is effective in reducing crime and victimization.

Q15. Police use of technology has a positive impact on the efficiency and effectiveness of police practices.

[Stimuli Here]

Q16. Police technology is used and deployed in a legitimate manner.

Q17. Police technology is used and deployed in a transparent manner.

Q18. Police technology is used and deployed in an effective manner.

Q19. Police agencies utilize and deploy technology in a manner that serves the best interest of the public.

Q20.I trust police officer decision-making as it pertains to the use and deployment of innovative technology.

Q21.I consider myself informed on the vast array of technology utilized by police officers and law enforcement personnel.

Q22.Police agencies are transparent about their use of innovative technologies.

Q23. The use of innovative technology within police agencies has the potential to improve the efficiency and effectiveness of police practices dramatically.

Q24. Police technology is effective in reducing crime rates and improving public safety.

[Post-Test]

Q25. Overall, on a scale of 1 to 10, indicate your approval of police use and deployment of technology.

Q26.I would like more technologies deployed by police in my neighborhood and places where I work.

Q27.I would like less, or no technologies deployed by police in my neighborhood and places where I work.

Q28.I support police use of technologies with proper oversight and procedure in my neighborhood and places where I work.

Q29.I support the absolute and unregulated use of technology by police agencies in my neighborhood and places where I work.

APPENDIX D

Vignettes Deployed as Stimuli Within the Three Respective Treatment Group

Positive Vignette	Police agencies across the country are beginning to implement units known as real-time crime centers. These units are equipped with innovative technologies such as closed-circuit television (CCCTV), facial recognition software, license plate readers, and gunshot detection systems, amongst others. These technologies are deployed to high-crime areas based on analysis of historical crime data. Agencies are careful to not violate citizen privacy rights. Police agencies often publish information on the use of technologies on social media or other outlets and ensure that the technologies are deployed in ways that are open and able to be seen by community members. Real-time crime centers offer assistance to responding officers, provide information to officers and detectives after incidents have occurred and can monitor live community events (such as protests of entertainment locations). The primary purpose of real-time crime centers is to improve public safety and reduce criminal activity. For example, surveillance cameras combined with gunshot detection systems can alert patrol officers immediately after shots are fired, allowing them to respond to incidents faster. In most cases, officers are able to either assist victims and/or arrest the offender who committed the crime more quickly.

Negative Vignette	Police agencies across the country are beginning to implement units known as real-time crime centers. These units are equipped with innovative technologies such as video cameras in neighborhoods, facial recognition software, license plate readers, and gunshot detection systems, amongst others. These technologies are deployed within high-crime, minority communities and based on police discretion. They are also used to monitor live community events such as protests and entertainment locations. There are no ways to ensure that certain technologies such as facial recognition software and searches of social media platforms are done in a manner that ensures citizen privacy rights. Police agencies often do not publish information, nor do they make known how they use these technologies. For operational security, police have decided to not reveal when, where and why these technologies are being used. There is no evidence that these systems are effective in reducing crime or improving community safety.
Neutral Vignette	*Police technology*: Police technology refers to the broad range of electronic devices, computer equipment and specialized software used to perform police work. As technology has evolved, so has the reliance on technology by police officers and law enforcement officials to prevent crime and apprehend criminals. Closed-circuit television (CCTV), facial recognition software, gunshot detection systems, and automated license plate readers (ALPRs) are all examples of innovative technologies acquired by police agencies nationally. (1) *CCTV (surveillance cameras):* Typically used to identify suspects in offenses that are committed within view of the camera. (2) *Automated license plate readers (ALPRs)*: Computer-controlled camera systems attached to either police cars or street polls that capture all license plate numbers that come into view, along with the location, date, and time. (3) *Gunshot detection system (ShotSpotter):* ShotSpotter consists of audio sensors implemented throughout a targeted area (generally on top of buildings of similar locations) that detect the sound of gunfire and identify the location of an incident. If it is confirmed to be gunfire, the relevant information (including time, place, and a recording of the incident) is sent to local police so that they can quickly go to the scene. (4) *Facial recognition software:* Allows officers to submit images of people's faces, taken in the field or lifted from photos or video, and instantaneously compares them to photos in government databases – mugshots, jail booking records, driver's license.

APPENDIX E

One-way ANOVA with Bonferroni post hoc group comparison of post-test police legitimacy scores

	Control (83)	Positive (87)	Negative (91)	Neutral (84)	One-way ANOVA Welch's *F*	Post Hoc Analysis (Bonferroni)		
	Mean(SD)	Mean(SD)	Mean(SD)	Mean(SD)		Group Comp.	Mean Diff.	*p*
Variables of Interest								
						0 vs. 1	-.510	1.000
						0 vs. 2	.157	1.000
						0 vs. 3	-.709	.670
						1 vs. 0	.510	1.000
						1 vs. 2	.667	.736
Legitimacy	10.41(2.98)	10.92(2.71)	10.25(3.12)	11.12(2.65)	$F(3,189)=1.760$, $p<.156$	1 vs. 3	-.200	1.000
						2 vs. 0	-.157	1.000
						2 vs. 1	.667	.736
						2 vs. 3	-.866	.283
						3 vs. 0	.709	.670
						3 vs. 1	.200	1.000
						3 vs 2	-.866	.283

Note. Minimum score of three, indicating lowered perceptions of police legitimacy, maximum score of 15 indicating heightened perceptions of police legitimacy.

One-way ANOVA with Bonferroni post hoc group comparison of post-test police transparency scores

	Control (83)	Positive (87)	Negative (91)	Neutral (84)	One-way ANOVA Welch's *F*	Post Hoc Analysis (Bonferroni)		
	Mean(SD)	Mean(SD)	Mean(SD)	Mean(SD)		Group Comp.	Mean Diff.	*p*
Variables of Interest								
						0 vs. 1*	-1.268	.011
						0 vs. 2	-.172	1.000
						0 vs. 3	-.876	.193
						1 vs. 0*	1.268	.011
						1 vs. 2*	1.096	.035
Transparency	9.42(2.59)	10.69(2.33)	9.59(2.99)	10.30(2.55)	*F*(3,189)= 4.745, *p* .003	1 vs. 3	.392	1.000
						2 vs. 0	.172	1.000
						2 vs. 1*	-1.096	.035
						2 vs. 3	-.704	.467
						3 vs. 0	.876	.193
						3 vs. 1	-.392	1.000
						3 vs. 2	.704	.467

Note. Minimum score of three, indicating lowered perceptions of police transparency, maximum score of 15 indicating heightened perceptions of police transparency.

One-way ANOVA with Bonferroni post hoc group comparison of post-test police effectiveness scores

	Control Group (83) Mean(SD)	Positive Treatment (87) Mean(SD)	Negative Treatment (91) Mean(SD)	Limited Intervention (84) Mean(SD)	One-way ANOVA Welch's *F*	Post Hoc Analysis (Bonferroni)		
						Group Comp.	Mean Diff.	*p*
Variables of Interest								
						0 vs. 1	-.363	1.000
						0 vs. 2	.647	.561
						0 vs. 3	-.261	1.000
						1 vs. 0	.363	1.000
						1 vs. 2*	1.010	.049
Effectiveness	11.11(2.26)	11.47(2.50)	10.46(2.73)	11.37(2.61)	$F(3,189)=$ 2.582, p .050	1 vs. 3	.102	1.000
						2 vs. 0	-.647	.561
						2 vs. 1*	-1.010	.049
						2 vs. 3	-.908	.111
						3 vs. 0	.261	1.000
						3 vs. 1	-.102	1.000
						3 vs. 2	.908	.111

Note. Minimum score of three, indicating lowered perceptions of police transparency, maximum score of 15 indicating heightened perceptions of police effectiveness.

VITA

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PUBLICATIONS AND PRESENTATIONS

2022 Przeszlowski, K., Guerette, R.T., Lee-Silcox, J., Rodriguez, J., Ramirez, J., & Gutierrez. A. The centralization and rapid deployment of police agency information technologies: An appraisal of real-time crime centers in the U.S. *The Police Journal: Theory, Practice and Principles.*

2022 Guerette, R.T., Przeszlowski, K., Major Jose Rodriguez, Lieutenant Jaime Ramirez, Sergeant Alejandro Gutierrez, and Joelle Lee-Silcox. "The Miami Real-Time Crime Center (MRTCC) Violence Response Initiative: An Embedded Police-Academic Partnership. Report submitted to the Smart Policing Initiative, Bureau of Justice Assistance, U.S. Department of Justice.

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